

SciVelyx Graphs

User manual

Science in Motion. Data in Focus.

To empower researchers worldwide with fast, precise and intuitive tools for scientific data analysis and visualization, transforming complex datasets into clear insights and accelerating the path from data to discovery.

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This manual is also on the web, where it is updated with each version:

https://laribeirojr.github.io/scivelyx/graphs/docs/

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SciVelyx Graphs user manual

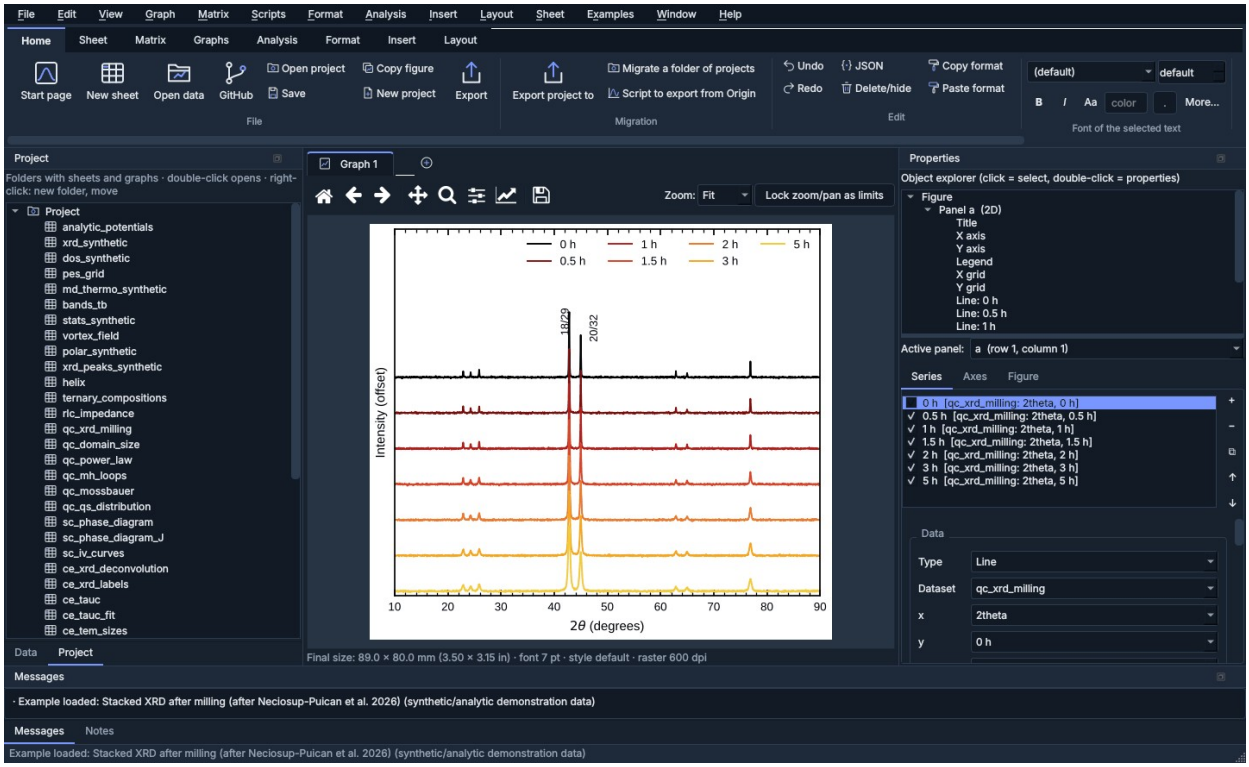
Science in Motion. Data in Focus.

SciVelyx Graphs turns measured or simulated data into publication figures and analyses them. It has:

- sheets with formulas;
- 176 graph types, in which every element can be edited;
- curve and peak fitting, spectroscopy and X-ray diffraction tools;
- statistics, matrices and Python scripts;
- the migration of Origin, SciDAVis and QtiPlot projects.

Our mission: to give researchers everywhere fast, precise and intuitive tools for analysing and visualising scientific data, so that complex datasets become clear insights and the path from data to discovery is shorter.

The PDF has the same chapters as these pages, for reading offline or printing. Short video tutorials, filmed in the program, are on the SciVelyx channel on YouTube.



SciVelyx Graphs with a figure being edited

Where to start

If you want to...	Read
install the program	Installation
know the parts of the window	The window
follow complete examples (XRD, Raman, band gap, implicit fit)	Worked examples
make a figure from a data file in five minutes	Your first graph
prepare data: import, formulas, units, masks	Sheets and data
change anything on a figure	Editing graphs
fit curves, analyse peaks, run tests	Fitting and analysis
X-ray diffraction, Raman, instrument files	Spectroscopy and diffraction
PCA, clusters, ROC, wavelets, ODR, read data from a picture	Multivariate, wavelets and the digitizer
use the Origin Apps (Tafel, logistic regression, Taylor and Venn diagrams...)	Origin Apps
work with grids, images and surfaces	Matrices and surfaces
automate with Python	Scripts
bring your Origin, SciDAVis or QtiPlot projects	Coming from Origin, SciDAVis and QtiPlot
deliver the figure to a journal	Export and publication
draw figures without the window	Command line

The window at a glance

- **Ribbon** (Home, Sheet, Matrix, Graphs, Analysis, Format, Insert, Layout) and the same commands in the menus.
- **Project** dock: sheets, matrices, graphs and folders. Double-click opens; right-click creates folders and moves items.
- **Graph tabs** in the middle, each graph with its own undo history.
- **Properties** dock: the object explorer (every element of the figure), the active panel and the series.
- **Messages** and **Notes** docks below; notes are saved in the project.

Press **F1** in the program for a short summary of the mouse and keyboard actions.

Installation

SciVelyx Graphs runs on Windows 10 and 11 (64-bit), macOS 11 or later (Apple silicon and Intel) and Linux x86_64. Download the file for your system from the latest release. During the testing phase the release page is open to invited testers.

System	File	Notes
Windows	`SciVelyx-Graphs-<version>-windows-x64-setup.exe`	Installs for your user without administrator rights (or for all users). Start menu, optional desktop icon, optional `scivelyx` on the PATH, uninstaller.
Windows, no installation	`SciVelyx-Graphs-<version>-windows-x64-portable.zip`	Unzip anywhere and run `SciVelyx-Graphs.exe`.
macOS, Apple silicon	`SciVelyx-Graphs-<version>-macos-arm64.dmg`	M1, M2, M3, M4…
macOS, Intel	`SciVelyx-Graphs-<version>-macos-x86_64.dmg`	
Linux	`SciVelyx-Graphs-<version>-linux-x86_64.AppImage`	One file for most distributions (glibc 2.35 or newer).
Linux, plain folder	`SciVelyx-Graphs-<version>-linux-x86_64.tar.gz`	No FUSE needed.

Windows

1. Run the `…-setup.exe`.
2. The test builds are not code-signed yet. If Windows SmartScreen says “Windows protected your PC”, click **More info → Run anyway**.
3. Open SciVelyx Graphs from the Start menu. To remove it: Settings → Apps.

macOS

1. Open the `.dmg` and drag **SciVelyx Graphs** onto **Applications**.
2. The test builds are not signed with an Apple Developer ID yet. The first time, Control-click (or right-click) the app in Applications and choose **Open**, then **Open** again. On macOS 15 and later, if that is not offered: System Settings → **Privacy & Security → Open Anyway**.

Linux

```
chmod +x SciVelyx-Graphs-*-linux-x86_64.AppImage
./SciVelyx-Graphs-*-linux-x86_64.AppImage                    # the program
./SciVelyx-Graphs-*-linux-x86_64.AppImage scivelyx --help     # the command line
```

If the AppImage asks for FUSE, install FUSE 2 or use the `.tar.gz` (extract it and run `SciVelyx-Graphs` inside the folder):

Distribution	Command
Ubuntu 24.04 or later	`sudo apt install libfuse2t64`
Debian, Linux Mint, older Ubuntu	`sudo apt install libfuse2`
Fedora, RHEL, Rocky, Alma	`sudo dnf install fuse-libs`
openSUSE	`sudo zypper install libfuse2`
Arch, Manjaro	`sudo pacman -S fuse2`

Testers with access to the source repository can install with git instead. `./install.sh` finds the package manager (apt, dnf, zypper or pacman) and installs the libraries Qt needs on Ubuntu, Debian, Linux Mint, Fedora, RHEL and its rebuilds, openSUSE and Arch. Where the system Python is older than 3.10 (RHEL 8–9, openSUSE Leap), it adds Python 3.11:

```
git clone https://github.com/laribeirojr/scivelyx-graphs.git ~/scivelyx
cd ~/scivelyx && ./install.sh     # later: ./update.sh
```

Checking a download

Every release has a `SHA256SUMS.txt`. Compare it with the checksum of your file:

```
sha256sum -c SHA256SUMS.txt --ignore-missing        # Linux
shasum -a 256 -c SHA256SUMS.txt --ignore-missing    # macOS
certutil -hashfile SciVelyx-Graphs-<version>-windows-x64-setup.exe SHA256    # Windows
```

Licence

SciVelyx Graphs is **freeware**:

- it is free of charge for any use, including research, teaching and work in companies;
- you may give the unmodified installers to others, free of charge;
- you may not modify, decompile or sell it;
- the figures and data you make with it are yours.

The installer shows the full licence, which is also in `LICENSE.txt` (English) and `LICENSE.pt-BR.txt` (Portuguese) in the program folder. The licences of the third-party components are in `THIRD_PARTY_NOTICES.txt`.

Checking the installation

Help → Run the self-test... checks what depends on your computer:

- the numerical libraries, figure drawing and the export formats, including the exact page size of a PDF;
- the fonts, the file formats and the settings folder;
- in the real window: the screen and its scaling, the clipboard, printing and the themes.

It ends with a report in the browser and a `selftest.zip`. Attach the zip to a problem report.

From a terminal, the same check runs with:

```
scivelyx selftest --gui            # report in ~/SciVelyx-selftest
```

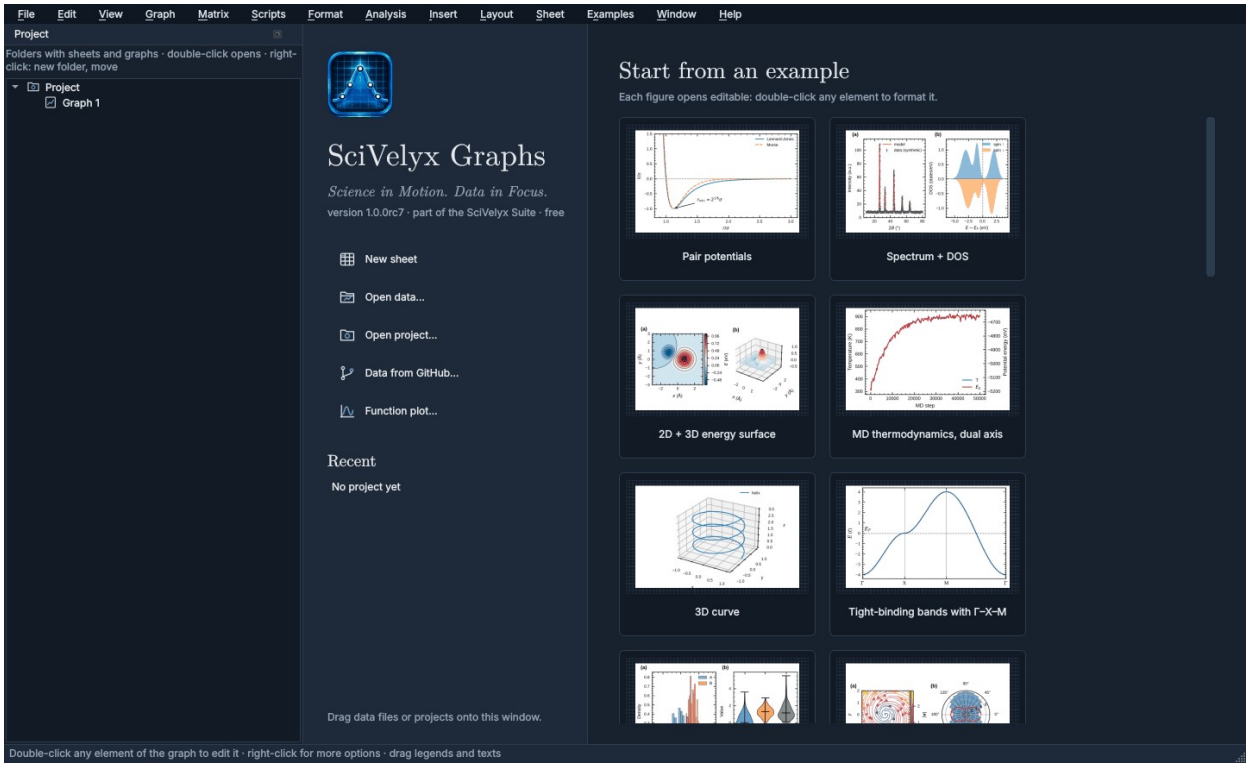
Help → Report a problem... opens a problem report on GitHub, already filled in with the version and the system.

The window

This chapter shows the parts of the program window and what each one is for. The other chapters refer to them by the names used here.

The start screen

SciVelyx Graphs opens on the start screen.



The start screen: actions on the left, the gallery of examples on the right

- **New sheet**, **Open data...** and **Open project...** start the usual work. You can also drag data files or projects onto the window.
- **Data from GitHub...** reads a file straight from a GitHub repository.
- **Function plot...** draws $y = f(x)$, a parametric curve or a surface without any data.
- **Recent** lists the last projects. A click opens one.
- **Start from an example** opens a ready figure, with its data, as a normal project. Every element can then be edited, so the examples are also templates.

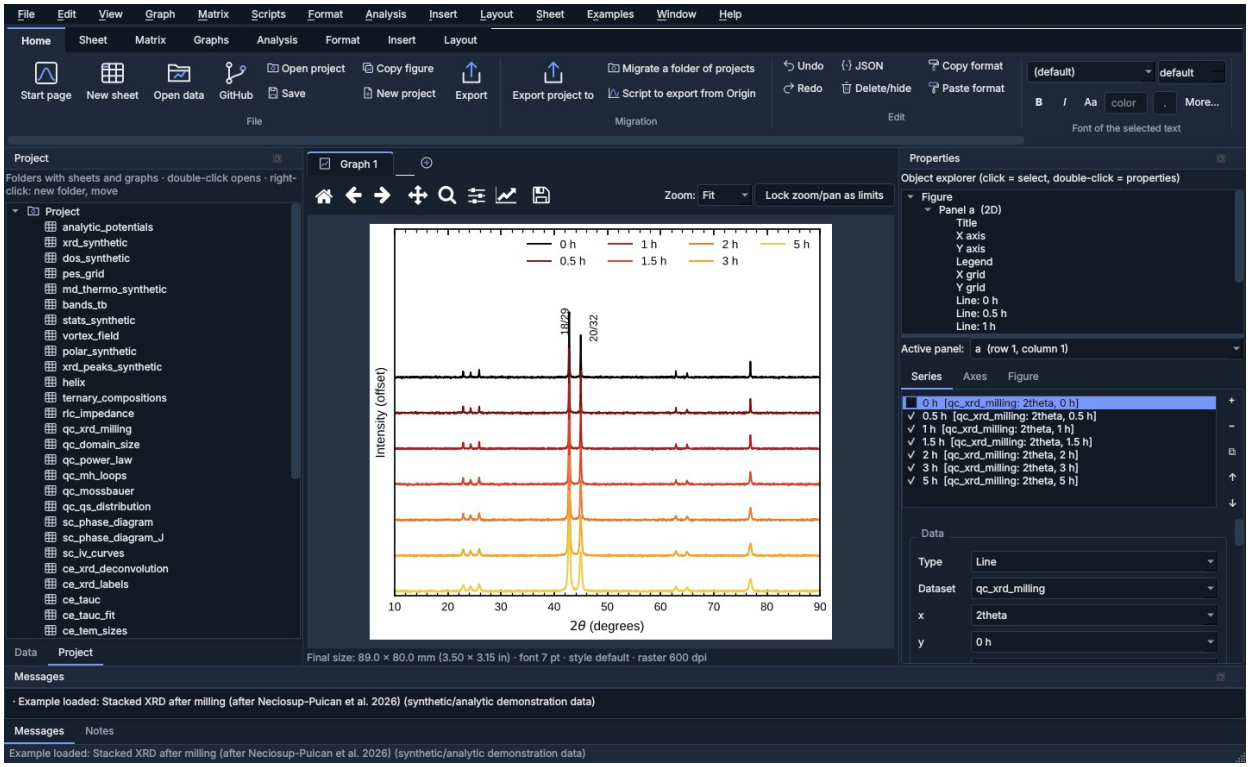
The gallery has two groups. The first ten examples use analytic or synthetic data. The next fourteen recreate figure types from five 2026 papers of colleagues at the University of Brasília:

- X-ray diffraction, magnetization loops and Mössbauer spectra of quasicrystals;
- the phase diagram and the current–voltage curves of a superconductor;

- XRD deconvolution, a Tauc plot and a particle-size histogram of CeO_2 nanoparticles;
- thermogravimetry, bar charts with significance marks and PCA scores of dentin samples;
- depth profiles and a FORC diagram of a tropical soil.

Their data are illustrative: they were made to look like the published figures, and each dataset note says so. **Help → Start page** (Ctrl+Shift+H) comes back to this screen.

The working window



The working window with an example figure

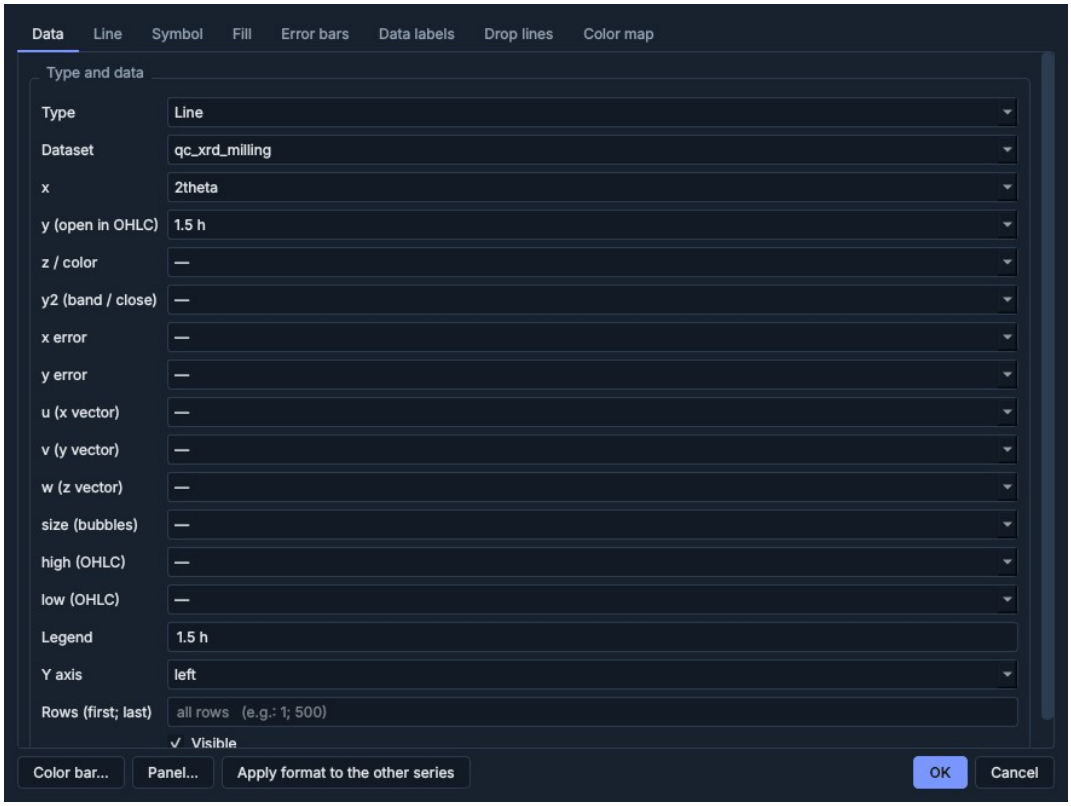
Part	Where	What it does
Menus	top line	Every command of the program, grouped as in Origin: File, Edit, View, Graph, Matrix, Scripts, Format, Analysis, Insert, Layout, Sheet, Examples, Window, Help.
Ribbon	below the menus	The most used commands as buttons, in tabs: Home, Sheet, Matrix, Graphs, Analysis, Format, Insert, Layout.
Project	left	Sheets, matrices, graphs and folders of the project. Double-click opens an item; right-click creates folders and moves items.
Graph tabs	middle	One tab per graph. Each graph has its own undo history. The + tab creates a graph.
Graph toolbar	above the figure	Home, back and forward, pan, zoom, subplot settings, axis limits, save. **Zoom** sets the screen scale of the page (it does not change the exported size).
Figure	middle	The page as it will be exported. The line below it gives the final size in millimetres, the font size and the raster resolution.
Properties	right	The object explorer (every element of the figure), the active panel, and the list of series with their data.
Messages and Notes	bottom	Results of analyses and warnings; the notes are saved with the project.

Part	Where	What it does
Status bar	last line	What is selected and what the mouse can do now.

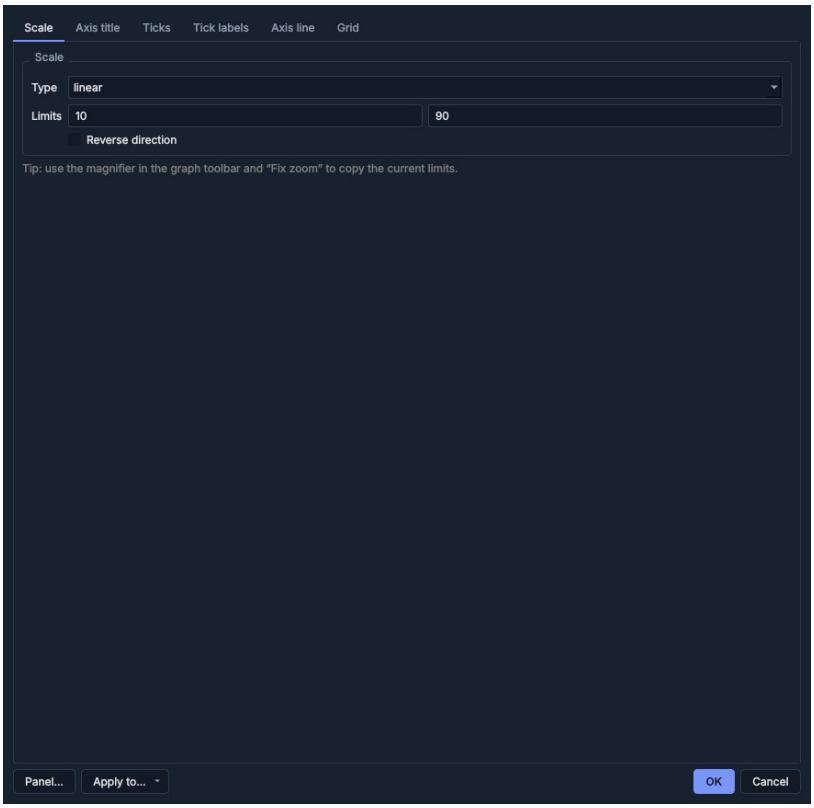
Docks can be closed, moved or made floating with the mouse; **View** shows them again.

Selecting and editing

- **Click** an element of the figure (a curve, an axis, the legend, a text) to select it. The status bar names it and the object explorer highlights it.
- **Double-click** opens its properties.
- **Right-click** opens a menu with the actions for that element: properties, copy and paste the format, hide, delete, bring to front…
- **Drag** moves texts, legends, insets and drawing objects.
- **Delete** removes the selected element; **Ctrl+Z** undoes.



Properties of a curve: data, line, symbol, fill, error bars, labels, drop lines and colour map



Properties of an axis: scale, limits, ticks, labels, grid and breaks

Sheets

Data live in sheets. Click a sheet in the **Project** dock, or its name in the **Data** tab, to see it. Columns have a role (X, Y, Z, error…), a long name and units; formulas fill a column from the others.

	2theta (X)	0 h (Y)	0.5 h (Y)	1
1	10	3.3199225	2.7711494	2.
2	10.05	3.3216624	2.7754651	2.
3	10.1	3.321337	2.7665442	2.
4	10.15	3.3192581	2.7721661	2
5	10.2	3.3257114	2.7737075	2.
6	10.25	3.3048869	2.7622085	2
7	10.3	3.3197567	2.7723884	2.
8	10.35	3.3198373	2.7752772	2.
9	10.4	3.3178921	2.7802461	2.
10	10.45	3.3220394	2.7646272	2
11	10.5	3.3256173	2.7804953	
12	10.55	3.3164982	2.7654279	2.
13	10.6	3.3166966	2.7636545	2.
14	10.65	3.3216768	2.7599613	2.
15	10.7	3.328584	2.7676261	2.
16	10.75	3.3291487	2.7785766	2
17	10.8	3.3226209	2.7743743	2.

A sheet: the first column is X and the others Y

The chapter Sheets and data explains import, formulas, masks and the sheet operations.

Keyboard

Key	Action
F1	Summary of the mouse and keyboard actions
Ctrl+N, Ctrl+O, Ctrl+S	New project, open, save
Ctrl+Z, Ctrl+Shift+Z (Ctrl+Y on Windows)	Undo, redo (of the active graph)
Ctrl+T	New graph
Ctrl+E	Export the figure
Ctrl+Shift+C	Copy the figure as a picture
Ctrl+Alt+C, Ctrl+Alt+V	Copy and paste the format of an element
Ctrl+G	Data from GitHub
Delete	Remove the selected element
Ctrl+Shift+H	Start page
Ctrl+Shift+P	Script window

Your first graph

This page takes a data file to a finished, fitted and exported figure. It uses only the window; the same steps can be scripted later (Scripts).

1. Open the data

File → Open data (Ctrl+O) reads CSV, TSV and text in columns (with `#` comments and Fortran `D` exponents), Excel, LibreOffice (.ods), NumPy, JSON, Parquet, GROMACS `.xvg`, LAMMPS logs, WAV and SQLite, and the files of diffractometers and spectrometers (Spectroscopy and diffraction). For files with an unusual layout — a decimal comma, header lines to skip, names on another line — use **File → Import wizard** (Ctrl+Shift+I), which shows the table while you adjust the options.

The data open as a **sheet**. Each column has a *designation* (X, Y, Z, X error, Y error, Label), a *long name* and *units*; long name and units become the axis titles, for example "Energy (eV)". Change them with a right click on the column header (**Set as X**, **Set as Y**, **Column properties**).

2. Plot

Select the columns (Ctrl+click on the headers) and click a button on the **Graphs** tab of the ribbon — *Line*, *Scatter*, *Error bars*, *Columns*... The graph follows the designations: each Y uses the nearest X to its left, and a selected Y-error column becomes error bars. Without a selection, the button opens a small setup where you choose the columns and the target panel.

3. Edit

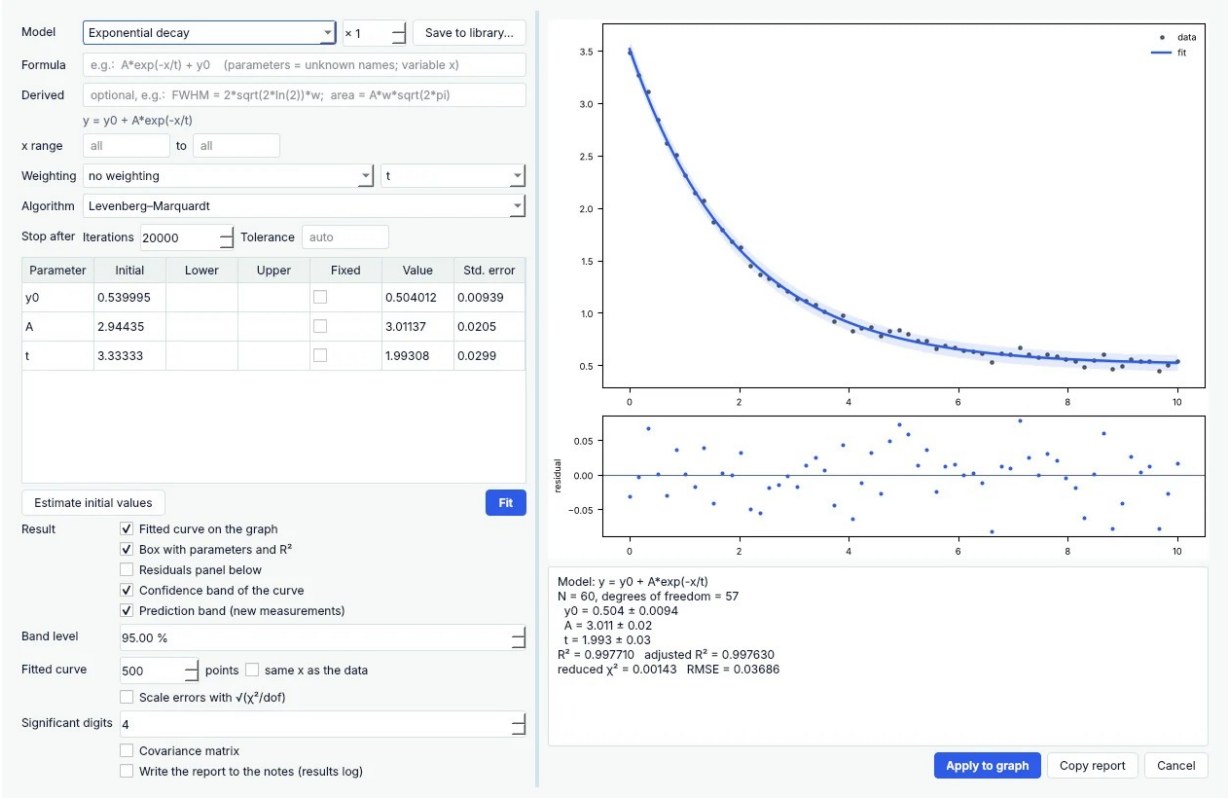
Every element of the figure is editable:

- **double-click** a title, an axis, a curve, the legend or the panel background and its properties open;
- a single click selects (blue outline) and the **Font** bar of the Home and Format tabs changes the selected text;
- drag the legend and texts; drag a 3D panel to rotate it;
- **Delete** removes the selected element, **Ctrl+Z** undoes.

Editing graphs lists what each dialog contains.

4. Fit

Click the curve, then **Analysis → Curve fitting** (Ctrl+Shift+F). Choose a model (linear, Gaussian, exponential decay, Voigt, several peaks, your own formula…), press **Fit**, look at the parameters, the residuals and the bands, and **Apply to graph**. The fitted curve, the confidence band and a box with the parameters appear on the graph and are recalculated whenever the data change.



The fit wizard

5. Export

Choose the journal size in **Format → Journal** (e.g. *Nature — 1 column (89 mm)*, *APS — 1 column*), then **File → Export** (Ctrl+E). PDF, SVG and EPS keep the text editable at the exact physical size; PNG and TIFF use the resolution you set. **File → Save** keeps everything — data, graphs, notes and scripts — in one `.scivelyx.json` project.

Try the examples

The **Examples** menu and the start screen open ready-made figures (band structure with Γ–X–M ticks, spectrum with density of states, energy surface in 2D and 3D, molecular-dynamics thermodynamics on two y axes, ternary and Smith charts, and the figures modelled on 2026 papers…). Each one is fully editable and shows a technique you can reuse.

Worked examples

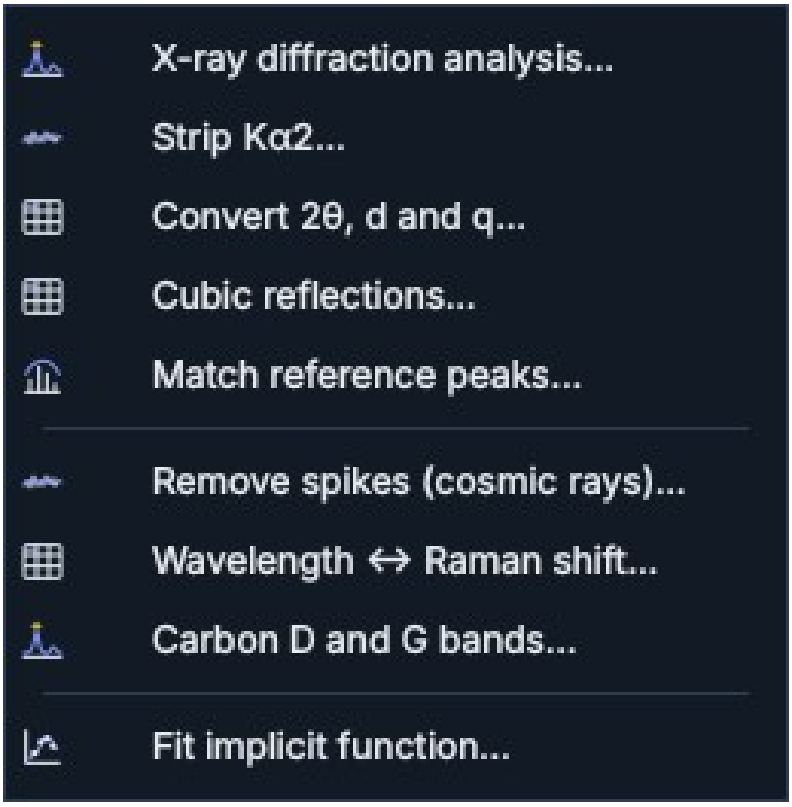
Each example goes from the data to the result, step by step, with the windows you will see. The data are either in the program (the **Start from an example** gallery) or easy to make yourself. Values in the pictures come from those data; yours will differ.

1. Crystallite size from an X-ray diffractogram

Video (1 minute): this example on YouTube.

Goal: the d-spacings, the Scherrer size of each peak, the Williamson–Hall size and strain, and the lattice constant of a cubic powder.

1. Open the diffractogram: **File → Open data** reads `.xrdml` (Panalytical), `.ras` (Rigaku), `.uxd` (Bruker) and plain two-column files. The wavelength stored in an instrument file is kept with the sheet.
2. Plot 2θ against the intensity (select both columns, **Graphs → Line**).
3. Click the curve, then **Analysis → Spectroscopy and diffraction**:

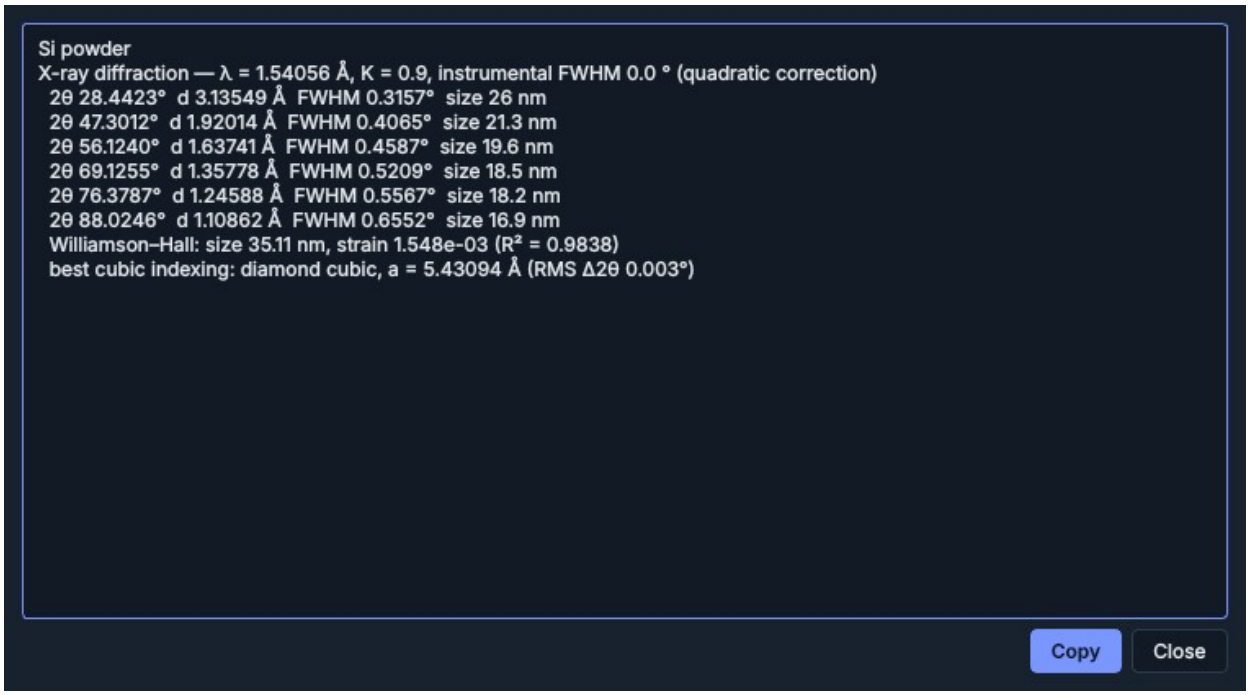


The Spectroscopy and diffraction menu

4. Choose **X-ray diffraction analysis...**. In the dialog:

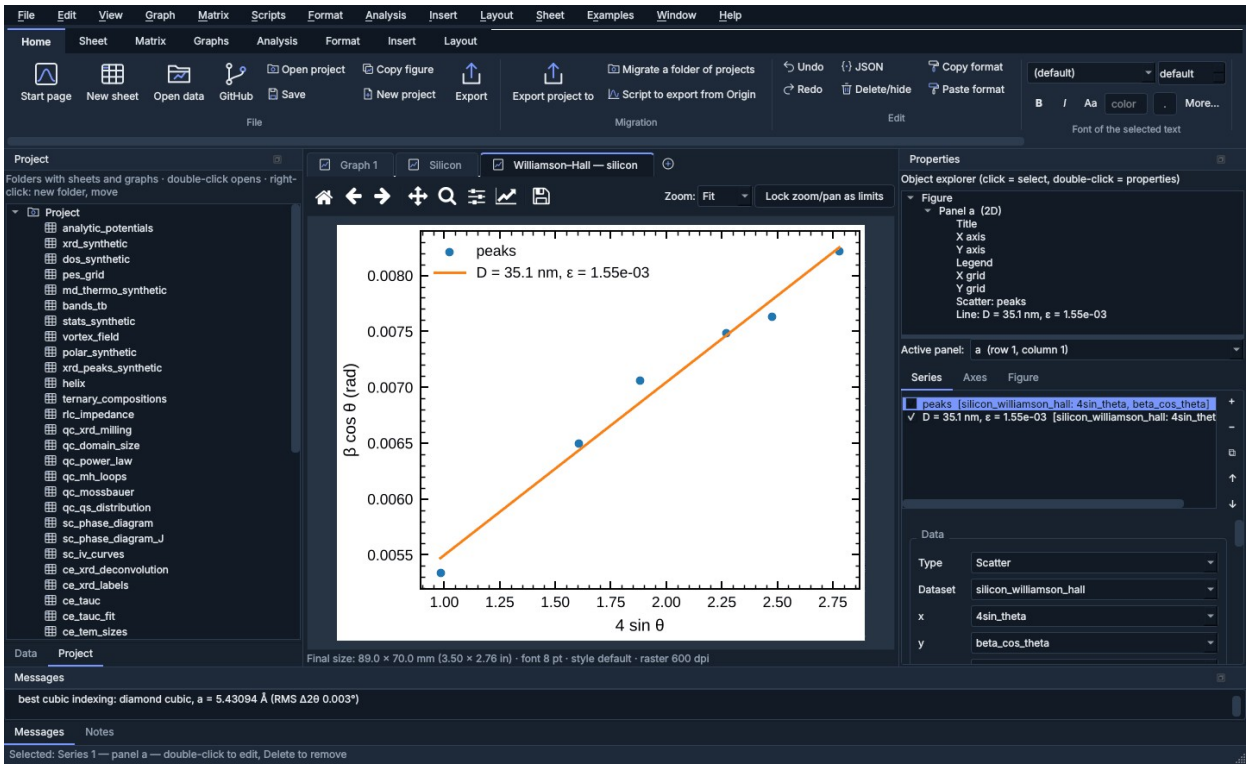
- **Wavelength**: pick the line (Cu Kα1 = 1.540562 Å for most laboratory instruments) or type a value in Å;
- **Scherrer constant K**: 0.9 for spherical crystallites;
- **Instrumental FWHM**: the width of a standard (LaB_6, Si) measured with the same set-up, in degrees of 2θ; leave 0 if you do not have one;
- **Instrumental correction**: quadratic for Gaussian peaks, linear for Lorentzian, geometric for Voigt;
- **Peak shape** of the fit (pseudo-Voigt is the usual choice), and **strip Kα2 first** if the doublet is visible at high angles.

5. **OK** finds and fits the peaks. The report lists, for each peak, 2θ, d, the FWHM and the size, then the Williamson–Hall result and the best cubic indexing:



Report of the X-ray diffraction analysis of a silicon powder

6. Two sheets and one graph are added: `<data>_xrd` (the peak table with d, q and size), `<data>_williamson_hall` and the Williamson–Hall plot, β cos θ against 4 sin θ. Its intercept gives the size and its slope the strain:



Williamson–Hall plot made by the analysis

Check the result. The Scherrer equation is reliable below about 100 nm. Above that, the instrumental width dominates. A negative or scattered Williamson–Hall slope usually means that the strain is below what the data can resolve. In the silicon example the indexing finds the diamond lattice with a = 5.4309 Å.

Related tools in the same menu:

- **Strip Kα2** removes the doublet from the whole pattern.
- **Convert 2θ, d and q** adds the columns d and q, or 2θ for another X-ray tube.
- **Cubic reflections** draws the hkl lines of a lattice on the graph.
- **Match reference peaks** pairs your peaks with a reference card typed in a sheet.

The example *Stacked XRD after milling* in the gallery shows a set of patterns offset vertically. *Power-law fit with a log–log inset* plots the crystallite size against the milling time, with the values of the paper's Table 1.

2. D and G bands of a carbon Raman spectrum

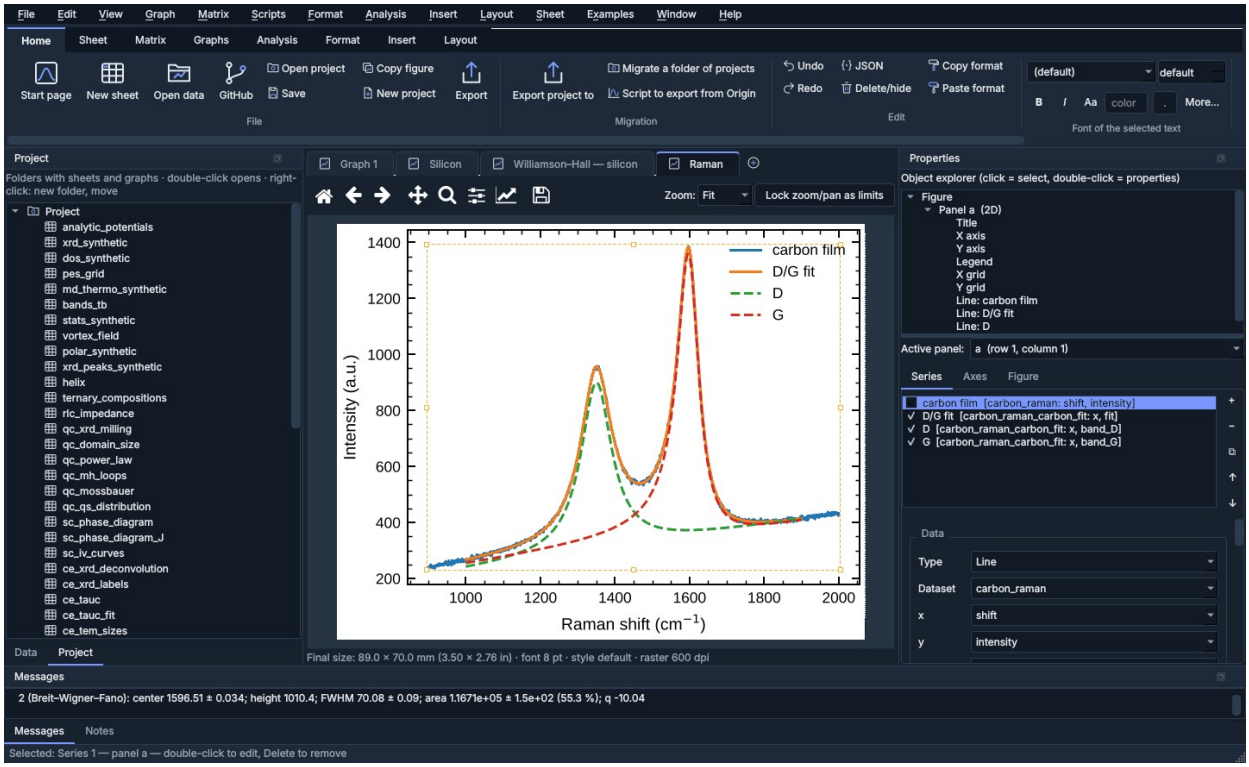
Video (1 minute): this example on YouTube.

Goal: I_D/I_G, A_D/A_G and the crystallite size L_a of a disordered carbon.

1. Open the spectrum (Raman shift in cm^{-1} against intensity). JCAMP-DX (`.jdx`) and Galactic SPC (`.spc`) files open directly.
2. If the spectrum has cosmic-ray spikes, click the curve and run **Spectroscopy and diffraction → Remove spikes**. It adds a clean copy of the curve, and the `spike` column marks the points that were replaced.

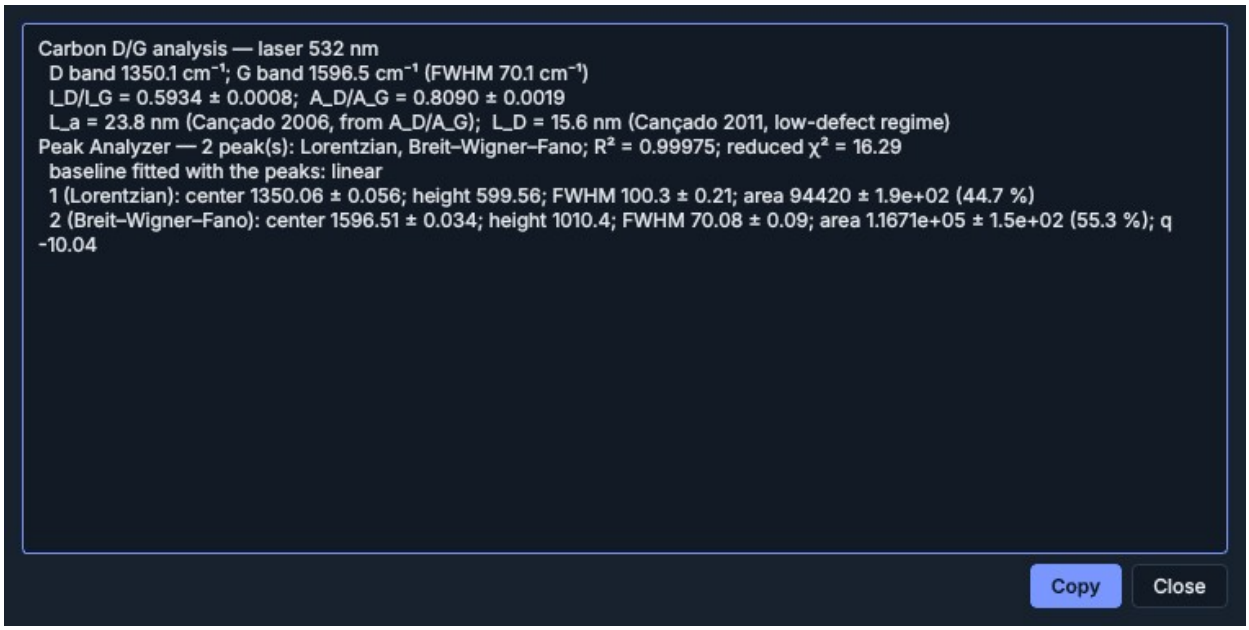
3. **Carbon D and G bands...** asks for the laser wavelength and fits a Lorentzian D band and a Breit–Wigner–Fano G band on a linear baseline. Tick **also D″ and D′** for amorphous carbons.

4. The fit and the bands are drawn on the graph:



D and G bands fitted on the spectrum

5. The report gives the band positions, the ratios with their errors, L_a = 2.4×10^{-10} λ^4 $(A_D/A_G)^{-1}$ (Cançado et al. 2006) and the defect distance L_D (Cançado et al. 2011):



Report of the carbon analysis

The same fit by hand. Analysis → Peak Analyzer fits any combination of peaks. Each peak can have its own shape, and the baseline can be fitted with the peaks:



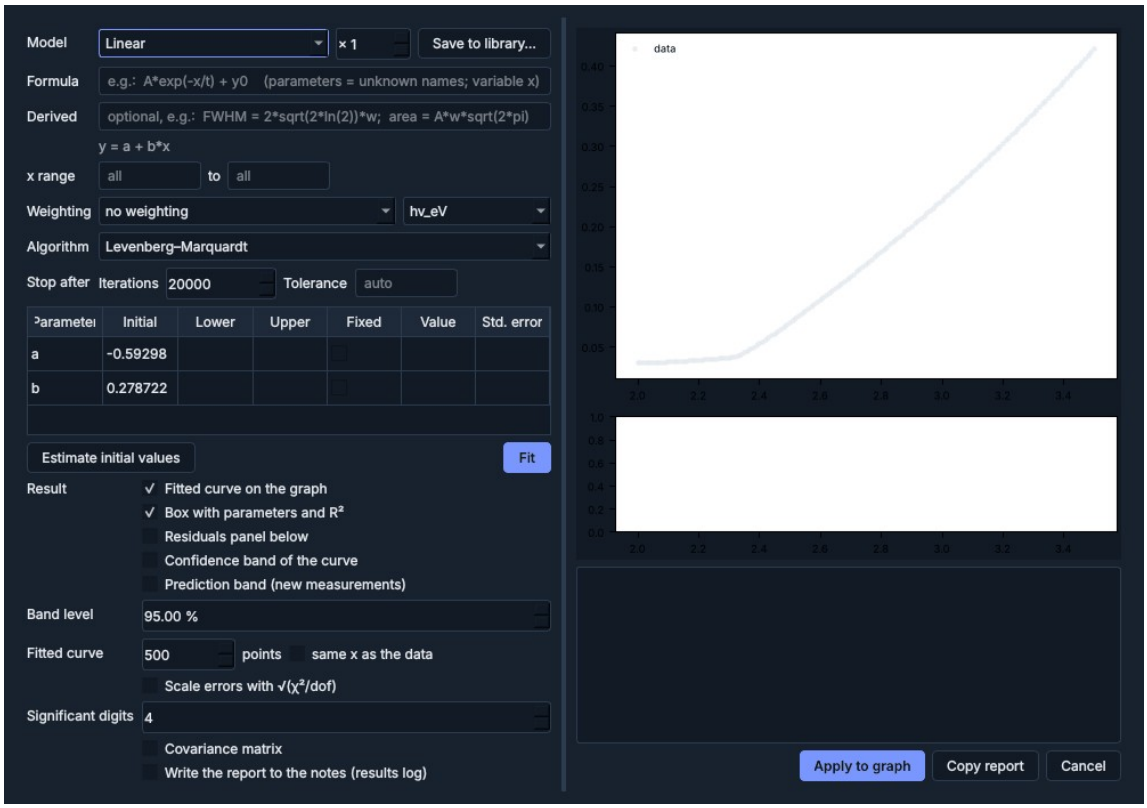
Peak Analyzer with a Lorentzian D band, a BWF G band and a linear baseline fitted together

- **Positions**: `1350; 1590` (or leave empty and let it find the peaks; *Minimum prominence* controls how many).
- **Shapes per peak**: `lorentz; bwf`. The keys are gauss, lorentz, voigt, pvoigt, bwf, pearson7 and bigauss.
- **Baseline fitted with peaks**: linear. Step 1 can still subtract an ALS or polynomial baseline first.
- **Analyze**, look at the residuals, then **Apply to graph**. The peak table goes to a new sheet. Its columns are centre, height, FWHM, area and area %, each with its error, plus the shape parameter (q for BWF).

3. Optical band gap from a Tauc plot

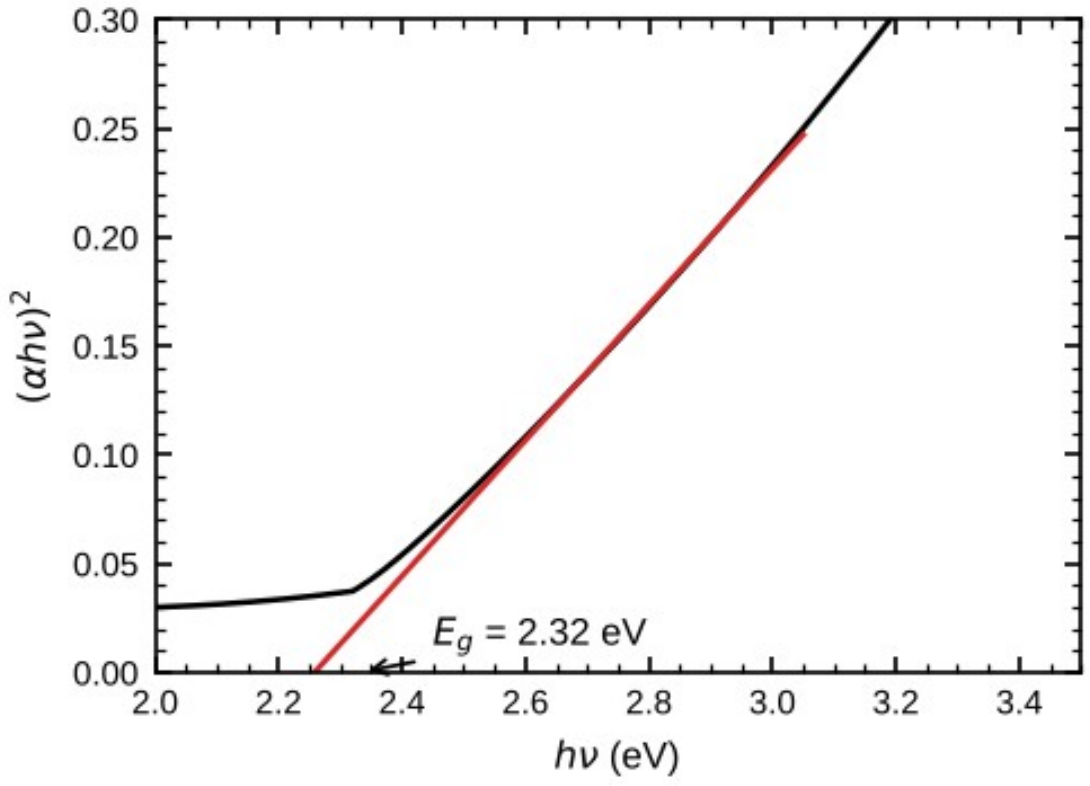
Goal: E_g of a semiconductor by linear extrapolation of $(\alpha h\nu)^2$ against $h\nu$.

1. Compute the Tauc variable in the sheet. Add a column with a formula, for example `(2.303 * A * hv)^2` for a direct allowed transition, where `A` is the absorbance and `hv` the photon energy in eV (`hv = 1239.84 / wavelength_nm`). In formulas the decimal separator is the point.
2. Plot $(\alpha h\nu)^2$ against $h\nu$. The gallery example *Tauc plot* has such a curve.
3. Click the curve and open **Analysis → Curve fitting**. Choose the **Linear** model and set the **x range** to the straight part above the absorption edge (2.6 to 3.0 eV in the example).



Curve fitting dialog with the linear model

4. **Fit**, then **Apply to graph**. E_g is where the line crosses zero, E_g = −a/b for y = a + b·x. Type it in **Derived** (`Eg = -a/b`) and the fit report shows it with its error.

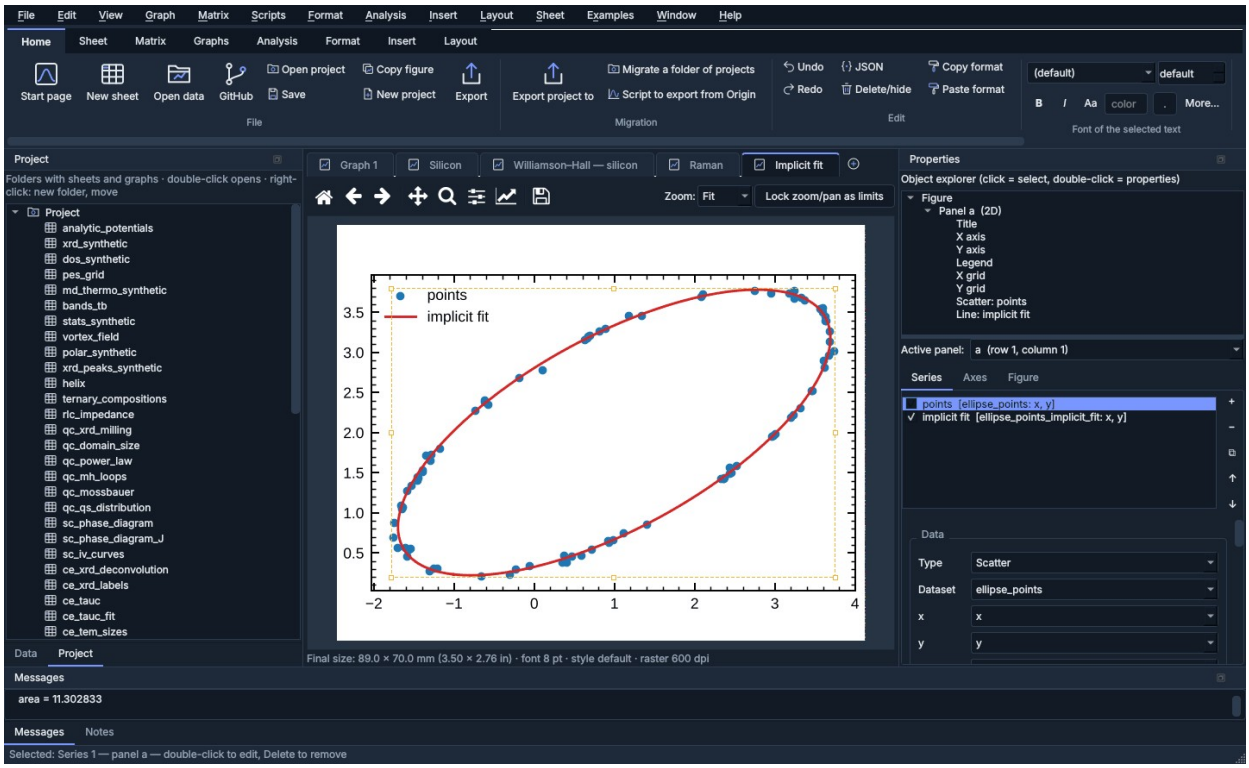


The finished figure: data, the fitted line and E_g marked

4. Ellipse through scattered points (implicit fit)

Goal: the centre, semi-axes and orientation of an ellipse, or of any curve given as $f(x, y) = 0$.

1. Plot the points as a scatter.
2. Click them and choose **Spectroscopy and diffraction → Fit implicit function...**.
3. Choose **Ellipse** (or circle, conic, straight line with errors in x and y, or type your own function, such as `x^2/a^2 + y^2/b^2 = 1` with initial values `a = 1, b = 1`).
4. The fitted curve is drawn and the parameters with their errors go to a new sheet. The fit minimises the distance of each point to the curve, not the vertical distance, so the result does not depend on which variable is called y.

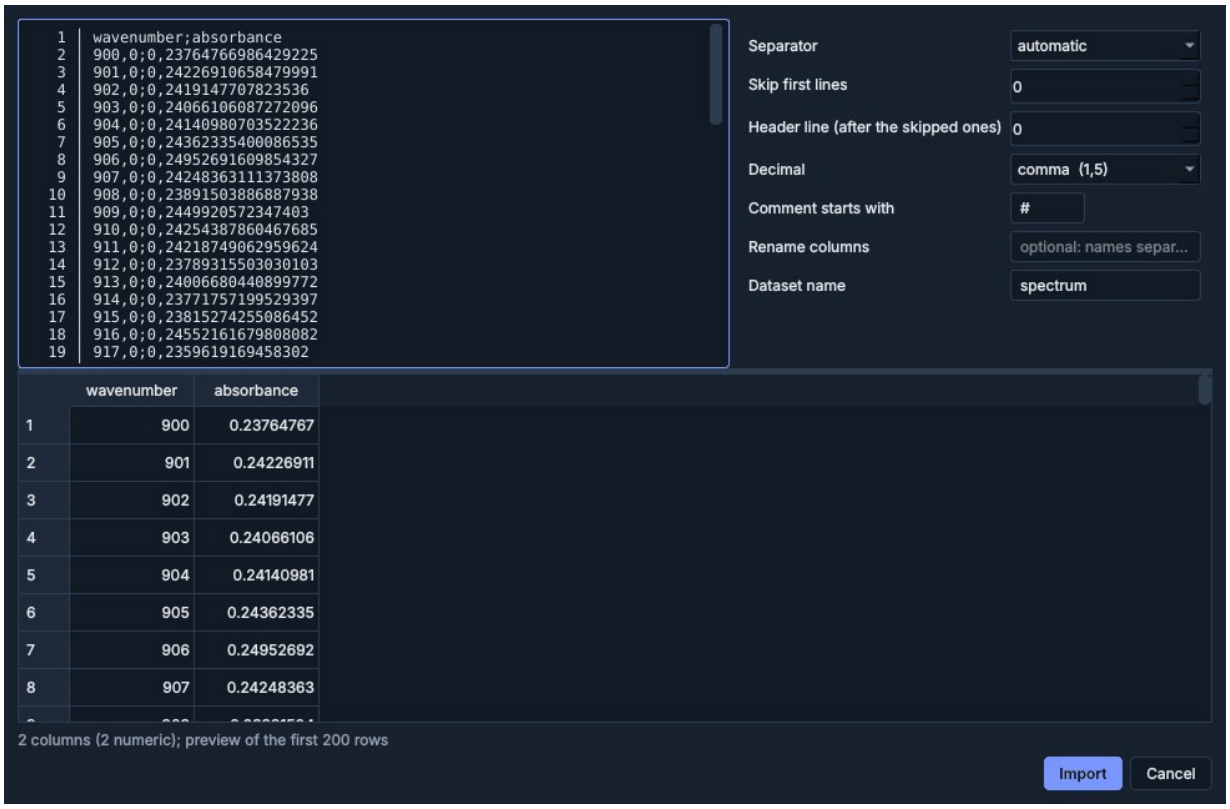


Ellipse fitted to 90 points

5. A file with a decimal comma

Goal: read data written by software set to Portuguese, German or French, with ; between the columns and a comma for the decimals.

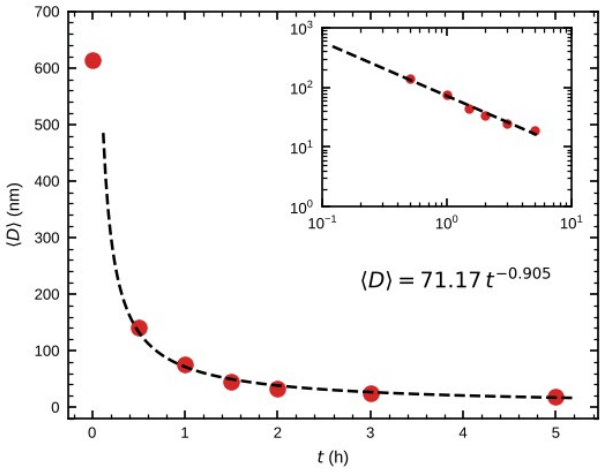
1. **File → Import wizard** (Ctrl+Shift+I) and choose the file.
2. The wizard shows the raw text and the table it will make. Set **Separator** to automatic or ; and **Decimal** to `comma (1,5)`. If the file starts with notes, set **Skip first lines** and the **Header line**.
3. **Import** creates the sheet.

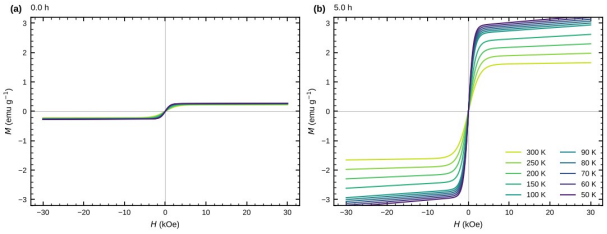
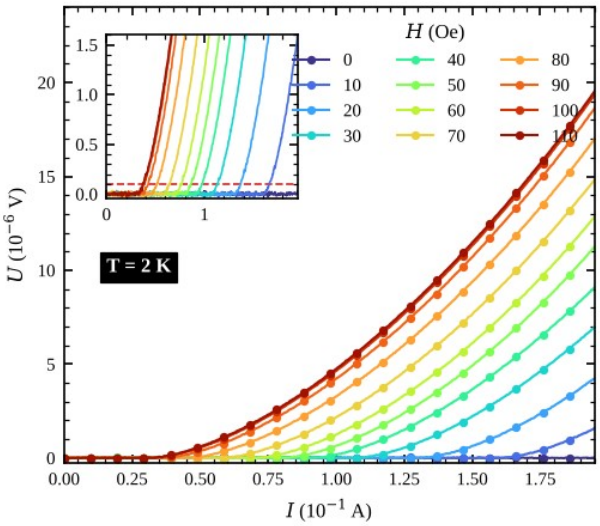
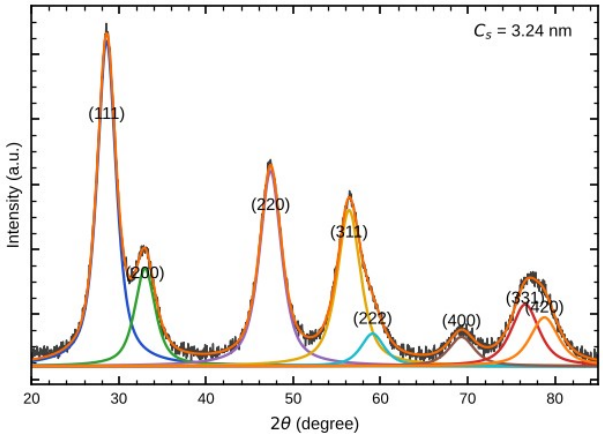
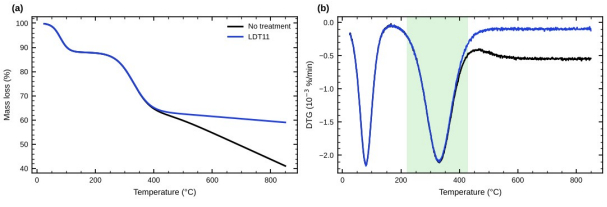


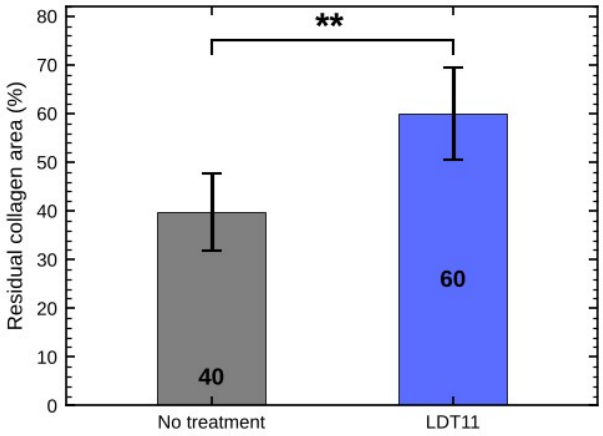
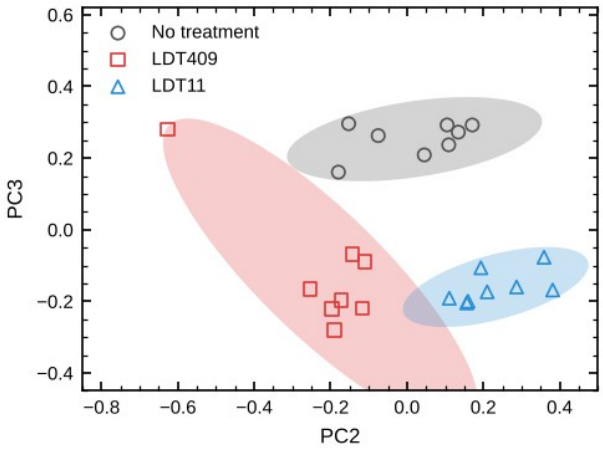
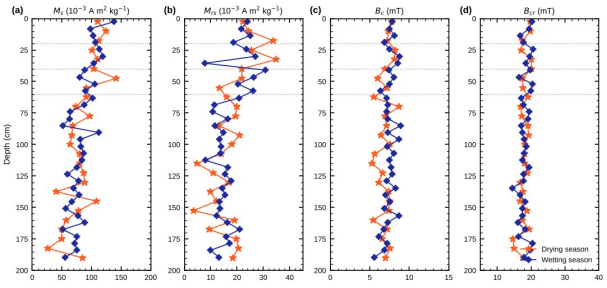
Import wizard reading a CSV file with a decimal comma

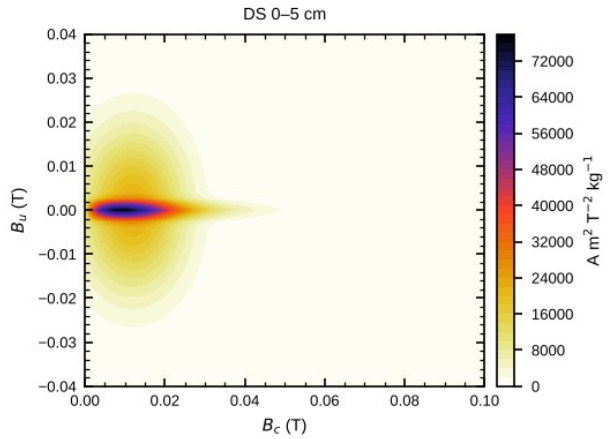
6. Figures modelled on published papers

The gallery examples recreate figure types from 2026 papers. Open one, look at how it is built in **Properties** (series, axes, insets, reference lines, drawing objects), and replace its data with yours: in the series list, change **Dataset**, **x** and **y**.

Example	What it shows how to do
	An inset with its own logarithmic axes (**Insert → Zoom (inset)**, then set both scales to log), a dashed fitted curve and an equation in the panel.

Example	What it shows how to do
	Two panels sharing a colour per temperature taken from a colour map, zero lines, one legend for both panels.
	Twelve curves with markers every 20 points, a legend with a title in three columns, a zoom inset and a cut-off line.
	Fitted components in colour, the data in grey and hkl labels taken from a text column.
	A shaded temperature range (a reference band) and two panels with labels (a), (b).

Example	What it shows how to do
	Bars with standard-deviation error bars, values inside the bars, and a bracket with ** drawn with the drawing tools.
	Hollow markers per group and filled 95 % confidence ellipses.
	Four panels with depth increasing downwards, the axis titles on top and dotted lines at the soil horizons.



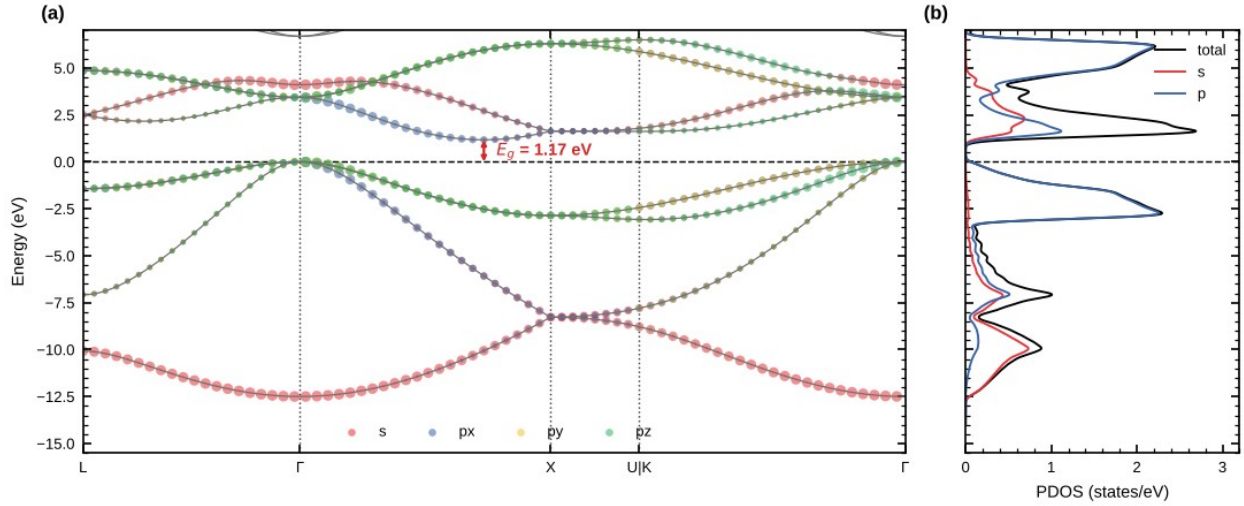
A filled contour map of a distribution with a colour bar and its units.

Sources of the figure types: Neciosup-Puican et al., *J. Alloys Compd.* 1080 (2026) 190504; de Almeida et al., arXiv:2610.36482 (2026); da Silva et al., *Micro Nanostruct.* 221 (2027) 208963; Endres et al., *Clin. Oral Investig.* 30 (2026) 484; do Vale et al., EarthArXiv preprint (2026).

7. Figures for theoreticians: fat bands with the projected DOS, and ELF maps

Two examples of the Start page show how to build the figures of electronic-structure papers. Both are computed from textbook models in the program, so they are physically sound but **illustrative**, not DFT results.

Fat bands and projected DOS of silicon. Bands of the sp3s* tight-binding model of Vogl, Hjalmarson and Dow (1983) along L–Γ–X–U|K–Γ, with the weight of the s, px, py and pz orbitals as bubbles, and the total, s and p DOS in a narrow second panel that shares the energy axis.



Fat bands and projected DOS of silicon

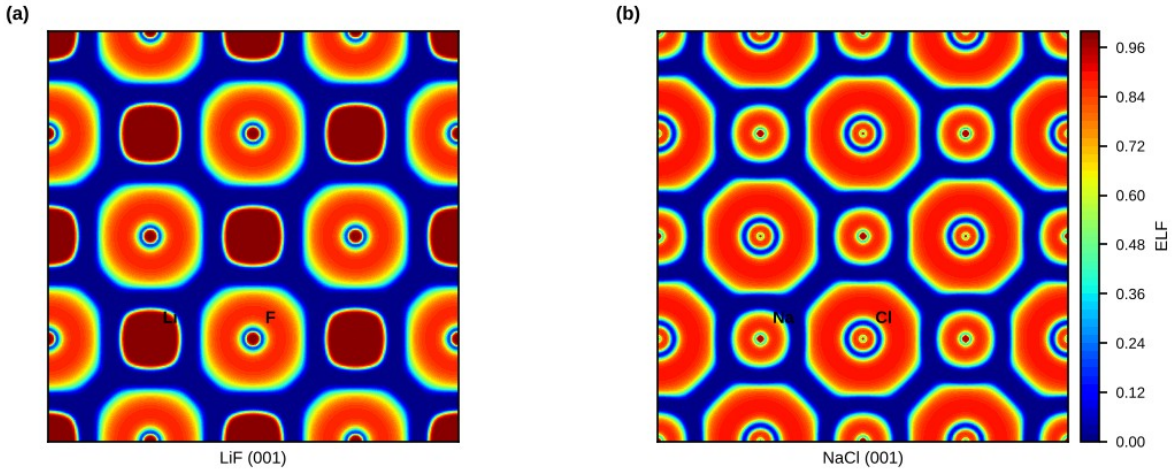
How it is built (open the example and look in **Properties**):

1. **Two panels side by side** with *width ratios* 3 : 1 and a shared y axis; the DOS panel hides its y tick labels.

2. **The bands** are one line per band (columns `band 1` … `band 10`), thin and grey.
3. **The fat bands** are *bubble* series: x = k, y = energy, size = the weight of one orbital, one series and one colour per orbital, without edges and half transparent. Bubble max sets the size of a weight of 1.
4. **The k path:** the x axis has explicit ticks at the high-symmetry points with the labels L, Γ, X, U|K, Γ, and dotted vertical reference lines there. A dashed horizontal line marks the valence-band maximum (0 eV).
5. **The gap:** a double-headed arrow (**Insert → Arrow (drawing)**) from the valence-band maximum to the conduction-band minimum, and a text with E_g.
6. **The DOS panel** plots the DOS on x and the energy on y.

To use your own calculation, put the bands in a sheet (k in the first column, one column per band) and the projections in another (k, energy and one weight column per orbital). Then change **Dataset**, **x** and **y** of each series.

Electron localization function (ELF). Two maps of the (001) planes of LiF and NaCl, with a shared 0–1 colour scale. The ELF is computed from closed-shell ions with Slater-type orbitals (the promolecule), which shows the shell structure of each ion.



ELF of the (001) planes of LiF and NaCl

Each panel is a *filled contour* of a sheet with x, y and ELF columns (50 levels, colour map *jet*, fixed range 0 to 1, colour bar on the right panel only). The axes have no ticks, the aspect is equal, and the ion names are text labels. A VASP `ELFCAR`, or a plane exported by Quantum ESPRESSO (`pp.x`) or VESTA as x, y, value, plots the same way.

Sheets and data

Importing

Source	How
CSV, TSV, text in columns	**File → Open data** (Ctrl+O); `#` comments at the top are kept, Fortran `D` exponents are read
Unusual text files	**File → Import wizard** (Ctrl+Shift+I): separator, decimal comma, lines to skip, names and comment lines, with a live preview
Excel, LibreOffice (.ods), NumPy, JSON, Parquet	**File → Open data**
GROMACS `.xvg`, LAMMPS log	**File → Open data**
Diffractometers and spectrometers (XRDML, RAS, UXD, JCAMP-DX, SPC)	**File → Open data** — see Spectroscopy and diffraction
Sound (.wav)	**File → Open data**: one column per channel
SQLite database	**File → SQLite database...**: a table or a `SELECT` query (read-only)
HDF5, NetCDF, MATLAB (.mat)	**File → HDF5, NetCDF or MATLAB file...** (or open/drop the file): variables as columns or as a matrix — see Multivariate, wavelets and the digitizer
A picture of a graph	**Analysis → Digitizer...** reads the points back
Several files into one table	**File → Several files into one table...**: side by side or stacked, with the file name as a column
GitHub	**File → GitHub** (Ctrl+G): a repository, folder or file URL; private repositories with a `GITHUB_TOKEN`
A matrix (grid of numbers)	**File → Matrix** or **Sheet → Matrix**

Date columns become a date axis automatically.

Columns

- **Designation**: X, Y, Z, X error, Y error, Label or none (right click on the header). Graphs follow them.
- **Long name and units**: they become the axis titles.
- **Type**: number, text or date/time; **numeric format**: decimal, scientific or engineering (10k), with decimal places; **read-only**; **hidden**. All are kept in the project.
- Move, swap, add, fill (row numbers, uniform or normal random values) and sort columns — also by several keys.

Formulas — Set values

Sheet → Set values (Ctrl+Q) computes a column from a formula, with a preview:

```
A^2 + i                       # i is the row number (1-based)
col("E (eV)") * 1000          # another column by its name
hc_eVnm / A                   # physical constants: kB, kB_eV, h, hbar, h_eV, c0, qe, NA, me, eps0,
x > 0 ? x : 0                 # SciDAVis and QtiPlot (muParser) syntax works: ?:, &&, ||, if(c, a,
```

Special functions (erf, gamma, Bessel, Voigt...) and the muParser functions of SciDAVis and QtiPlot (avg, min/max/sum with several arguments, rint, w0, wm1, chi2cdf, invt, rnd...) are available. The evaluator is safe: a formula cannot run arbitrary code. Formula columns **recalculate by themselves** when the columns they use change; typing into a formula column turns it into ordinary data.

In formulas the decimal separator is always the point (`2.303`), as in every programming language, even when your files use a comma.

Masks and undo

Sheet → Mask rows strikes rows through in red: they stay out of graphs, fits, tests and statistics but remain in the file. **Unmask** brings them back. With the sheet in focus, **Ctrl+Z** undoes data edits.

Reorganizing

The **Sheet** menu also has: transpose; stack and unstack columns; filter rows by a condition (`T > 300 and k < 50`); join sheets (by a key or side by side); pivot table; row statistics; frequency count; stem-and-leaf; convert to a matrix and back. Each operation creates a new sheet, so the original stays intact.

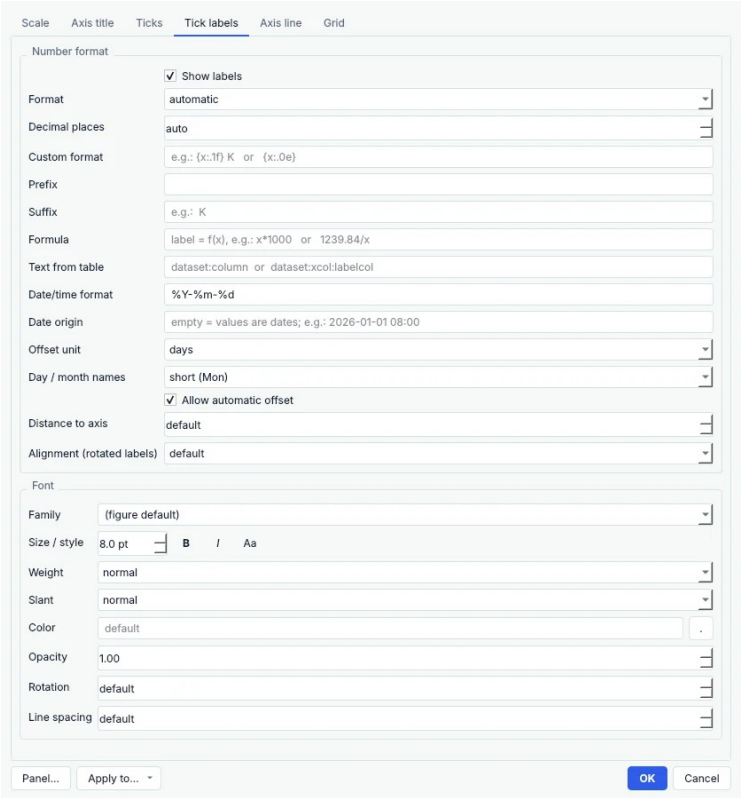
Sheet → Export sheet writes Excel (with long names and units), LibreOffice, CSV or TSV.

Editing graphs

A figure is made of **panels** (rows and columns of axes) that hold **series** (curves, bars, maps...), texts, shapes and a legend. Everything you see is linked to an editable element.

What a double click opens

Clicked element	What you can change
Titles, axis titles, texts	Text with a symbol palette, 37 Greek letters, sub- and superscripts, partial bold or italic, font, size, colour, rotation, frame and background, with a preview. Invalid math notation is pointed out before it reaches the figure.
Axis numbers or axis line	Scale (linear, log, symlog, logit, asinh), limits, inversion; ticks by step, count or explicit positions and labels (Γ, X, M...); number format (scientific $\times 10^{n}$, fixed, percent, engineering, custom), prefix, suffix, formula, text from a table, weekday and month names, date/time; axis line; grid.
Curve, symbols, bars, maps	Type and columns; line (dash, caps, spline, steps, connect across missing data); symbols (filled, half-filled, hollow, every N points); fill under the curve; error bars (plus and/or minus) or an error band; data labels (also from a text column); box and whisker ranges; pie labels and ring; vector heads; colour map, levels and contour labels; 3D styles.
Legend	Position (also free), columns, title, font, frame, transparency, order, entries.
Colour bar	Title, ticks and format, map, normalization, position, size.
Panel background	Projection (2D, 3D, polar, ternary, Smith), background, aspect, grid (also on top of the data), 3D view, zoom and resolution.
Outside the panels	The figure: journal, size in mm, DPI, rows and columns of panels, spacing, base font, math font, Matplotlib style sheets, colour cycle, (a)/(b) labels, overall title.



The axis dialog

Mouse and keyboard

Action	Result
Single click	Selects the element; the object explorer follows.
Right click	Element menu: properties, hide, duplicate, delete, highlight a point, insert text or a line.
Drag the legend or a text	Moves it (the position is stored).
Drag a 3D panel	Rotates it (elevation and azimuth are stored).
Delete / Ctrl+Z / Ctrl+Y	Delete the selected element / undo / redo. **Edit → Undo history** lists every step.
Ctrl+Alt+C / Ctrl+Alt+V	Copy and paste the format between series, axes, texts, legends or shapes.
Alt+Enter	Properties of the selected element.

Panels and layout

The **Layout** tab adds and removes columns and rows of panels, sets the projection, shares x or y between panels, breaks an axis (// marks, the range left out) and adds a **secondary axis** with a unit conversion (eV↔nm, cm^{-1}↔nm, eV↔cm^{-1}, eV↔THz, K↔°C, rad↔degrees, Hartree and Rydberg→eV, a·x+b, a/x). **Insert → Zoom (inset)** creates an enlarged inset of a region you drag. **Format → Several curves** offers waterfall, stacked panels, varying symbols and dashes, a scatter matrix and marginal histograms.

Drawing and annotation

Insert: text (click where it goes), arrow with text, horizontal and vertical lines, reference bands, rectangle, ellipse, line, arrow, polygon, freehand curve, image, legend, colour bar, panel labels (a)(b) and figure title; **Graph → Equation...** (LaTeX notation) and **Date and time stamp**. **Insert → Data reader** (Ctrl+Shift+R) shows the coordinates of the nearest point; a double click annotates it.

Data tools on the graph

Graph → Data tools: move points with the mouse (the sheet changes too), drag a whole curve, remove bad points with a click (the row is masked), select the range used by fits and operations, and read free coordinates. All of them can be undone.

Themes and templates

Format → Save theme... stores the look of a figure (fonts, sizes, colours, ticks, grid); **Apply theme...** and the **Theme organizer** (with a preview of the current graph) reuse it. **Save as template** keeps a whole graph, including its curve types; **Graph from template** builds a new one with other columns.

3D

3D line, scatter, surface, mesh, triangulated surface, contour, bars, ribbons and the SciDAVis/QtiPlot styles (surface with mesh, hidden lines, polygons, points, crosses, cones). Surfaces come from matrices, from x, y, z columns, from a function z = f(x, y) or from a parametric definition x(u,v), y(u,v), z(u,v). **Graph → 3D rotation animation...** writes a GIF (or MP4 with ffmpeg).

Fitting and analysis

Select a curve (click it on the graph) and choose an operation in the **Analysis** tab or menu. Results go to the same panel, the right y axis, a new panel below or a new graph, and they are stored as sheets with their recipe.

Curve fitting

Analysis → Curve fitting (Ctrl+Shift+F).

Option	Choices
Model	Linear, slope (y = a·x), polynomials of degree 2–9, Gaussian (also by area), Lorentzian, Voigt, exponential decay (1st, 2nd and 3rd order) and growth, logistic, Boltzmann, power, logarithmic, Arrhenius, exponential saturation, ExpLin, rationals, Planck, 2–6 Gaussian or Lorentzian peaks, your own formula (`A*exp(-x/t) + c`) or a function saved in the **library**.
Parameters	Initial value, **lower and upper bounds**, fixed or free; *Estimate initial values* fills them from the data.
Range	Only part of the x axis (also from the range selector on the graph).
Weighting	None, instrumental ($1/\sigma^2$ from an error column), statistical ($1/y$), arbitrary dataset, direct weights.
Algorithm	Levenberg–Marquardt (trust region when there are bounds), Simplex (Nelder–Mead) or **orthogonal distance regression** (ODR: errors in x and y); iterations and tolerance.
Output	Fitted curve (with N points or at the x of the data), box with the parameters and R^2, residuals panel, **confidence and prediction bands** at the level you choose, errors scaled with $\sqrt{(\chi^2/\mathrm{dof})}$, significant digits, **covariance matrix**, report written to the notes.

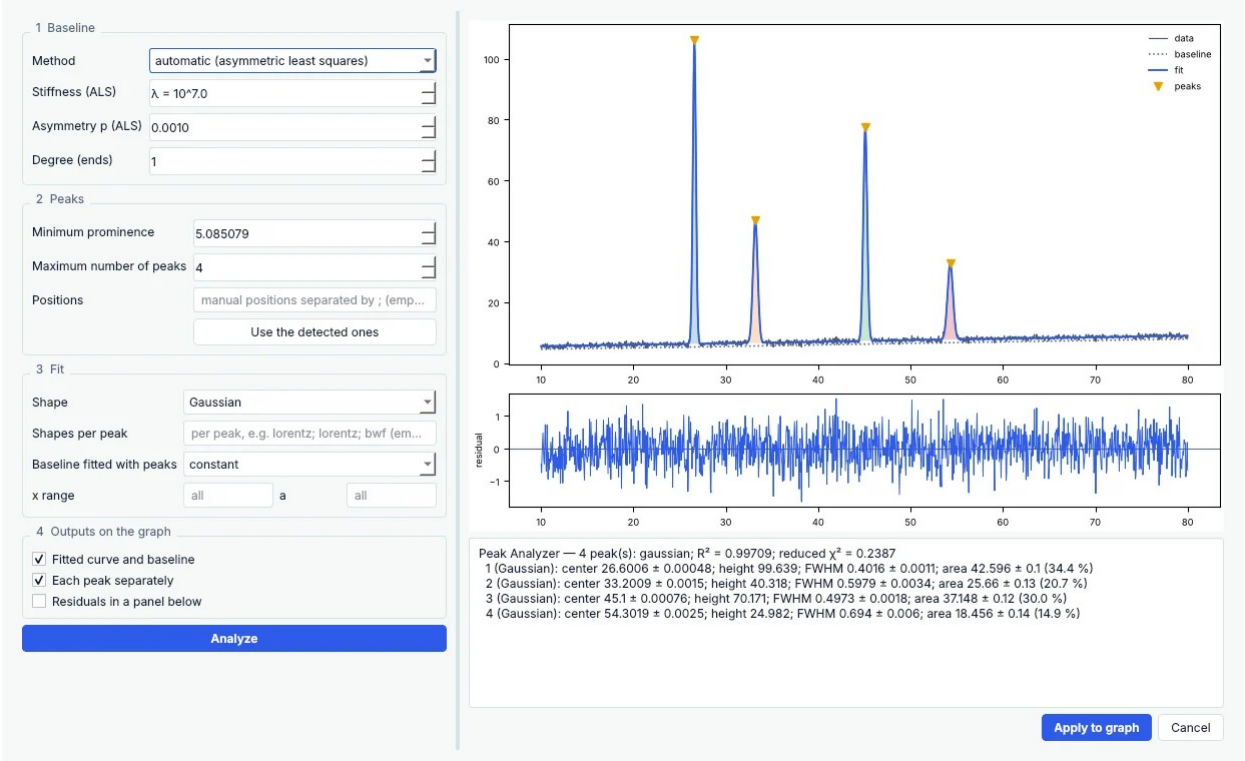
The report gives each parameter with its standard error, R^2, adjusted R^2, reduced χ^2 and RMSE. Everything the fit created — curve, bands, parameter table, box — is **recalculated in a chain** when the data change.

Analysis → Global fit fits one model to several curves at once and lets you choose which parameters are shared (the same width in every spectrum, for example).

For curves given as f(x, y) = 0 — circles, ellipses, conics — use **Analysis → Spectroscopy and diffraction → Fit implicit function...** (Worked examples, section 4).

Peaks

Analysis → Peak Analyzer (Ctrl+Shift+K) goes through four steps: baseline (asymmetric least squares or a polynomial through the ends), detection (prominence, maximum number, or positions typed by hand), the fit of Gauss, Lorentz, Voigt, pseudo-Voigt, Breit–Wigner–Fano, Pearson VII or bi-Gaussian peaks (one shape for all or one per peak, with an optional baseline fitted together), and a table with center, height, FWHM, area and area % — all with uncertainties. **Find peaks** is the quick version (positions and FWHM). The tools for Raman and X-ray diffraction are in Spectroscopy and diffraction.



The Peak Analyzer

Signal

Smoothing (Savitzky–Golay, adjacent averaging, FFT low-pass, LOWESS), baseline, FFT (amplitude, power, phase; Hann, Hamming, Blackman windows), interpolation (spline, Akima, PCHIP, linear), Butterworth low-, high-, band-pass and band-stop filters and the median filter, correlation and autocorrelation, convolution and deconvolution (Richardson–Lucy, Wiener), Hilbert envelope, derivative, integral (net and absolute area), integrate a function from a to b, normalize, translate a curve, subtract a line or a reference curve.

Statistics

- **Statistics** of columns (N, mean, standard deviation and error, quartiles, median, skewness, kurtosis) and of rows; **Curve statistics**.
- **Distribution fit**: normal, lognormal, Weibull, gamma, exponential, logistic and Cauchy, ranked by AIC and BIC with the Kolmogorov–Smirnov test, drawn over the histogram.

- **Hypothesis tests**: t (one sample, two samples, paired), ANOVA, Shapiro–Wilk, Kolmogorov–Smirnov, Mann–Whitney, Wilcoxon, Kruskal–Wallis, Levene, chi-square for independence and for the variance — with the decision at your level α. **Two-way ANOVA** with interaction.

Principal components, cluster analysis, ROC curves, kernel density, the spectrogram, wavelets and the digitizer have their own page: Multivariate, wavelets and the digitizer.

Recalculation and batches

Analysis → Automatic recalculation is on by default. Turn it off to work on large data, and press **Recalculate now** (Ctrl+F5) when you want the results updated; out-of-date results are marked.

Analysis → Batch processing runs the same fit, Peak Analyzer, statistics, integral or extremes over a folder of files or a list of datasets and collects a summary table — with the number in each file name (a temperature, for example) as the x axis — and its graph.

Spectroscopy and diffraction

Version 1.0.0rc5 added tools for Raman, infrared and UV-Vis spectra and for X-ray diffractograms. They are in **Analysis → Spectroscopy and diffraction**, next to the Peak Analyzer (**Analysis → Peak Analyzer**), which they use. Tools that work on a curve use the curve you clicked on the graph; tools that work on columns use the columns you select in the sheet.

Instrument files

File → Open data reads these files directly, like any table:

File	Instruments
`.xrdml`	Malvern Panalytical diffractometers (Empyrean, X'Pert, Aeris)
`.ras`	Rigaku diffractometers (SmartLab, MiniFlex, Ultima)
`.uxd`	Bruker DIFFRACplus text export
`.jdx`, `.dx`, `.jcamp`	JCAMP-DX spectra (IR, Raman, UV-Vis, 1D NMR), including the compressed forms
`.spc`	Galactic/Thermo GRAMS SPC (new format; several spectra with one x become several columns)

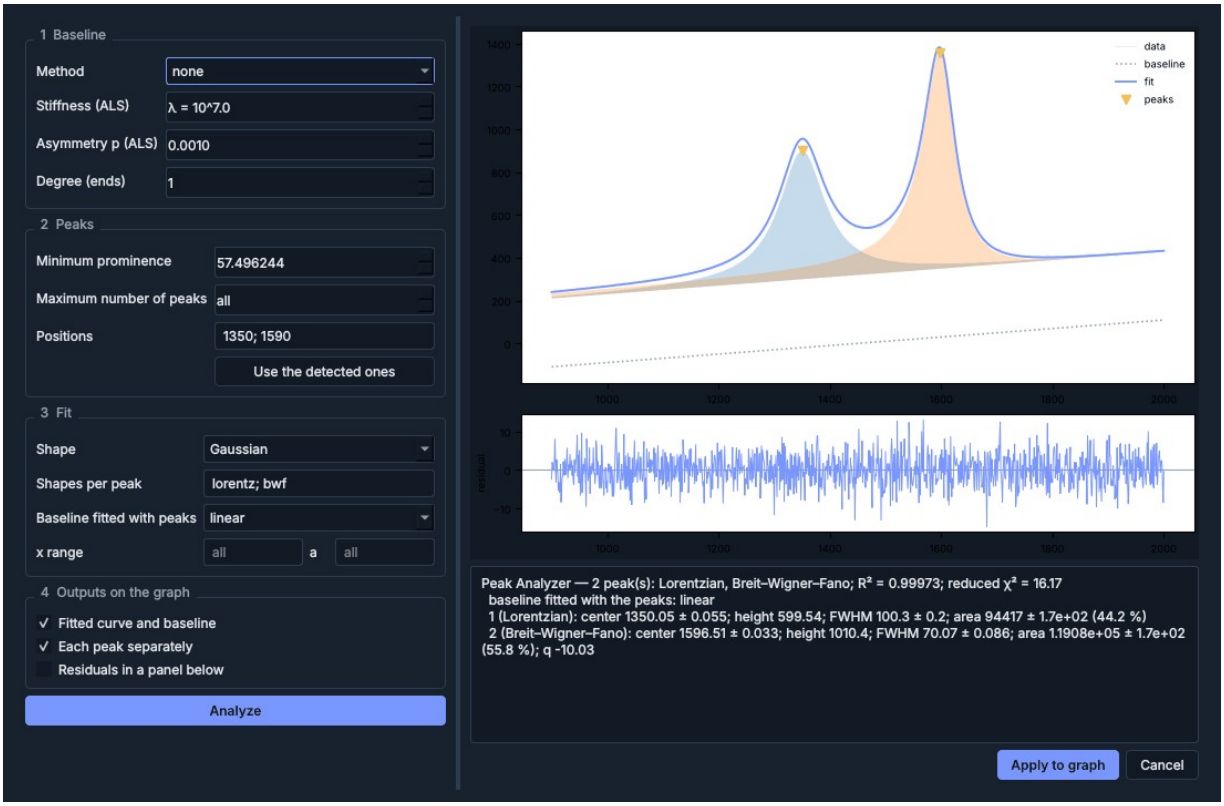
The wavelength, anode, laser line or sample name stored in the file is kept with the sheet. The X-ray tools then start with that wavelength.

Peaks with different shapes

In the Peak Analyzer, step 3 has two new fields.

- **Shapes per peak:** one shape key per peak, separated by `;`. For example, `lorentz; bwf` is a Lorentzian D band and a Breit–Wigner–Fano G band. The last shape repeats for the remaining peaks; left empty, the shape chosen above is used for all of them. The keys are:
  - `gauss`, `lorentz`, `voigt` and `pvoigt` (pseudo-Voigt);
  - `bwf` (Breit–Wigner–Fano);
  - `pearson7` (Pearson VII);
  - `bigauss` (asymmetric Gaussian, with its own width on each side).
- **Baseline fitted with peaks:** none, constant, linear or quadratic. It is fitted at the same time as the peaks, on top of the baseline subtracted in step 1. Choose *none* in step 1 to fit only this polynomial.

The table shows the shape of each peak and its shape parameter: η for the pseudo-Voigt, m for Pearson VII, q for BWF, and the left and right widths of a bi-Gaussian. A BWF tail never reaches zero, so its area is integrated over the fitted range, and its center is the position of the maximum.



Peak Analyzer with a shape per peak and a linear baseline fitted with the peaks

X-ray diffraction

Tool	What it does
X-ray diffraction analysis	Finds and fits the peaks of the selected diffractogram (x = 2θ in degrees). The new sheet has d, q and the Scherrer size of each peak. The instrumental width (from a standard such as LaB_6 or Si) is removed quadratically (Gaussian peaks), linearly (Lorentzian) or geometrically (Voigt). It also draws a Williamson–Hall graph with size and microstrain, and reports the cubic lattice (sc, bcc, fcc or diamond) and lattice constant that index the peaks best.
Strip Kα2	Removes the Kα2 doublet (Rachinger correction) and adds the Kα1-only curve to the graph.
Convert 2θ, d and q	Adds columns d (Å) and q ($Å^{-1}$) to the sheet, and optionally 2θ for another X-ray line.
Cubic reflections	The hkl, multiplicity, d and 2θ of a cubic lattice, drawn as vertical lines on the active panel.
Match reference peaks	Pairs observed 2θ values with a reference sheet that has a `d` or `two_theta` column (and optionally `hkl` and `intensity`).

Wavelengths are built in for the Kα1, Kα2, Kα and Kβ lines of Cu, Mo, Co, Cr, Fe and Ag. You can also type any value in ångström.

Step-by-step examples: Worked examples, sections 1, 2 and 4.

Raman

Tool	What it does
Remove spikes (cosmic rays)	Finds narrow spikes from the jumps between neighbouring points and replaces them by interpolation. Raman bands rise smoothly, so they are kept.
Wavelength ↔ Raman shift	Converts an absolute wavelength column (nm) into the Raman shift (cm^{-1}) for a laser line, or the reverse.
Carbon D and G bands	Fits the D band (Lorentzian) and the G band (BWF) with a linear baseline. It reports I_D/I_G (heights), A_D/A_G (areas), the crystallite size $L_a = 2.4\times10^{-10}\,\lambda^4\,(A_D/A_G)^{-1}$ (Cançado et al. 2006) and the defect distance L_D (Cançado et al. 2011, low-defect regime). Optionally it adds the D″ (≈1500 cm^{-1}) and D′ (≈1620 cm^{-1}) bands.

Fitting implicit functions

Fit implicit function fits a curve $f(x, y) = 0$ to the selected points. It minimises the distance of each point to the curve (orthogonal distance regression), not a vertical distance. The presets are:

- a circle;
- an ellipse (centre, semi-axes and angle, with eccentricity and area);
- a general conic;
- a straight line with errors in both x and y.

You can also type your own function, such as `x^2/a^2 + y^2/b^2 = 1`, with initial values `a = 1, b = 1`. The fitted curve goes on the graph; the parameters and their errors go to a new sheet.

Multivariate, wavelets and the digitizer

These tools came with version 0.16. Each one reads the columns you select in the active sheet (Ctrl+click the column headers), opens a short dialog and writes its results as **new sheets** (or matrices) plus a **graph**. A report goes to the Messages dock and opens in a window you can copy from.

Array files: HDF5, NetCDF and MATLAB

File → HDF5, NetCDF or MATLAB file..., or open or drop a `.h5`, `.hdf5`, `.nc` or `.mat` file onto the window.

The dialog lists every numeric variable with its size, type, units and long name. Groups appear as `group/name`.

- **As a sheet:** one-dimensional variables of the same length become columns. Their units and long names become the column's units and long name, so the axis labels come out right. The sheet remembers the file and the variables, so the project reloads them.
- **As a matrix:** one two-dimensional variable becomes a matrix. The coordinate variables of its dimensions supply x and y: NetCDF coordinate variables or HDF5 dimension scales. For variables with more dimensions, the first slice is taken.

NetCDF files in the classic format and in NetCDF-4 are both read, and so are MATLAB files from version 4 to 7.3, fields of structs included. Fill values, scale factors and offsets are applied.

Principal components (PCA)

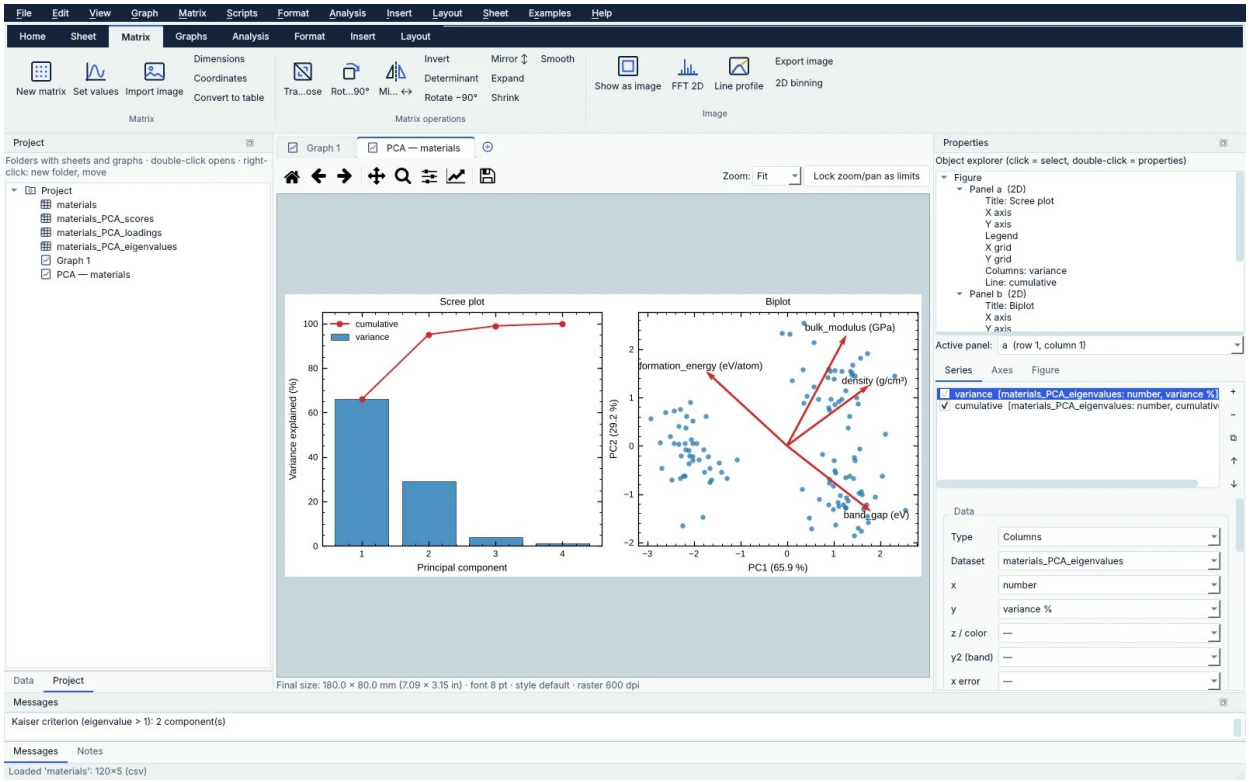
Analysis → Multivariate → Principal components. Select two or more numeric columns.

Option	Choices
Matrix	**correlation**: each variable standardized, as Origin does by default; or **covariance**, in the original units
Components to keep	1 up to the number of variables

The results are three sheets:

- **scores:** one row per observation;
- **loadings:** the eigenvectors;
- **eigenvalues:** with the percentage and the cumulative percentage of variance.

The graph shows the **scree plot** and the **biplot**: scores as points, loadings as arrows labelled with the variable names. The report also gives the Kaiser criterion, the number of eigenvalues above 1. Rows with a missing value are left out.



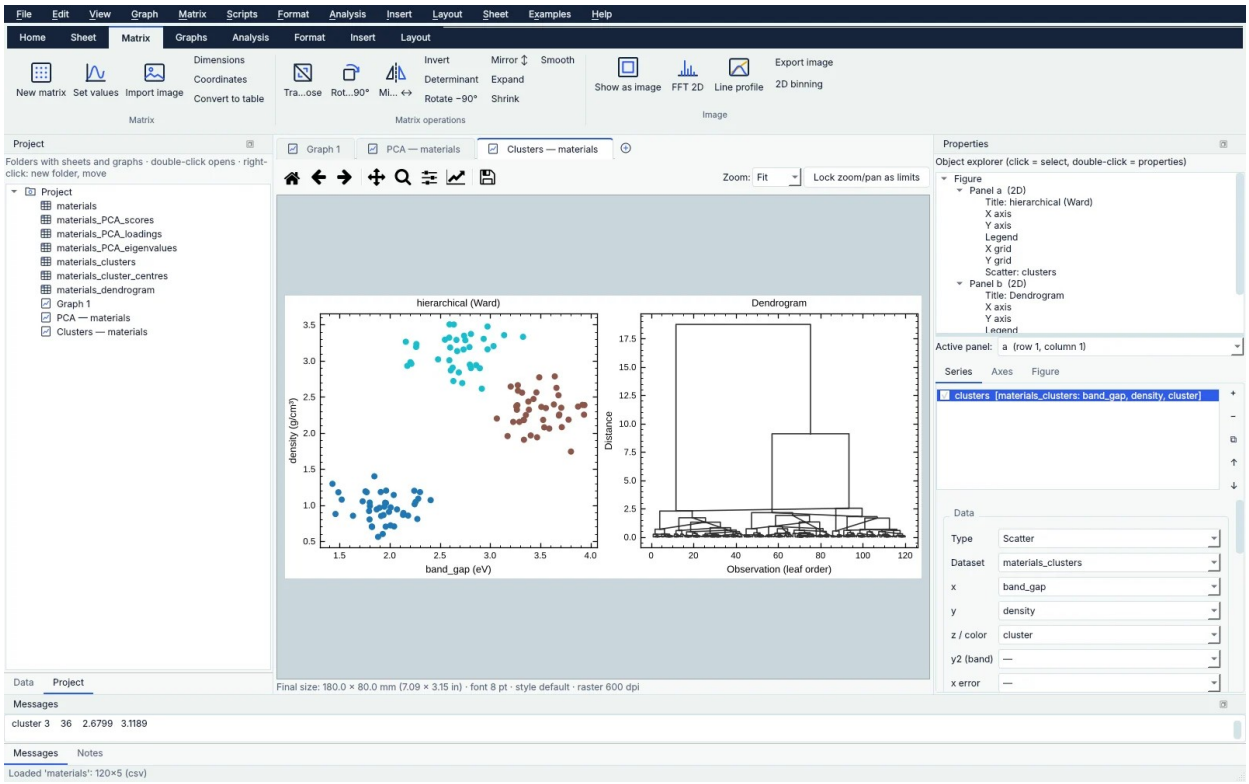
PCA: scree plot and biplot

Cluster analysis

Analysis → Multivariate → Cluster analysis.

- **k-means:** k-means++ starts; the best of ten runs is kept.
- **Hierarchical:** Ward, average, complete, single, centroid or median linkage. The tree is cut into the number of clusters you ask for.

Clusters are numbered by size. The new sheet holds the chosen columns, the cluster of each row and, for k-means, the distance to the centre. A second sheet holds the centres. The graph colours the points by cluster and, for the hierarchical method, draws the **dendrogram**. The report gives the mean silhouette width, from −1 to 1, where higher means better separated clusters, and for trees the cophenetic correlation.



Hierarchical clustering with the dendrogram

ROC curve

Analysis → Multivariate → ROC curve.

- Choose the score column (the result of a test or a classifier) and the column with the true class. The class column must have exactly two values, numbers or text.
- Choose which class is positive and whether a higher score means positive.

The sheet lists every threshold with the false-positive rate, the sensitivity, the specificity and Youden's J. The report gives:

- the **AUC** with its standard error (Hanley–McNeil) and 95 % interval;
- the p-value against AUC = 0.5;
- the best threshold.

Kernel density

Analysis → Multivariate → Kernel density.

- **One column:** a smooth estimate of its distribution with a Gaussian kernel. The kernel width comes from Scott's or Silverman's rule, or you fix it. The curve is drawn over the histogram, and the sheet also holds the cumulative distribution.
- **Two columns:** the 2-D density as a matrix and a colour map, with the points on top.

Spectrogram

Analysis → Time–frequency and wavelets → Spectrogram. Select x and the signal, or only the signal. The x values must be evenly spaced.

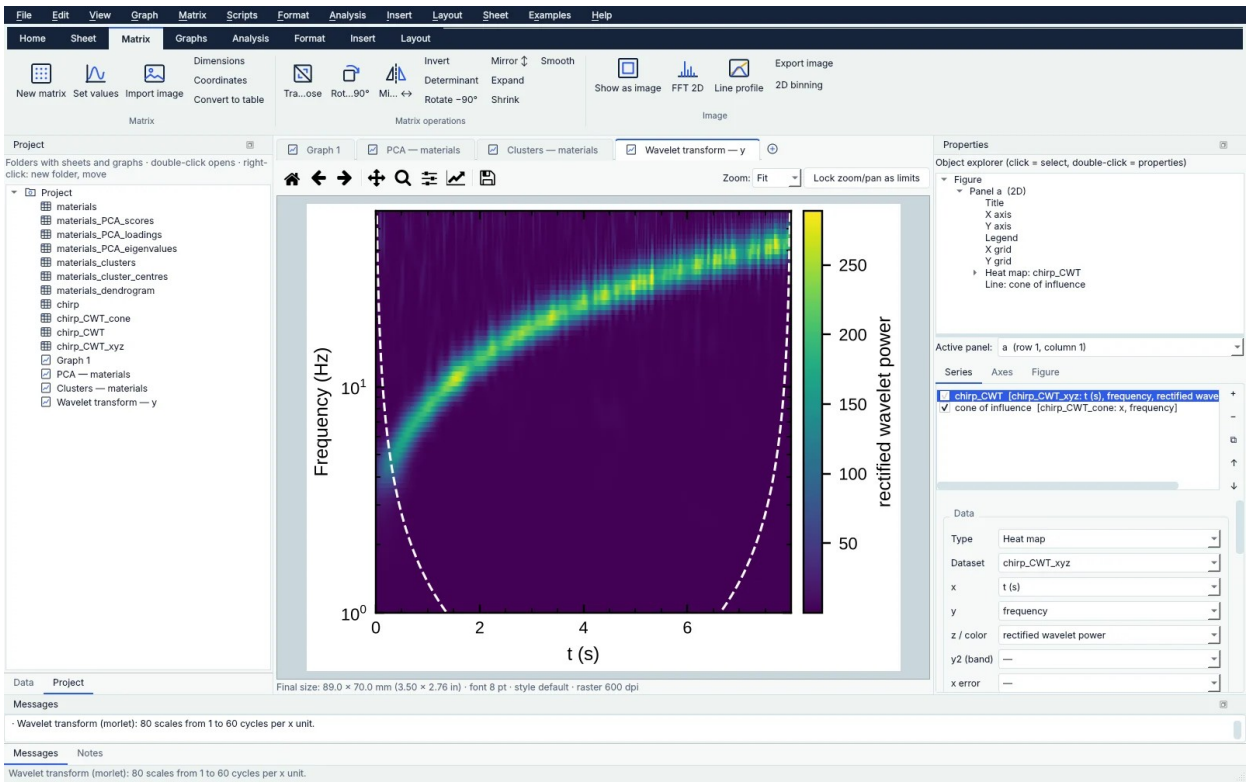
The settings are the window (Hann, Hamming, Blackman, rectangular, flat top, Bartlett), its length in points and the overlap. The result is a matrix of power density (in dB, as power or as amplitude) against x and frequency, drawn as a colour map. When x is in seconds, the frequency is in hertz.

Wavelets

Continuous wavelet transform:

- **Wavelets:** Morlet, Mexican hat or Paul.
- **Frequency range:** logarithmically spaced, with a default range.
- **Values:** power, **rectified power** (comparable across frequencies), amplitude or real part.

The scalogram is drawn on a logarithmic frequency axis. The dashed **cone of influence** marks where the edges of the record distort it.



Continuous wavelet transform of a chirp

Wavelet decomposition splits the signal into an approximation and the details at each level. This is the multiresolution analysis, with Haar, Daubechies, Symlets, Coiflets or biorthogonal wavelets. The parts add up to the signal again, and each one gets its own stacked panel.

Wavelet denoise / smooth:

- **Denoise:** shrinks the detail coefficients with the universal threshold $\sigma\sqrt{(2 \ln n)}$, soft or hard, where the noise σ comes from the finest details. You can also type your own threshold.

- **Smooth:** keeps only the approximation at a chosen level.

Fitting with errors in x and y (ODR)

In the **Fit Wizard**, choose the algorithm **Orthogonal distance (ODR, errors in x and y)**. It minimises the distances from the points to the curve, weighted by both error bars:

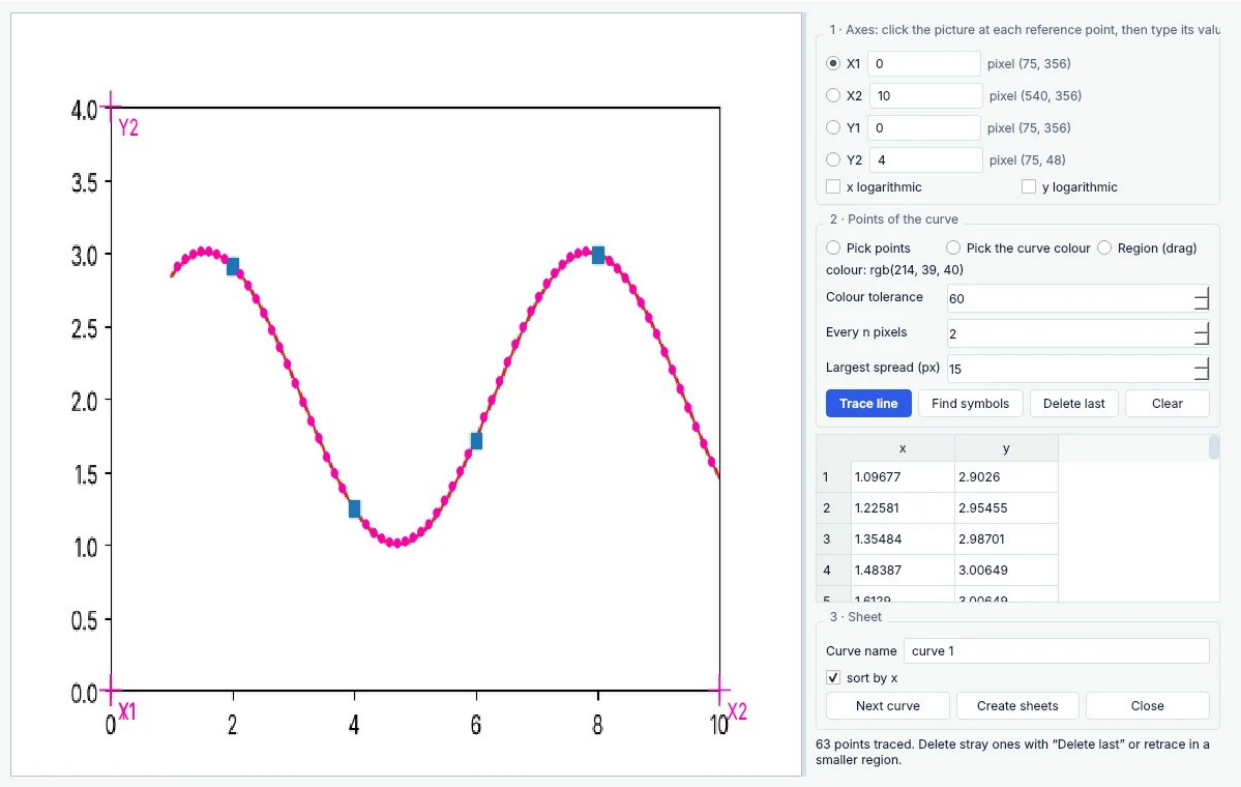
- the **x error bars** of the curve;
- the y errors from the *Weighting* option.

Without x error bars, x and y count equally. Bounds, fixed parameters, bands and chained recalculation work as with the other algorithms. The results match ODRPACK, the library behind Origin's and SciPy's ODR. In scripts: `fit("data", "T", "rho", "expdec", method="odr", sigma="drho", sigma_x="dT")`.

Digitizer

Analysis → Digitizer... (also in the File menu) opens a picture of a graph: a scan, a figure from a paper, a screenshot.

1. **Axes.** Choose X1 and click a labelled tick on the x axis, then type its value. Do the same for X2 and for Y1 and Y2 on the y axis. Tick *logarithmic* for log axes. Slightly rotated scans are handled.
2. **Points.**
   - *Pick points* adds a point per click.
   - *Pick the curve colour* followed by **Trace line** follows a line through the whole picture.
   - **Find symbols** finds markers of that colour.
   - *Region* (*drag*) limits the search and keeps the legend out.
   - The tolerance, the step and the largest spread tune the tracing.
3. **Sheet. Next curve** keeps the points and starts another curve. **Create sheets** makes one sheet per curve and a graph with all of them.



The digitizer tracing a curve

In scripts

`pca("data", ["a", "b", "c"])`, `kde("data", "a")`, `roc("data", "score", "class")`, `cluster("data", ["a", "b"], 3, method="ward")`, `spectrogram("signal", "t", "y")` and `cwt("signal", "t", "y", wavelet="morlet")` return the same results as dictionaries of arrays and tables.

Plot wizard, windows and presentations

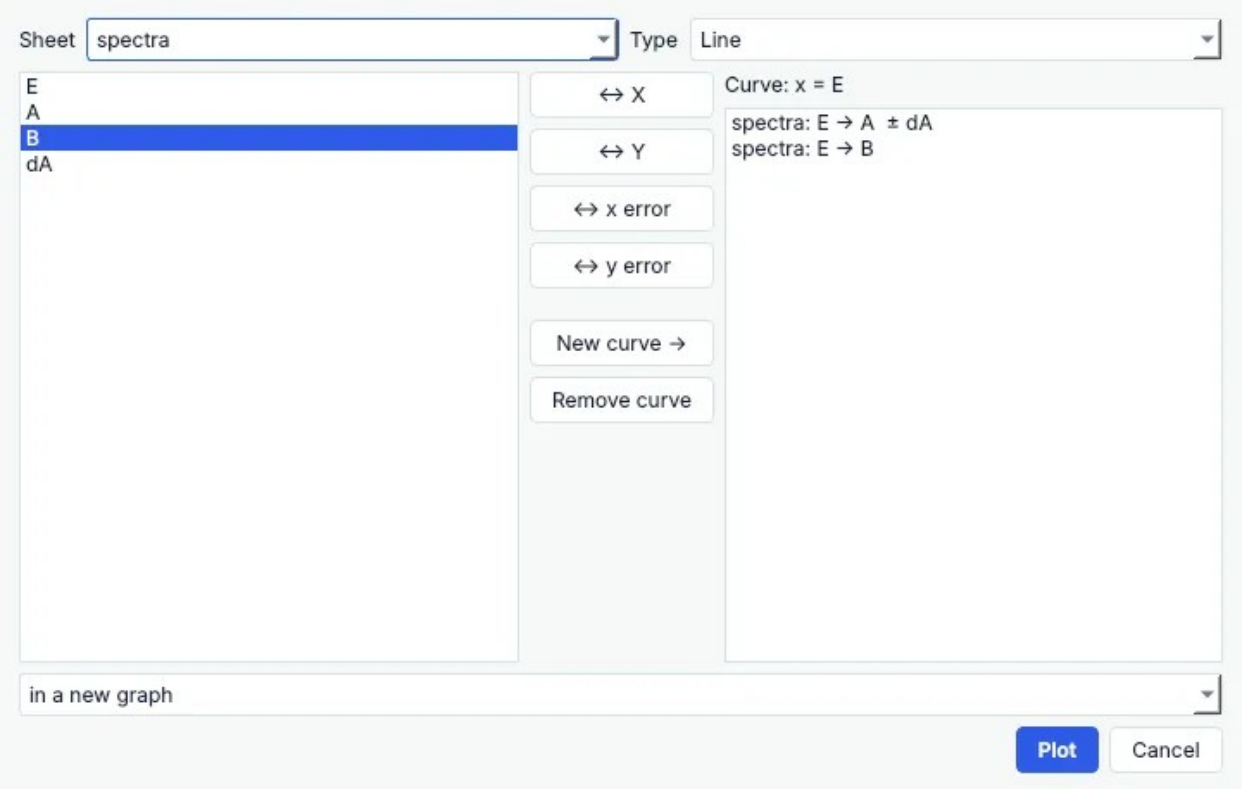
Version 0.17 added the last tools of SciDAVis and QtiPlot that SciVelyx Graphs did not have yet. They speed up everyday work with many sheets and graphs.

Plot wizard

Graph → Plot wizard... builds a graph curve by curve, without selecting columns in the sheet first.

1. Choose the **sheet** and the **type** (line, line + symbol, scatter, error bars, steps, columns, area, stems).
2. Click a column, then **↔ X**. Do the same for **↔ Y**, and for **↔ x error** or **↔ y error** if the curve has error bars.
3. Click **New curve →**. X stays chosen, so the next curve only needs a new Y.
4. Choose *in a new graph* or *in the active panel* and click **Plot**.

Curves with an error column are drawn with error bars.



The plot wizard with two curves

Curves and drawing order

- **Hide other curves** and **Show all curves** are in the right-click menu of a curve and in the Graph menu.
- **Edit → Bring to front** (Ctrl+Shift+]) and **Send to back** (Ctrl+Shift+) change the drawing order of the selected text, shape, image or curve. They are also in the right-click menu.
- **Series → Data → Rows (first; last)** draws a curve from part of its sheet only, for example `11; 60`.

New graph options

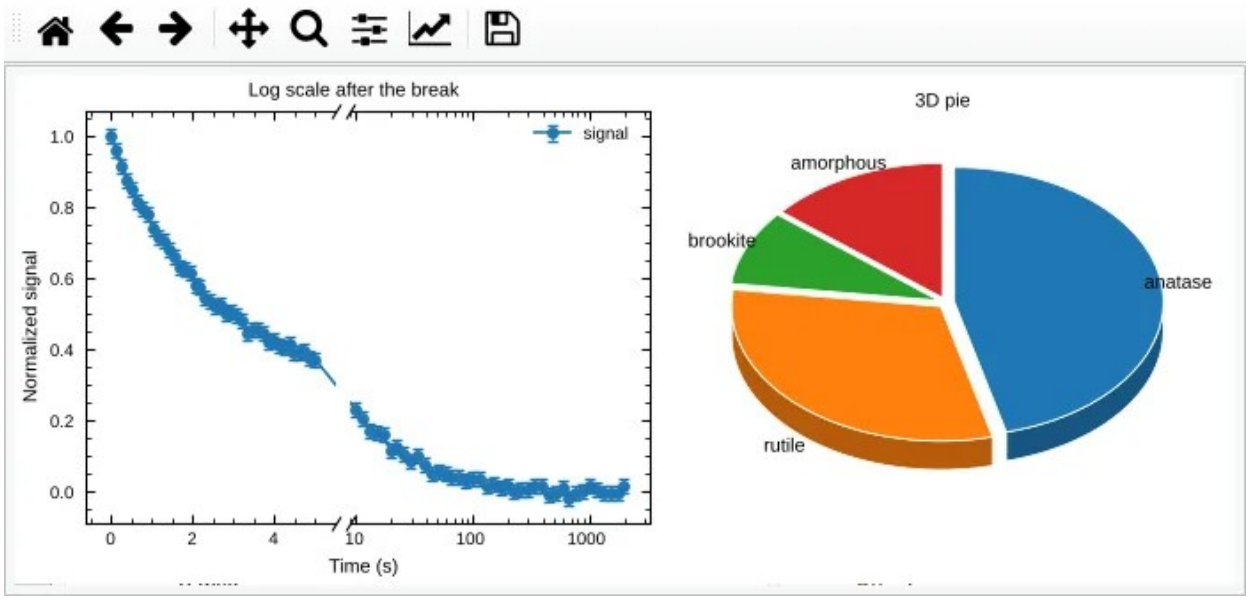
- **Axis break with another scale after it.** In the panel's *Axis break* section, choose *Scale after the x break* (or y) = logarithmic for a linear start followed by decades. You can also set the major steps on each side (`before; after`) and the position of the break along the axis.
- **Anti-aliasing:** Figure → *Anti-aliasing*, or Panel → *Anti-aliasing* for one panel.
- **Picture as the symbol:** Series → Symbol → *Picture as the symbol* (a PNG, JPEG…). The height of the picture follows the symbol size.
- **Error bars over the symbols:** Series → Error bars → *Through the symbol*.
- **3D pie:** Series → Pie → *3D pie*, with tilt and thickness.
- **Speed mode:** Series → Line → *Speed mode*. Curves longer than the number of points you give are simplified with the Douglas–Peucker algorithm. The tolerance is a fraction of the data range, and gaps are kept.
- **Logarithmic 3D axes:** in a 3D panel, *Logarithmic axes* = x, y, z or any combination.

Zoom and drag along one axis

View → Zoom and drag has *Zoom horizontally*, *Zoom vertically*, *Drag canvas horizontally* and *Drag canvas vertically*: the other axis stays as it is. *Normal pointer* ends the mode. To keep the result as the axis limits, use **Lock zoom/pan as limits**.

Projects and windows

- **File → Append project...** adds the sheets, graphs, notes, folders and scripts of another project to the open one. The project can come from SciVelyx, SciDAVis, QtiPlot or the Origin exporter. Sheets whose names are already in use are renamed, and the graphs follow the new names.
- **File → Save window as project...** saves the active graph with only the data it uses, including the sheets its analyses were computed from, as a project of its own.
- **File → All graphs as a presentation (.odp)...** writes one slide per graph, titled with the graph's name. The file opens in LibreOffice Impress and in PowerPoint.
- The **Window** menu lists every graph, sheet and matrix. **Next window** (Ctrl+PgDown) and **Previous window** (Ctrl+PgUp) cycle through the graphs. **Open graph in a separate window** shows the active graph in a window of its own, which is redrawn after every change; *Tile separate windows* arranges them side by side.



A graph in a separate window

Your own menu actions and forms

- **Scripts → Add custom script action...** puts one of your Python files in **Scripts → Custom actions** and on a toolbar, with an optional keyboard shortcut. The actions are kept between sessions.
- **Scripts → Open custom interface (.ui)...** shows a form made with Qt Designer. To make it do something, load it from a script:

```
w = load_ui("normalize.ui")
w.runButton.clicked.connect(lambda: print(table(w.sheetEdit.text())["y"].max()))
w.show()
```

The widgets of the form are available by their object names.

Stem-and-leaf plot

Select a numeric column and choose **Analysis → Stem-and-leaf plot**. The display goes to the project notes and the Messages dock, and a text graph is created. The leaf unit is chosen so there are about $\sqrt{n}$ stems. In scripts, `scivelyx.core.extras17.stem_leaf(values, leaf_unit=1, split=2)` splits each stem in two lines.

Small things

- **Edit → Copy status bar text**, or right-click the status bar.
- **Help → Search for updates...** reads a small file on this website with the latest version number. Nothing about you or your computer is sent.

Regression, statistics and gadgets

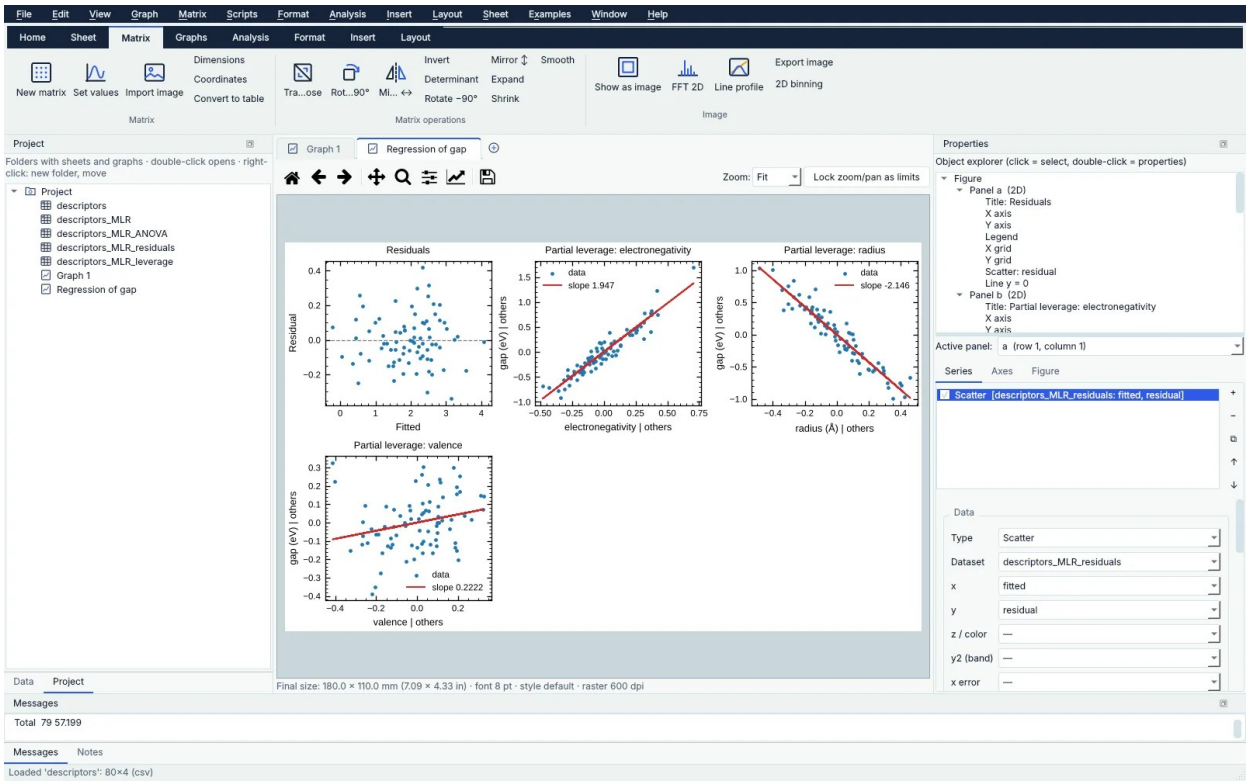
Version 0.18 added the analysis tools of Origin (standard edition) that were still missing. They are in the **Analysis** menu, in the submenus *Fitting* (*more*), *Peaks* (*more*), *Statistics* (*more*), *Signal and mathematics* (*more*) and *Gadgets*. Tools that work on columns read the columns you select in the sheet (Ctrl+click the headers). Tools that work on a curve use the curve you clicked on the graph. Results go to new sheets, to the Messages dock and, where it helps, to a graph.

Fitting

Multiple linear regression. Select the dependent column and the predictors, then choose which column is the dependent one. The report and the sheets give:

- each coefficient with its standard error, t, p, confidence limits and variance inflation factor (VIF);
- the ANOVA table, R^2, adjusted R^2 and RMSE;
- the fitted values and the residuals.

The graph shows the residuals against the fitted values and one *partial leverage* (added-variable) plot per predictor. The slope of each partial leverage plot is that predictor's coefficient.



Multiple regression with partial leverage plots

Confidence ellipse. Select two columns. The ellipse holds a new observation (*population*) or the mean with the probability you choose. The graph also shows the data and the straight-line fit.

In the **Fit Wizard**:

- **× N** repeats the model N times with a shared offset y0, for example three Gaussian peaks of the same shape. The parameters become `xc_1`, `xc_2`, …
- **Derived** takes expressions of the fitted parameters, such as `FWHM = 2*sqrt(2*ln(2))*w; area = A*w*sqrt(2*pi)`. Their errors come from the covariance matrix (delta method). They are recalculated with the fit when the data change.

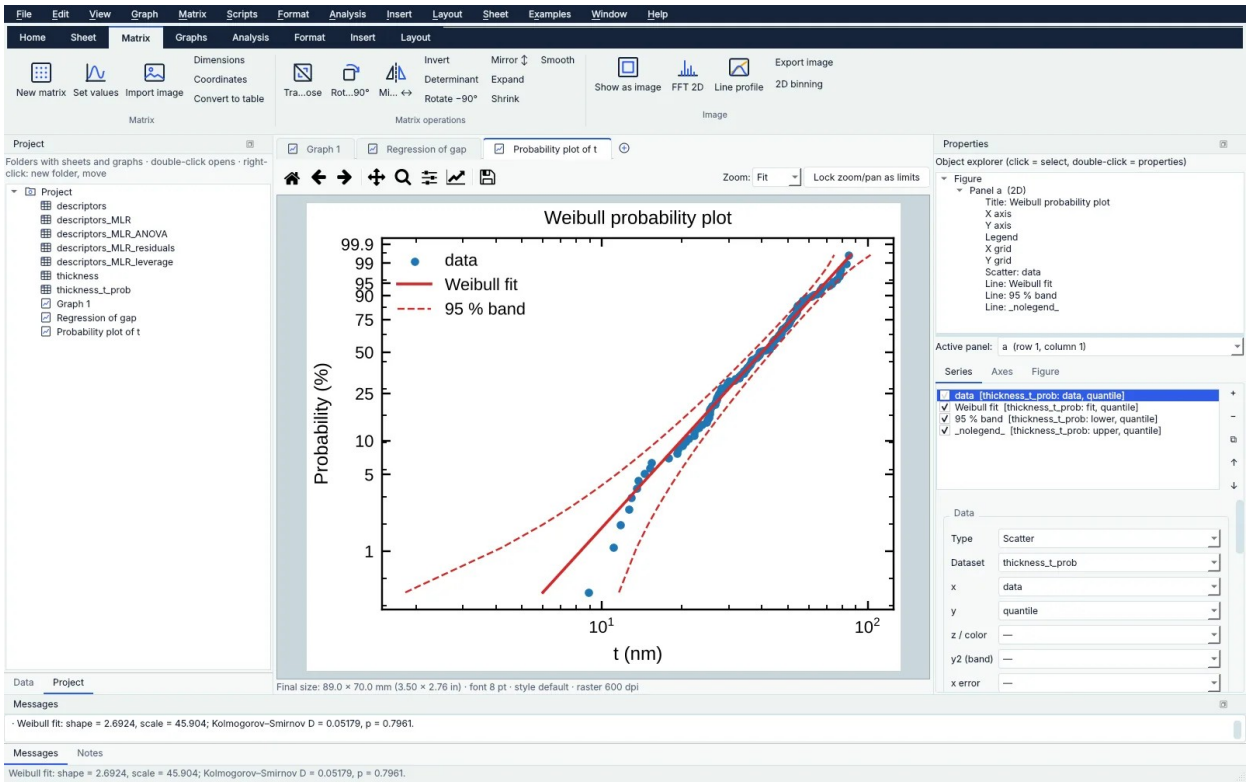
Peaks

- **Integrate peaks** finds the peaks of the selected curve and integrates each one between its boundaries, without fitting. It reports the area, centre, centroid, height, FWHM and area %. The baseline can be none, a straight line between the ends, or automatic.
- **Batch peak analysis** runs the same analysis, integration or a Gaussian fit, on every curve of the graph and writes one summary sheet.

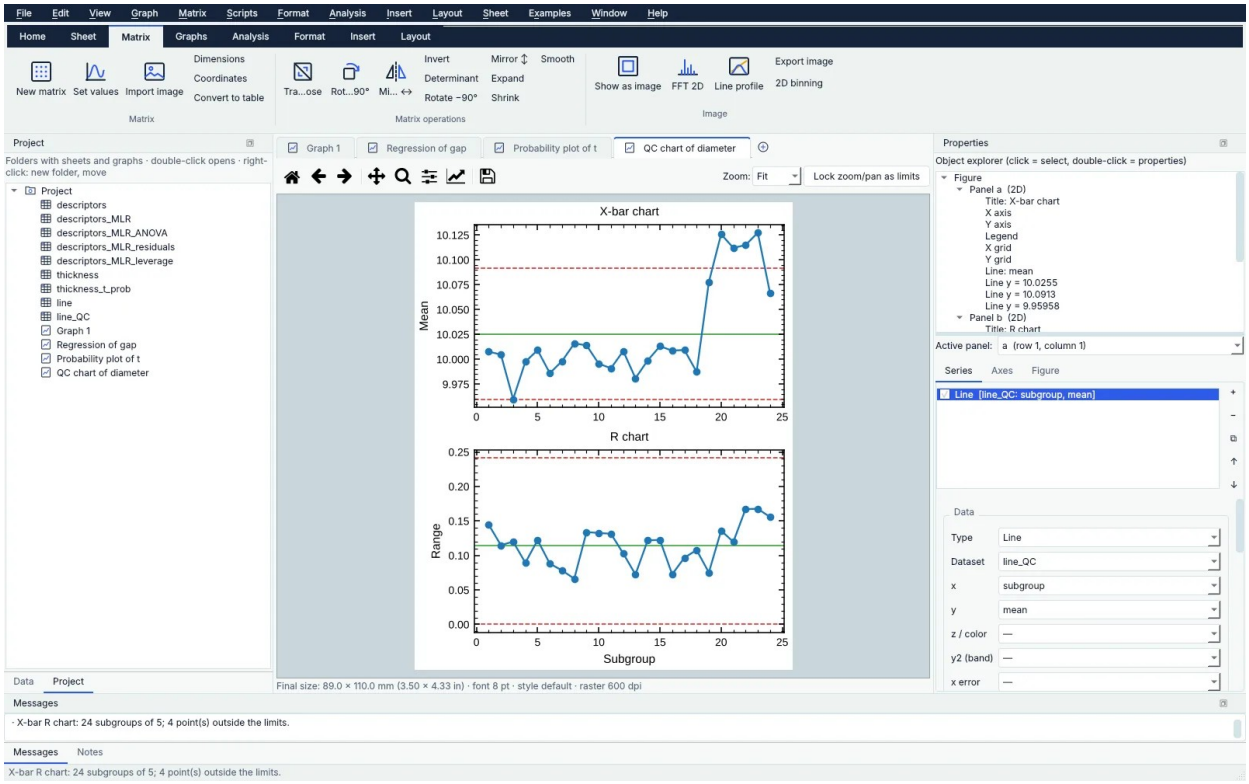
Statistics

Tool	What it gives
Normality tests	Shapiro–Wilk, Kolmogorov–Smirnov, Lilliefors, Anderson–Darling, D'Agostino K^2 and Chen–Shapiro, with the decision at your level
Probability plot	The ordered data on the probability scale of a normal, log-normal, Weibull, exponential, logistic or Gumbel distribution, with the fitted line and a confidence band
2D frequency count	Counts in a grid of x and y bins, as a matrix and a colour map
Discrete frequency	How often each distinct value or text occurs, with percent and cumulative values
QC chart (X-bar R)	Means and ranges of subgroups (consecutive values, or labels in a first selected column) with centre lines and control limits
Outlier tests	Grubbs (once or repeated) and Dixon's Q; optionally masks the outlying rows
Means comparison	After a one-way ANOVA (one column per group): Tukey, Bonferroni, Šidák, Fisher LSD, Scheffé, Holm–Bonferroni, Holm–Šidák or Dunnett against a control

The Lilliefors, Chen–Shapiro and Dixon p-values come from simulated normal samples with a fixed seed, so they are the same every time.



A Weibull probability plot



An X-bar R chart

Signal and mathematics

- **Percentile filter:** each point is replaced by a percentile of its neighbours (50 = median).
- **FFT filter: parabolic or threshold:** a low pass whose gain falls as $1 - (f/fc)^2$, or the removal of every Fourier component weaker than an amplitude.
- **Average multiple curves:** the curves of the active panel on a common x, with mean, standard deviation, standard error and count.
- **Trace interpolation:** points evenly spaced along the curve in the order of the data. **On XYZ**, z is read along a path that you give over scattered x, y, z data.
- **3D interpolation:** scattered x, y, z and a value become a regular grid. Each z level becomes a matrix.
- **Polygon area:** the area, centroid and perimeter of a closed curve.
- **Intersections:** where the curves of the panel cross each other, or a horizontal or vertical line.

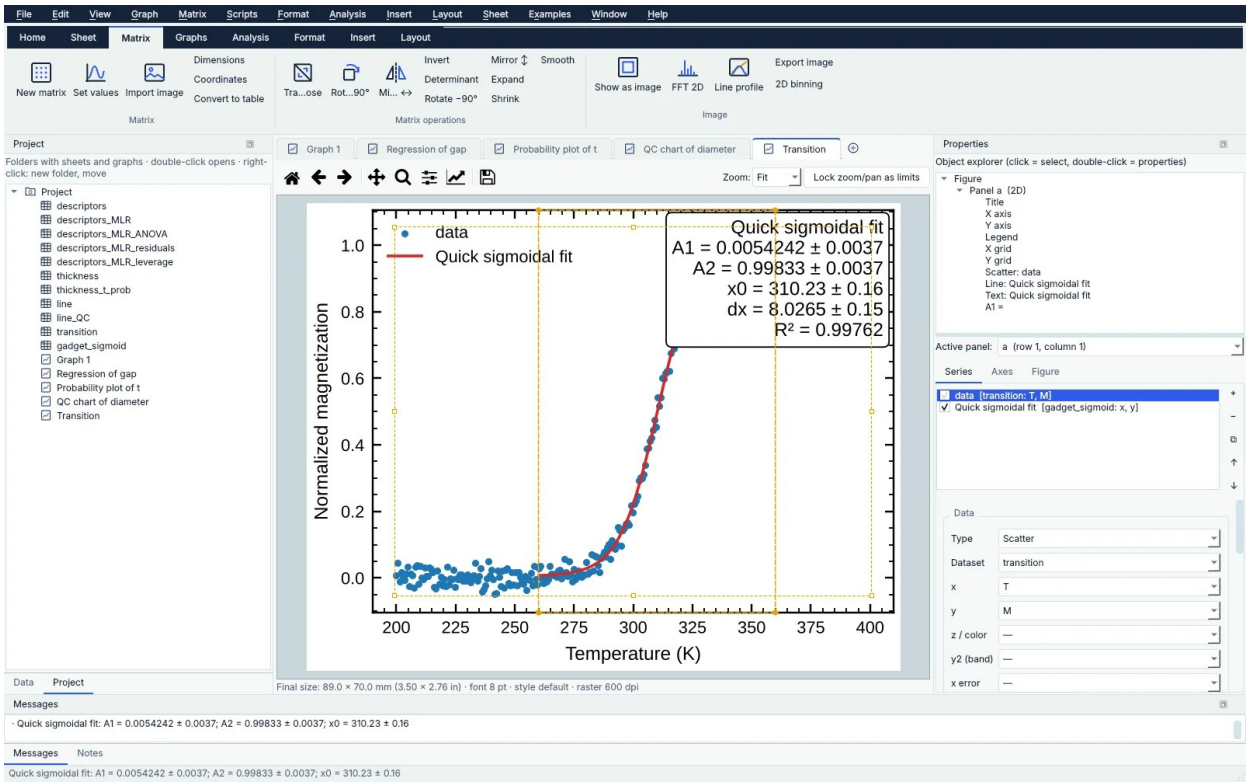
Gadgets

Vertical cursor (all graphs): click or drag on a graph. The status bar shows the value of every curve of every graph at that x.

The other gadgets work on the selected curve, inside the **range selector** (Analysis → Gadgets → Range selector, then drag along x):

- quick fit (straight line) and quick sigmoidal fit;
- peaks, differentiate, interpolate and integrate;
- FFT and statistics;
- rise time (10–90 % rise and fall times of the edges in the range);
- cluster (number of points, means, standard deviations and correlation of the points in the range).

The result appears in a box on the graph, and a curve or a second graph is added when there is one. Move the range and the gadget is computed again. **Stop the gadget** ends it.



The sigmoidal fit gadget

Added in 1.0.0rc7: more statistics, signal processing and mathematics

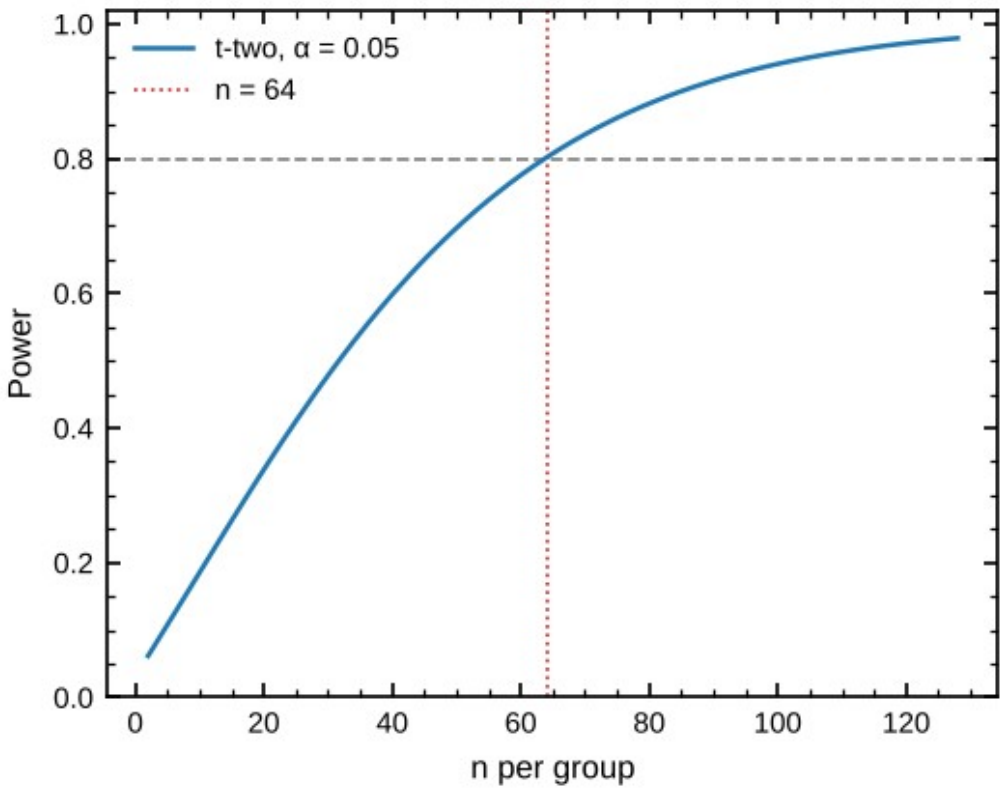
These are the OriginPro tools most used in physics. They are in the same submenus.

Statistics (more)

Tool	Select	You get
Partial correlation	three or more columns	the correlation of each pair with the other columns held constant (Pearson or Spearman), t and p
Cross tabulation tests	two columns of categories, or the columns of a table of counts	χ^2, likelihood ratio G, Fisher's exact test (2×2), φ, Cramér's V and the contingency coefficient; a warning when expected counts are below 5
Two-sample F test for variances	two columns	$F = s_1^2/s_2^2$, p and the confidence interval of the ratio
Proportion tests	nothing (type the counts)	one proportion against p_0 (exact binomial and z, Wilson and exact intervals) or two proportions (z test and Fisher's exact test)
Repeated-measures ANOVA, one way	one column per condition, one row per subject	the ANOVA table, partial η^2, Mauchly's test of sphericity, and the p values corrected by Greenhouse–Geisser and Huynh–Feldt

Tool	Select	You get
Repeated-measures ANOVA, two way	the a × b columns in the order A1B1, A1B2, ..., A2B1, ...	each effect with its own error term and Greenhouse–Geisser correction, and the cell means
Sign test	one column (against a median) or two (paired)	positive, negative and tied differences, and p
Wilcoxon signed-rank, one sample	one column	W and p against a hypothesised median
Two-sample Kolmogorov–Smirnov	two columns	D and p: do the samples come from the same distribution?
Mood's median test	two or more columns	χ^2, p and the counts above and not above the grand median
Friedman ANOVA	three or more columns (rows are subjects)	χ^2, p, mean ranks and Kendall's W
t tests on rows	the columns of group A, then those of group B	a t test on every row (Welch, paired or one-sample), with the Benjamini–Hochberg q values
Power and sample size	nothing	the power curve against n and the n that reaches the target power, for t tests, one-way ANOVA and two proportions

The power calculations agree with G*Power: n = 64 per group for a two-sample t test with d = 0.5, α = 0.05 and power 0.8.



Power of a two-sample t test against the number per group

Fitting (more)

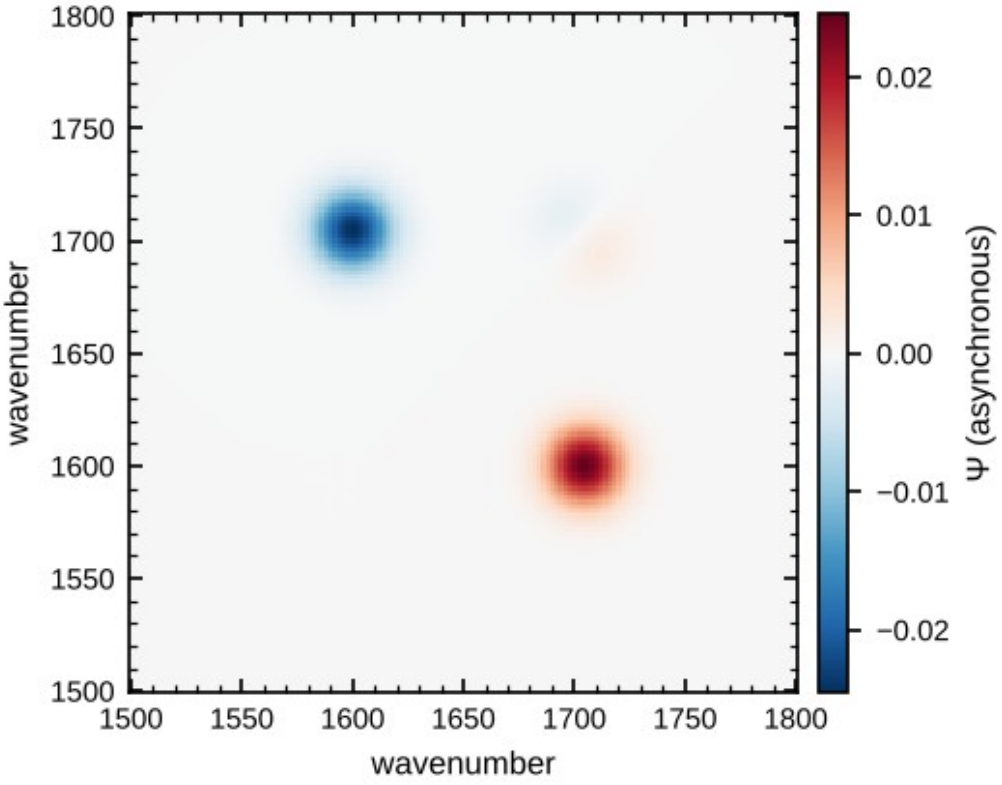
F test of nested models fits two models to the selected curve, a simpler one and one with more parameters that contains it (linear and quadratic, one and two Gaussians). The extra-sum-of-squares F test and the AICc of each model say whether the extra parameters are worth it. Both fitted curves are drawn.

Signal and mathematics (more)

- **Power spectral density (Welch):** the averaged periodogram of the selected curve, with segment length and window.
- **Coherence:** the magnitude-squared coherence (0 to 1) of two curves of the panel sampled on the same x.
- **Decimate:** keeps one point in n after an anti-aliasing filter.
- **Rise and fall times:** the 0 % and 100 % levels are the two most frequent values of the signal (IEEE 181). Each rising and falling edge is timed between the levels you choose (10 % and 90 % by default). Dotted lines mark the levels on the graph.
- **Hilbert transform:** the transform, the envelope, the unwrapped phase and the instantaneous frequency.
- **2D correlation (Noda):** select one column per step of a perturbation (temperature, time, pressure), with the spectral axis as their x column. You get the synchronous map Φ and the asynchronous map Ψ as matrices and colour maps. ν_1 is read on the y axis and ν_2 on the x axis: when $\Phi(\nu_1, \nu_2)$ and $\Psi(\nu_1, \nu_2)$ have the same sign, the band at ν_1 changes before the one at ν_2.
- **Continuous wavelet function:** the values of a Morlet, Mexican hat, derivative-of-Gaussian or Paul wavelet at a scale and shift, normalised to unit energy.
- **2D FFT filter** (also in the Matrix menu): low-pass, high-pass, band-pass or band-stop filter of the active matrix, with an ideal, Gaussian or Butterworth window.
- **2D volume integration and Surface area:** for three columns x, y, z (a regular grid or scattered points, which are triangulated) or for the active matrix.
- **Error propagation:** type the function, such as `a*b/c`, and the variables as `a = 2 ± 0.1; b = 3 ± 0.2`. You get the first-order uncertainty with the contribution of each variable, and a Monte Carlo check.
- **Convex hull and centroid** (also in the Graph menu): the hull of the selected scatter, its centroid and the lines from the centroid to every point.

Matrix → Eigenvalues and eigenvectors decomposes the active square matrix.

The digital filter (**Analysis → Filter (IIR / median)**) now offers the Chebyshev I and II, elliptic and Bessel designs besides Butterworth, with the pass-band ripple and the stop-band attenuation.



Asynchronous 2D correlation map of a heating series: the band at 1600 cm^{-1} grows before the one at 1700 cm^{-1}

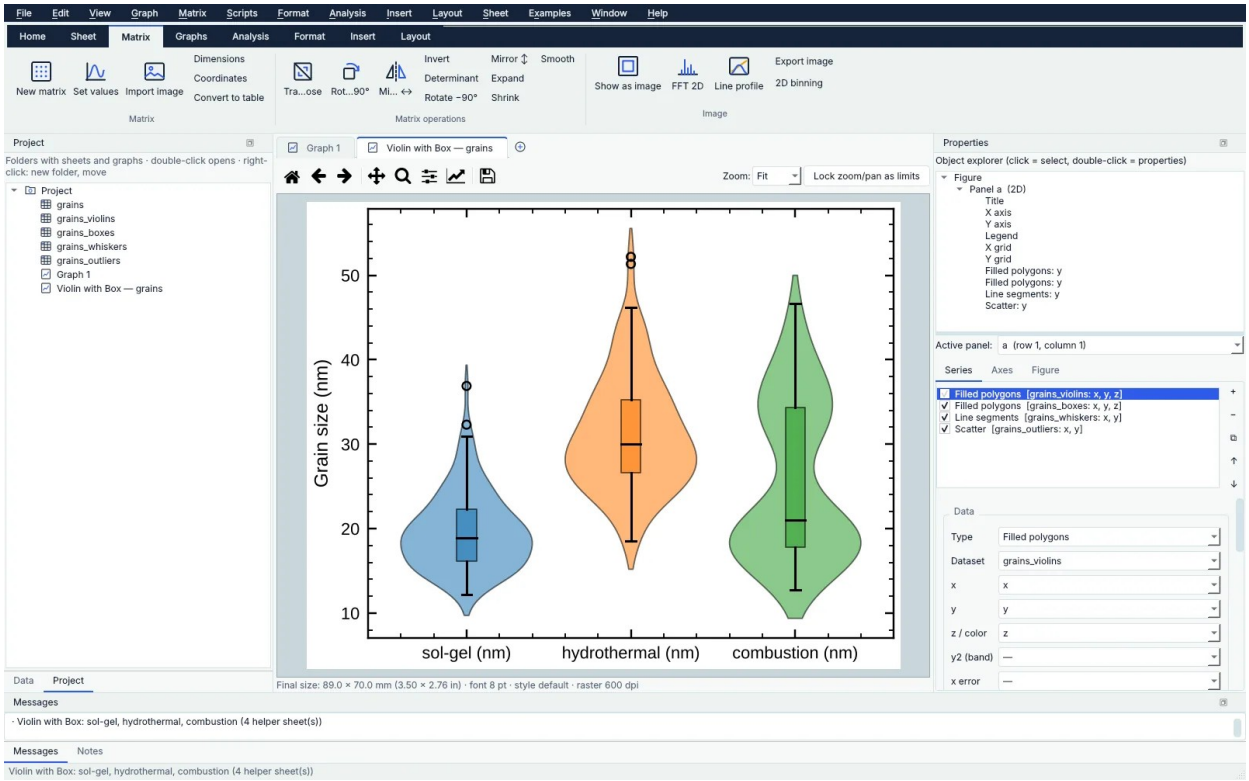
Origin graph types

Graph → Origin graph types has one entry for each graph type of Origin's *Basic 2D*, *Bar, Column*, *Pie, Doughnut*, *Area*, *Multi-axis, multi-panel* and *Statistical* chapters, under Origin's own names, and since 0.20 also the *Specialized*, *Financial*, *3D XYY*, *3D surfaces*, *3D symbol, bar, vector*, *Contour*, *Waterfall, grouped, image* and *Map* chapters, and since 0.21 the graphing *Apps* of Origin — 176 types in all.

To make one:

1. Select the columns in the sheet (Ctrl+click the headers), in the order the table below gives.
2. Choose the type in **Graph → Origin graph types**.

Some types need numbers that are not in your sheet, such as means and confidence intervals, kernel densities, box statistics, percentages or the polygons of a Sankey diagram. Those numbers are calculated and kept in small helper sheets named after your sheet. The graph is an ordinary graph: every curve, axis, text and colour can be edited, and the helper sheets can be inspected or exported.



Violin with box

Basic 2D

Graph type	Columns to select
Scatter Central	x and y
Column Scatter	one column per group
Error Band	x, y and the error
Scatter + Rug	x and y
Line Series	two or more columns; each row becomes a line across them
Row-wise (each row a curve)	the columns; their names are the x values if numeric
Before and After	the 'before' and the 'after' columns
2 Point Segment	x and y; rows 1–2, 3–4, … are joined
3 Point Segment	x and y; rows 1–3, 4–6, … are joined
Spline Connected	x and y
Lollipop	categories (or x) and values
Density Dots	x and y
Color Dots	x, y and optionally the column that sets the colour
Colormapped Line Series	x and two or more y columns
Stacked Lines by Y Offsets	x and two or more y columns

Bar, Column

Graph type	Columns to select
Column + Label	categories (or x) and values
100% Stacked Column	categories and two or more value columns
100% Stacked Bar	categories and two or more value columns
Kite Diagram	position along the transect, then one column per species/category

Pie, Doughnut

Graph type	Columns to select
3D Color Pie Chart	categories and values
Pie of Pie	categories and values
Bar of Pie	categories and values
Doughnut of Pie	categories and values
Doughnut with Total	categories and values
Pie with Different Radius	categories, values (angle) and radius
Doughnut with Different Radius	categories, values (angle) and radius

Area

Graph type	Columns to select
100% Stacked Area	x and two or more value columns
Fill Area with 2 Colors	x, y and optionally a second y (else y = 0)

Multi-axis, multi-panel

Graph type	Columns to select
Double-Y Column	x (or categories), left y, right y
2Ys Column	x (or categories), left y, right y
Double-Y Column – Line Symbol	x (or categories), columns, line
2Ys Y-Y (Double Y)	x, left y, right y
3Ys Y-YY	x and three y columns (one left, two right)
3Ys Y-Y-Y	x and three y columns (two left, one right)
4Ys Y-YYY	x and four y columns (one left, three right)
4Ys YY-YY	x and four y columns (two left, two right)
Multiple Y Axes	x and 2 to 7 y columns, each with its own axis
9 Panel	x and up to nine y columns
Multiple Panels by Label	the label (group) column, x and y
Double X	x1, y1, x2, y2 (the second pair uses the top x axis)

Statistical

Graph type	Columns to select
Interval Plot	one column per group
Bar Chart (mean ± SD)	one column per group
Bar Overlap	one column per group
Bar Normal	one column per group
Box Overlap	one column per group
Box Normal	one column per group
Half Box	one column per group
Scatter Interval	one column per group
Marginal Box Charts	x and y
Beeswarm	one column per group
Histogram + Rug	one column
Histogram + Probabilities	one column
Multi-Panel Histograms	two or more columns
Distribution (+ Rug)	one column
Histogram with Labels	one column
Population Pyramid	age groups, left values (e.g. men), right values
Marginal Plots (grouped)	x, y and the group column
Violin with Box	one column per group
Violin with Point	one column per group
Violin with Quartile	one column per group
Violin with Stick	one column per group
Split Violin	the two columns compared (left and right half)
Half Violin	one column per group
Ridgeline	two or more columns (one ridge each)
QC (X-bar R) Chart	one column of measurements
Bland–Altman	the two methods (A and B)
2Ys Box Chart	two columns with different scales
Parallel Plot	numeric columns (optionally a group column first)
Parallel Index	numeric columns on a common scale (optionally a group first)
Parallel Sets	categorical columns (optionally a weight column last)
Alluvial	categorical columns (optionally a weight column last)
Sankey	source, target and value

Graph type	Columns to select
Network Plot	source, target and optionally the weight
Chord Diagram	source, target and value
Hierarchical Edge Bundling	source and target (names 'group.node')
Probability Plot	one column
Stacked Dot Plot	one or more columns
Interaction Plot	response, factor A (x axis), factor B (lines)
Main Effects Plot	response and one or more factors
Multi-Vari Chart	response, factor A (panels of points), factor B
Variability Chart	response, factor A, factor B
Symmetry Plot	one column
Run Chart	one column in time order
CDF Plot	one or more columns
Cause-and-Effect (Ishikawa)	category and cause (text columns)
Frequency Chart	one column (values or texts)
Grouped Histogram	value and group
Grouped Scatter	x, y and group
Grouped Pareto	group, category and value

Specialized

Graph type	Columns to select
Bullet Chart	category, value, target, then 1 to 3 range limits
Polar Contour + Gradient Vector	angle (degrees), radius and value
Contour Line + Gradient Vector	x, y and z on a grid
Piper (Trilinear) Diagram	Ca, Mg, Na+K, Cl, SO4, HCO3+CO3 (meq/L)
Durov Diagram	Ca, Mg, Na+K, Cl, SO4, HCO3+CO3 (meq/L)
Stiff Diagram	sample name, Na, Ca, Mg, Cl, HCO3, SO4 (optionally Fe, CO3)
Polar Vector θrθr	start angle (°), start radius, end angle (°), end radius
Polar Vector θRAM	angle (°), radius, direction (°), magnitude
Compass Plot	x and y components (arrows from the centre)
Radial Bar	category and value
Radial Stacked Bar	category and two or more values
Right-Triangle Ternary	A, B and C (any scale)
Stacked Radial Plot	category and two or more values
Ternary Vector	start A, B, C and end A, B, C
Wind Rose (binned data)	direction (°), then one frequency column per speed class
Wind Rose (raw data)	direction (°) and speed
Window-view Browser Graph	x and y

Financial

Graph type	Columns to select
Bridge (Waterfall) Chart	category and change
Horizontal Bridge Chart	category and change
Stacked Bridge Chart	category and two or more changes
Stacked Bridge Chart with Total	category and two or more changes
High-Low-Close	date/x, high, low, close

Graph type	Columns to select
OHLC-Volume	date/x, open, high, low, close, volume
Line (stock)	date/x, close and optionally volume

3D XYY

Graph type	Columns to select
XYY 3D Bars	x (or categories) and one or more y columns
3D Side by Side Bars	x (or categories) and two or more y columns
3D Stacked Bars	x (or categories) and two or more y columns
3D 100% Stacked Bars	x (or categories) and two or more y columns
3D Bars with Error Bar	x, then pairs of y and its error
3D Wall	x and one or more y columns
Stacked Wall	x and two or more y columns
100% Stacked Wall	x and two or more y columns

3D surfaces

Graph type	Columns to select
Color Fill Surface	x, y and z on a grid
X and Y Constant with Base	x, y and z on a grid
3D Ternary Colormap Surface	A, B, C (fractions) and z
Surface with Error Bar (color fill)	x, y, z and the error of z on a grid
Surface with Error Bar (colormap)	x, y, z and the error of z on a grid
Multiple Surfaces	x, y and two or more z columns on a grid
3D Stacked Surfaces	x, y and two or more z columns on a grid
3D Stacked Heatmaps	x, y and one or more value columns on a grid

3D symbol, bar, vector

Graph type	Columns to select
3D Scatter with Z Error	x, y, z and the error of z
3D Bars with Categorical XY	x (text or numbers), y (text or numbers), z
3D Stacked Bars (XYZ)	x, y and two or more z columns
3D 100% Stacked Bars (XYZ)	x, y and two or more z columns
3D Floating Bars	x, y, z from and z to
3D Vector XYZ XYZ	start x, y, z and end x, y, z
3D Vector XYZ dXdYdZ	x, y, z and the components dx, dy, dz

Contour

Graph type	Columns to select
Contour – Categorical Z	x, y and a text (category) column
Polar Heatmap	angle (°), radius and value on a grid

Graph type	Columns to select
Split Heatmap	x, y (numbers or texts), value 1 (upper triangle), value 2
Heatmap with Split Tiles	x, y (numbers or texts) and 2 to 4 values
Ternary Contour	A, B, C (fractions) and value
Overlaid Contour	x, y and two z columns on a grid
Ternary Overlaid Contour	A, B, C (fractions) and two values

Waterfall, grouped, image

Graph type	Columns to select
Waterfall Color Mapped by Y	x and two or more y columns
Waterfall Color Mapped by Z	x and two or more y columns (numeric names = z)
3D Waterfall	x and two or more y columns
Grouped Column (by two factors)	factor A (groups), factor B (bars), value
Grouped Box (by two factors)	factor A (groups), factor B (boxes), value
Grouped Line + Symbol	x (or factor A), group, value
Grouped Scatter (index by two factors)	factor A, factor B, value
Image + Profiles	x, y and z on a grid

Map

Graph type	Columns to select
Pie Map	x, y, then two or more values (one pie per row)
Bar Map	x, y, then one or more values (one bar group per row)

Apps (0.21)

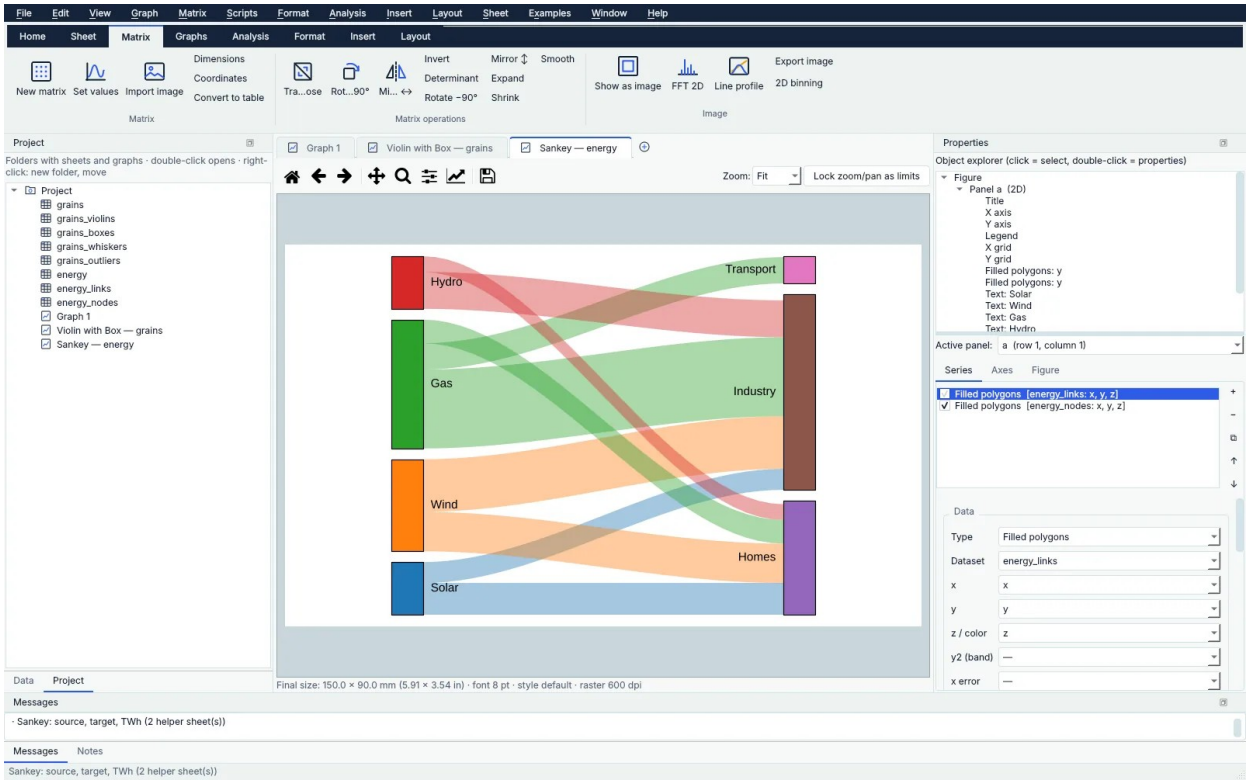
The graphing Apps of Origin that draw a new kind of graph. The other Apps (Graph Maker, Batch Plotting, Common Bar Width, Layer Stack Manager, Color Editor, LaTeX) are tools in **Graph → Apps**; see Origin Apps.

Graph type	Columns to select
Nyquist Plot (with Bode)	frequency, Z′ (real) and Z″ (imaginary)
Polar Contour Profile	angle (°), radius and value
Means Plot	one column per group
Paired Comparison Plot	the two paired columns (before, after)
Volcano Plot	log2 fold change, p-value and optionally a label column
Rectangle and Ellipse Profile	x, y and z (a grid or scattered)
Plot Sub Matrix	x, y and z on a grid
Color Spectrum Plot	wavelength (nm) and intensity
Chromaticity Diagram (CIE 1931)	x and y chromaticity, or X, Y, Z tristimulus
Hansen Solubility Parameter Plot	δD, δP, δH, good (1) or bad (0), optional solvent name
Ternary Contour Profile	A, B, C (fractions) and value
Kernel Density (Ternary)	A, B and C

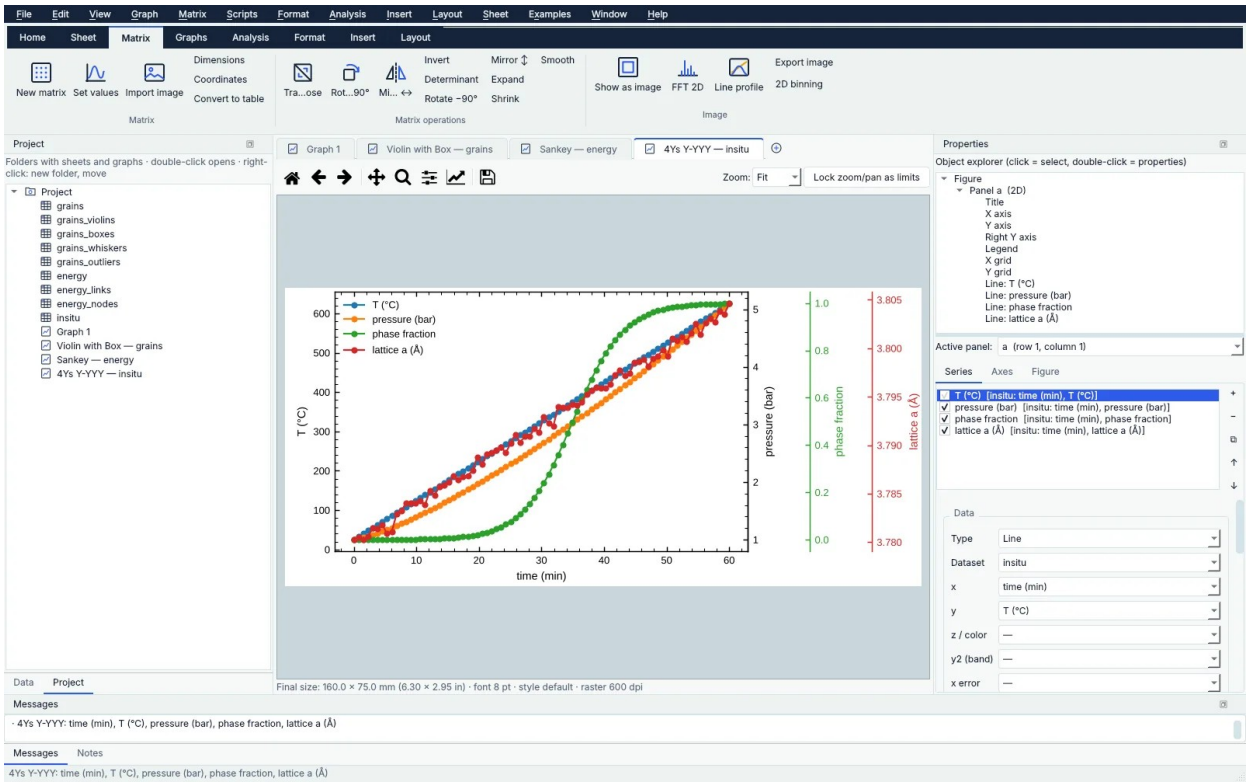
Graph type	Columns to select
Kernel Density (Polar)	angle (°) and radius
Soil Texture Triangle (USDA)	sand, silt and clay (%)
Sediment Classification (Folk)	sand, silt and clay (%)
Taylor Diagram	reference (observations), then one column per model
3D Polar Strain Map	angle (°), radius and strain (a polar grid)
3D Wind Rose	direction (°) and speed
Isosurface	x, y, z (a 3D grid) and the value
3D Wall Profile	x, y and z on a grid
Zoomed Inset	x and one or more y
Zoomed Inset Plus (two or three zooms)	x and one or more y
Venn Diagram	two or three columns of items (one item per cell)

New drawing tools behind them

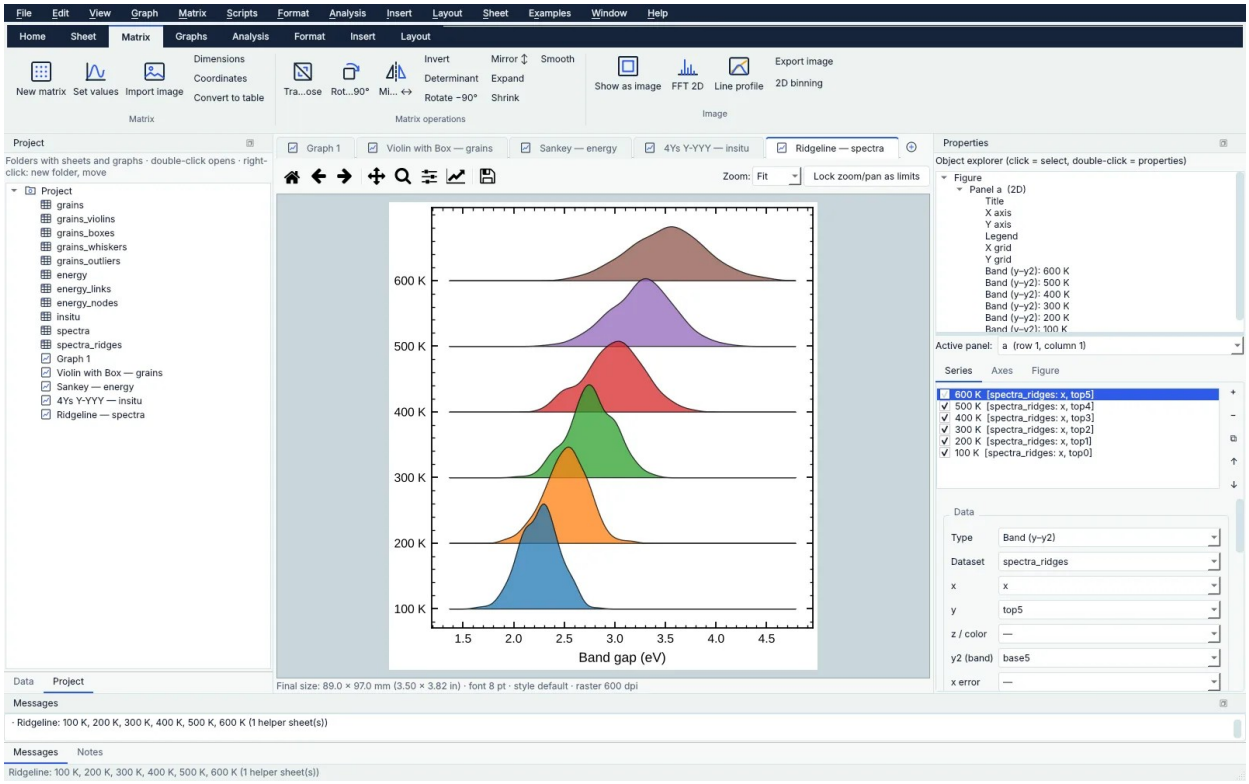
- **Filled polygons:** a curve kind whose polygons are separated by empty rows, each with its own colour or with a colour map. Kites, violins, Sankey links and chords use it.
- **Line segments:** lines broken at empty rows, for rugs, before/after pairs and parallel plots.
- **Colour-mapped line:** a line coloured by a third column.
- **3D polygons, 3D segments, 3D vectors and flat 3D heat maps** (0.20): 3D bars, walls, error bars, vectors with a *w* column, and heat maps stacked at different heights.
- **Lit 3D polygons and new colour maps** (0.21): isosurfaces are shaded by a light; the *visible_spectrum* colour map and your own colour maps (Graph → Apps → Color editor) appear in every colour-map list.
- **Extra axes:** in the series properties, *Y axis* can be the 2nd to 4th right axis, the 2nd or 3rd left axis (offset and coloured like the curve), or the top X axis (Double X).



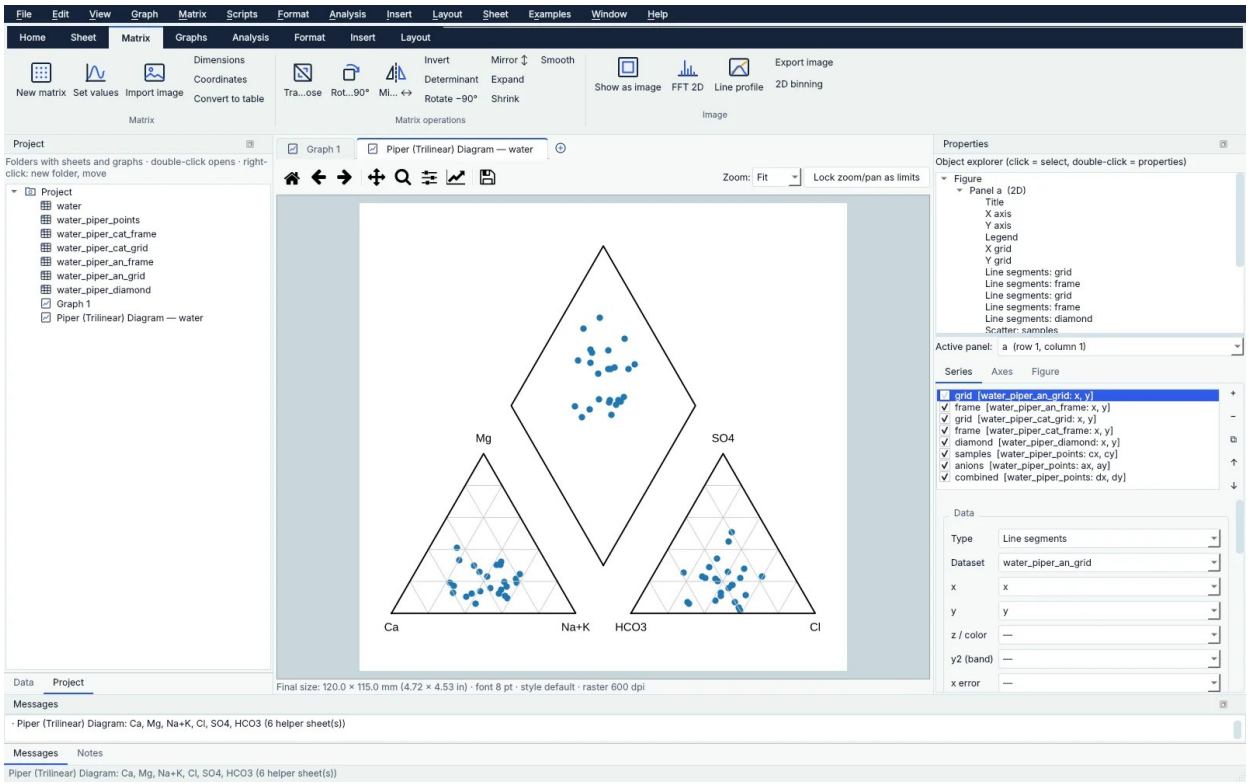
Sankey diagram



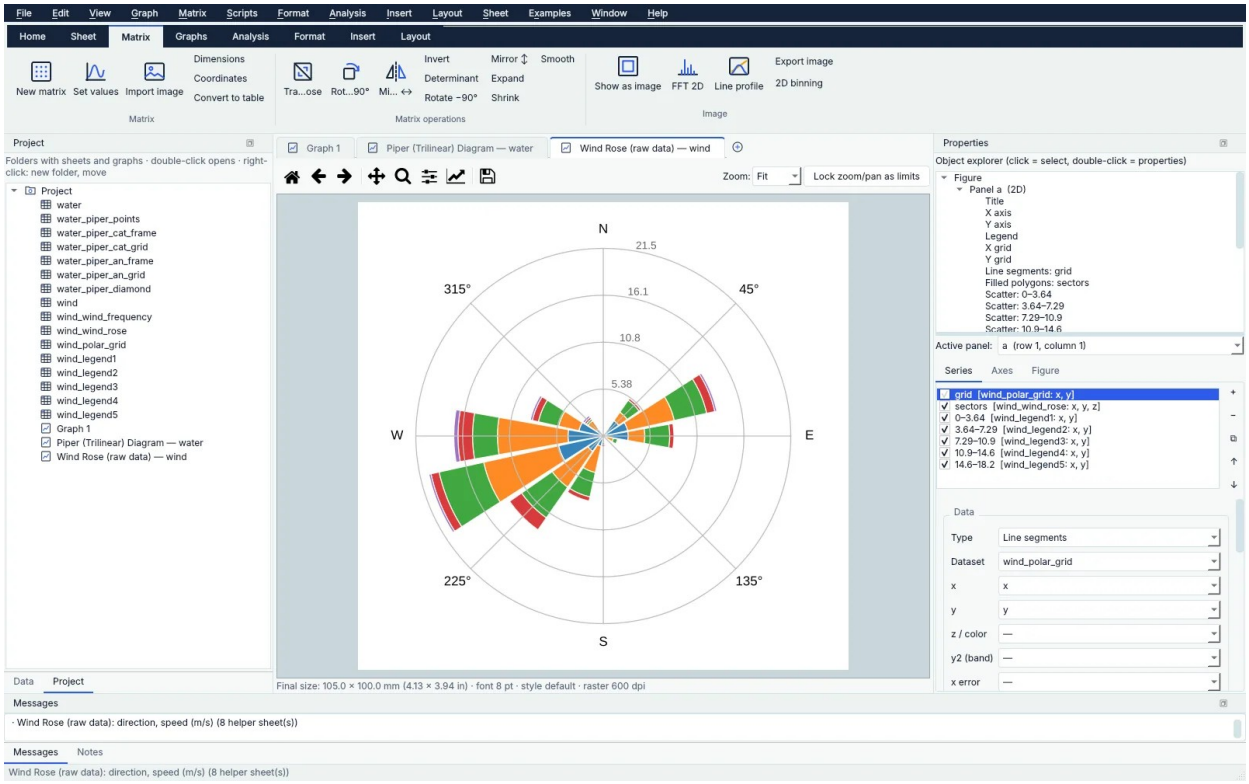
Four Y axes



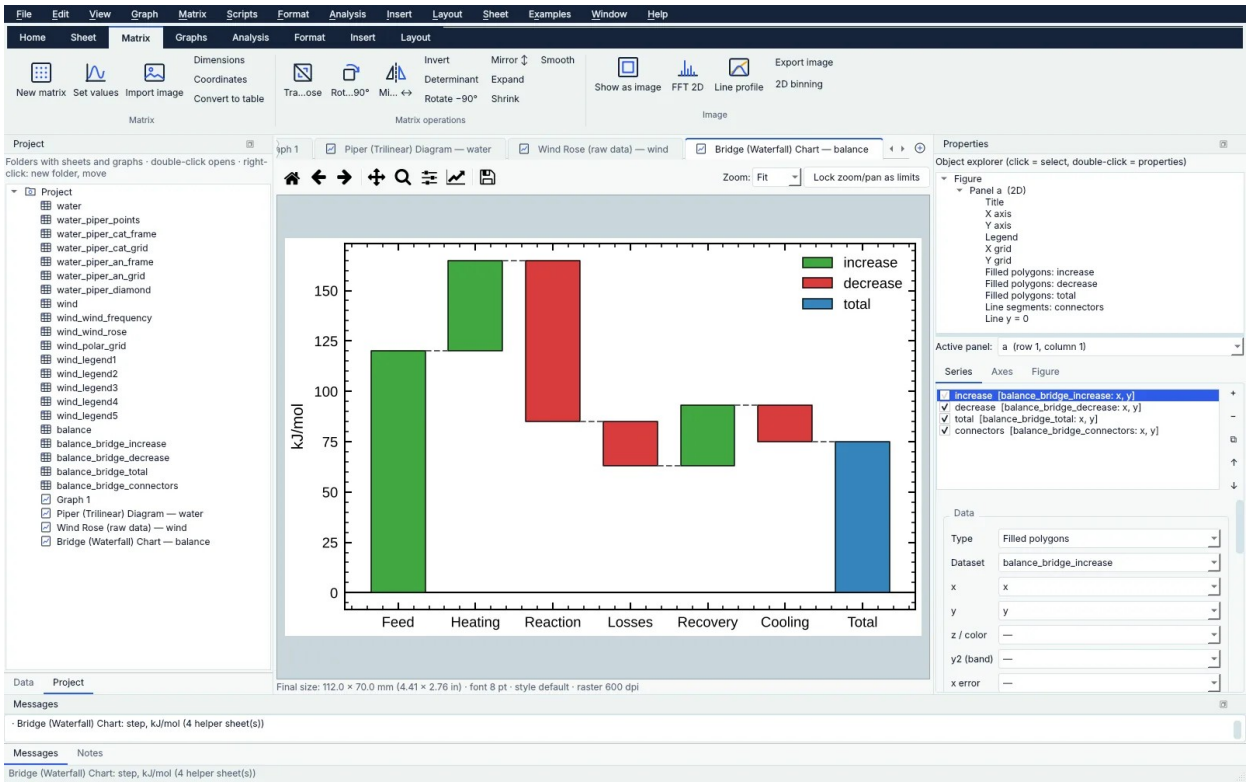
Ridgeline



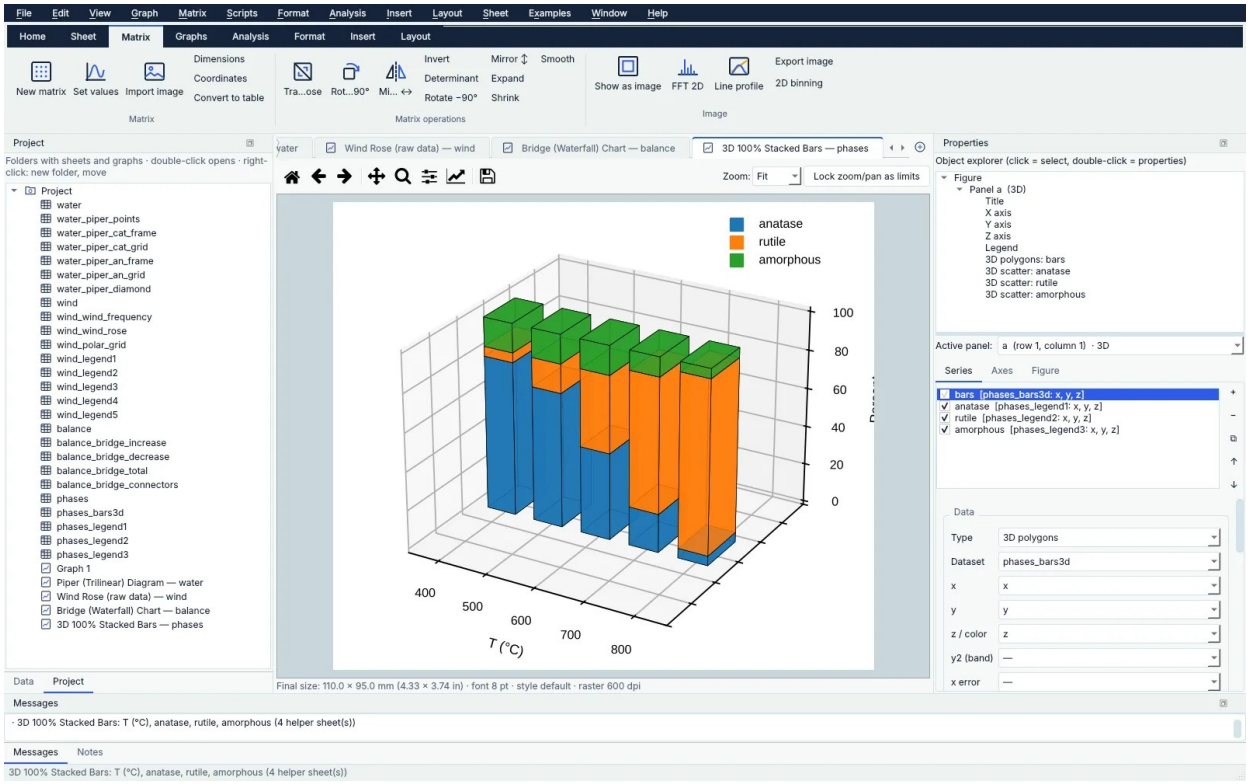
Piper diagram



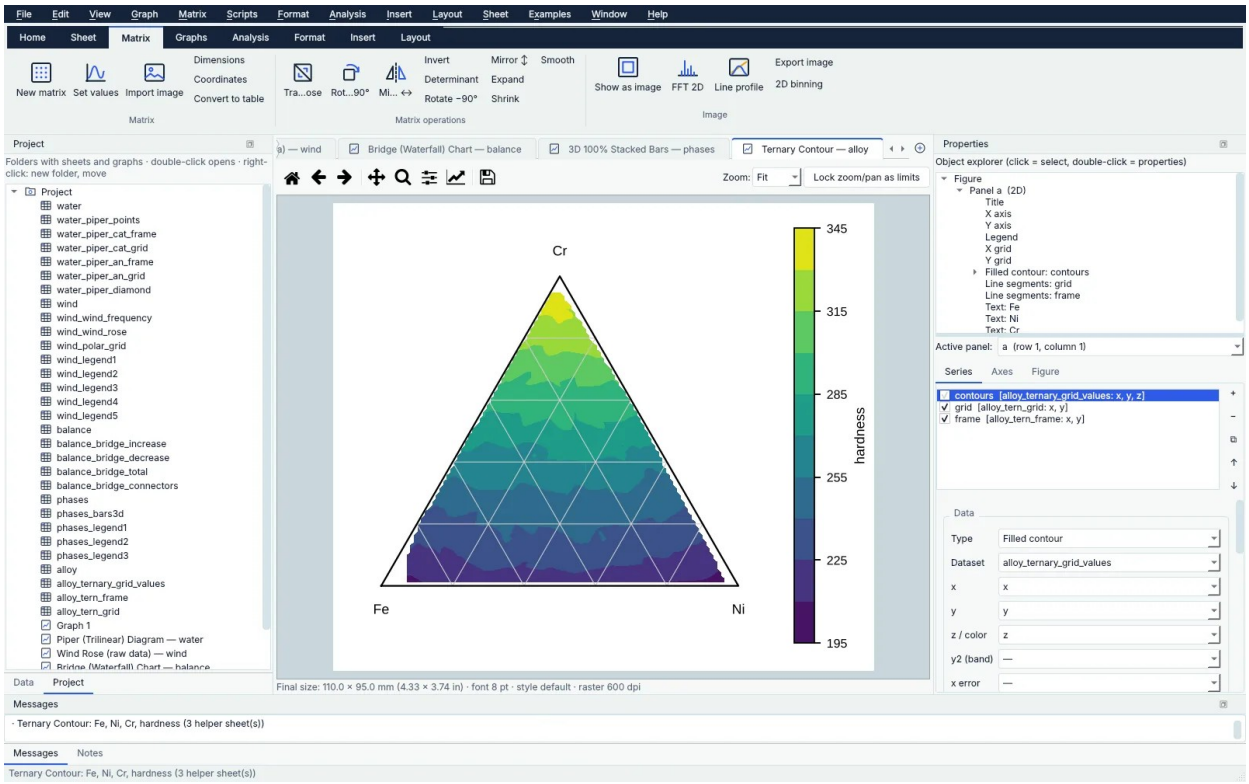
Wind rose



Bridge chart



3D 100 % stacked bars



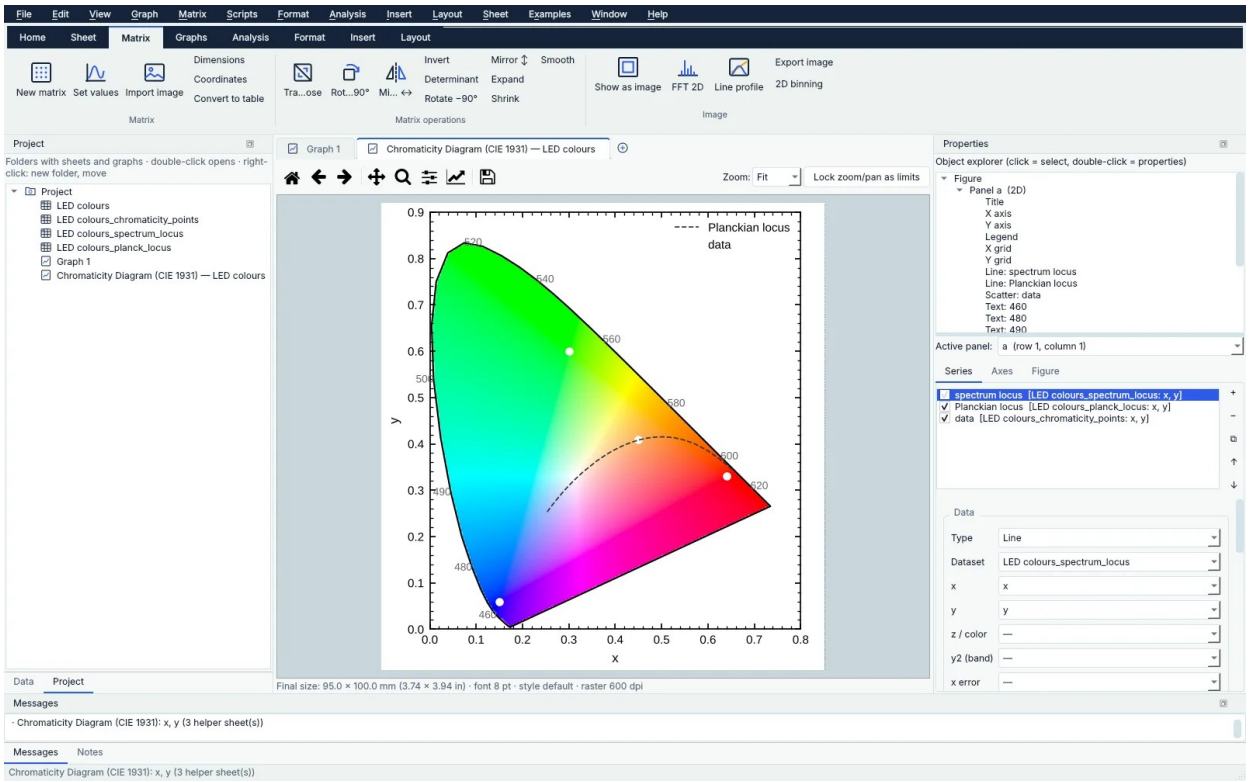
Ternary contour

Origin Apps

Version 0.21 added the free Apps of Origin (standard edition) for analysis and graphing. They sit in four places:

- **Analysis → Apps** for curve fitting, peaks, mathematics, signal and statistics;
- **Sheet → Apps** for data processing;
- **Graph → Apps** for tools that make or change graphs;
- **Graph → Origin graph types → Apps** for the 23 Apps that draw a new kind of graph (see Origin graph types).

Tools that work on columns read the columns you select in the sheet (Ctrl+click the headers). Tools that work on a curve use the curve you clicked on the graph. Tools that compare curves use every curve of the active panel. Results go to new sheets, to the Messages dock and, where it helps, to a graph.



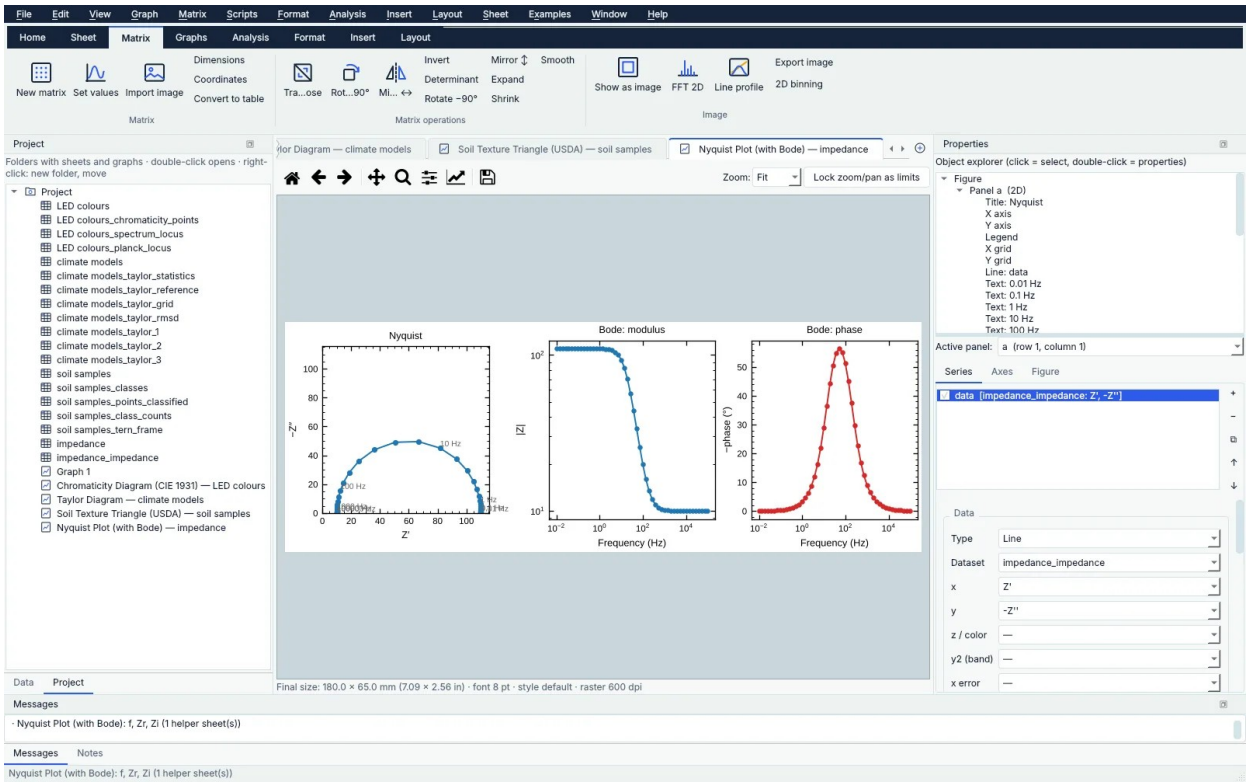
Chromaticity diagram

Curve fitting

App	What it does
Find a fitting function	Fits every built-in model to the selected curve and ranks them by the corrected Akaike criterion (AICc), with Akaike weights. You can have the best model fitted to the curve at once.
Simple fit	A quick fit with a model from a short list or your own expression `y = ...`. The fitted curve and a result box go on the graph and are refitted when the data change.
2D smoother	A smoothing surface through scattered x, y, z (radial basis functions). Smoothing 0 passes through the points. You get the surface as a contour map and the residuals at each point.
Compare datasets	Are the curves of the panel described by one set of parameters? Each curve is fitted alone and all of them together. The extra-sum-of-squares F test and the AIC decide.
Compare linear fit parameters and datasets	Tests whether the straight lines through the curves have equal slopes and, if so, equal intercepts (as in an analysis of covariance).

Peaks and mathematics

App	What it does
Align peaks	Shifts the curves of the panel along x so that their peaks line up with the first curve. Alignment can use the maximum, the centroid above half height or the cross-correlation.
Convert global to local coordinates	x, y in a frame with a new origin and a rotated x axis.
Popular constants	A sheet of CODATA constants with units and uncertainties.
Curvature radius	Curvature $\kappa = \lvert y'' \rvert / (1 + y'^2)^{3/2}$ and its radius along the curve.
Tangent	The tangent, and on request the normal, at an x, drawn on the graph.
Tafel extrapolation	Select a polarisation curve: x is the potential in V and y is the current. Straight lines through log|i| in an anodic and a cathodic window give Ecorr, icorr, βa and βc.
Distance between two points	Distance, dx, dy, angle and slope, with the segment drawn on the panel.
Level crossing	The x values where the curve crosses a level: rising, falling or both. Hysteresis ignores noise around the level.



Nyquist plot with Bode panels

Signal and statistics

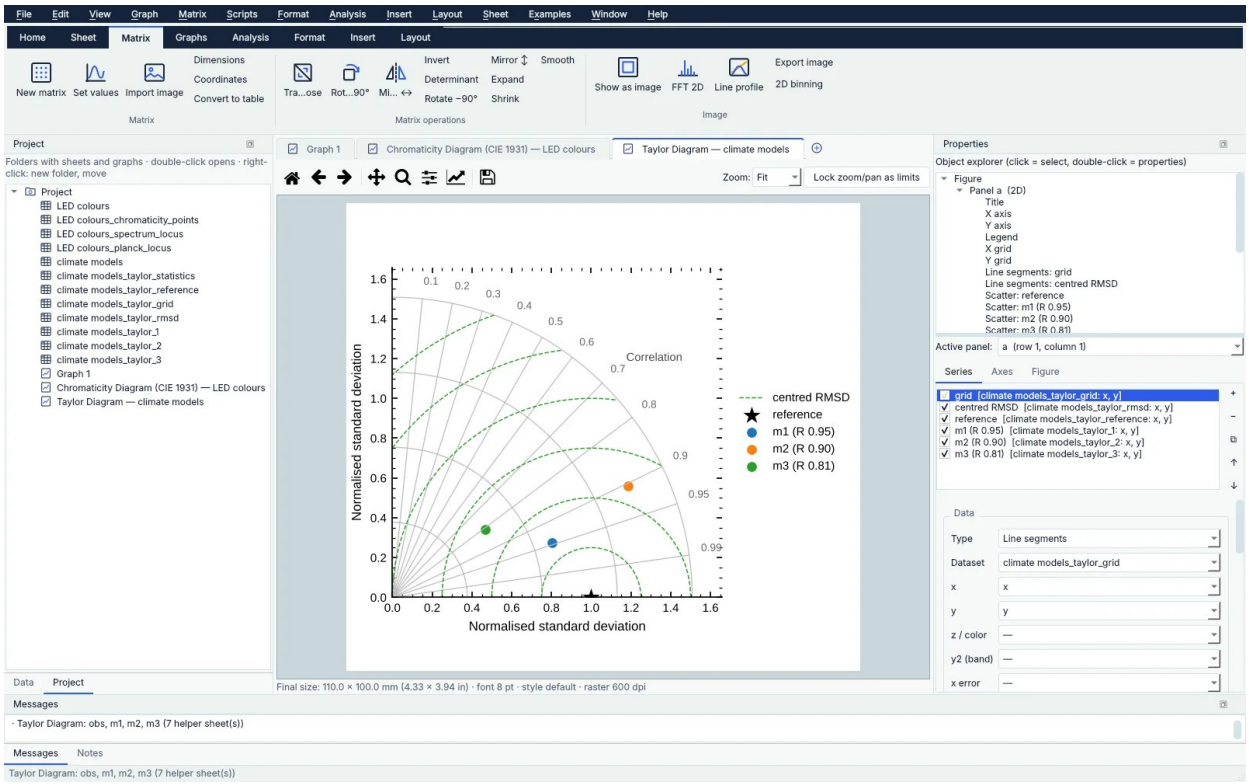
App	What it does
White noise	A sheet of Gaussian or uniform white noise, with a seed so the result can be repeated.
NetCDF data analysis	Choose a variable of a NetCDF or HDF5 file and the one or two dimensions to keep. The others are reduced by mean, sum, minimum, maximum, standard deviation or median. The result is a curve or a map.
Mann–Kendall test	Trend test with the tie correction, Kendall's τ and Sen's slope.
Asymmetric correlation matrix	Correlation (Pearson, Spearman or Kendall) of every column of group A with every column of group B, with p-values and a heat map.
Logistic regression	Binary logistic regression. The dependent column holds 0 and 1. You get coefficients, standard errors, Wald tests, odds ratios with 95 % limits, deviance, AIC, the likelihood-ratio test, McFadden's R^2, fitted probabilities and the confusion counts.
Hurst exponent	Rescaled-range (R/S) analysis with the log–log graph.
Hysteresis	For the selected loop: area, coercive fields, remanence, saturation and squareness.

Data processing (Sheet → Apps)

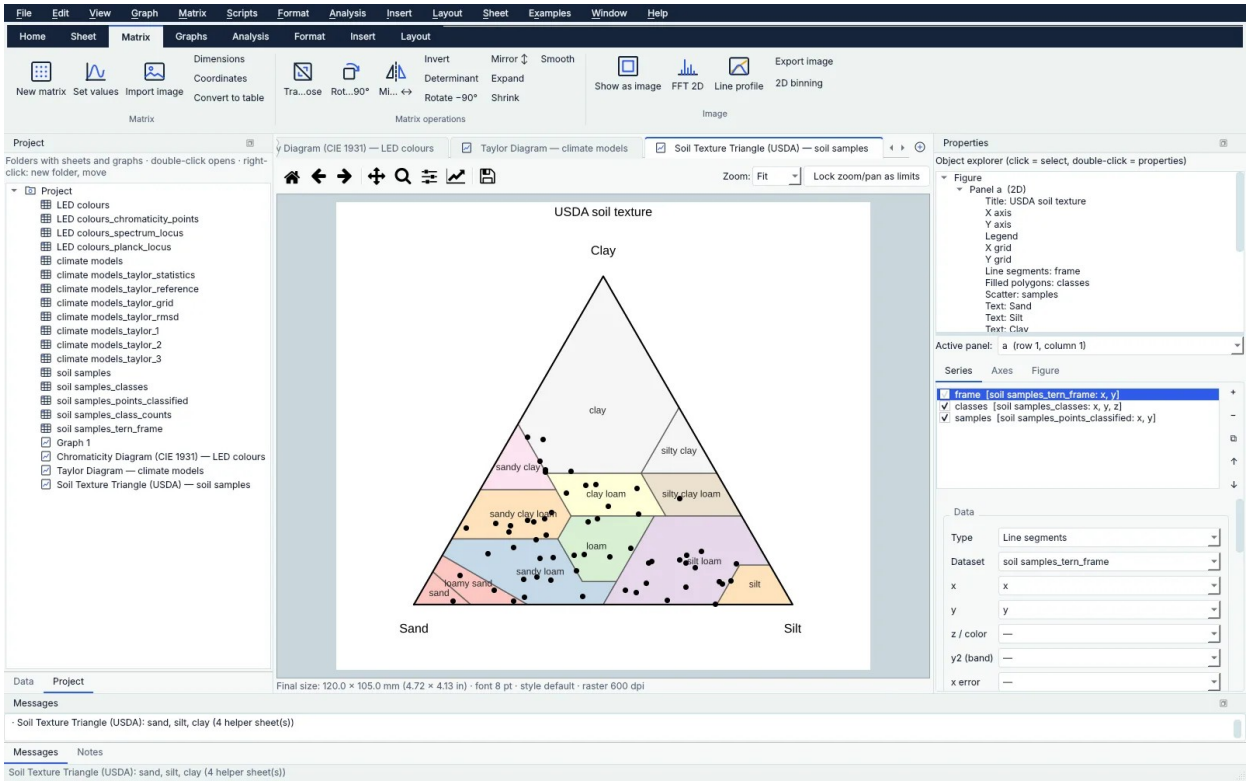
App	What it does
Text to columns	Splits a text column by a delimiter into new columns. Numeric parts become numbers.
Data slicer	Keeps the rows inside a numeric range or among chosen texts, either as a new sheet or by hiding the other rows.
Extract XY data from graph	The x, y of every curve of the panel that lie inside its current axis limits.
Periodic report	Re-reads a data file at an interval. Each time the file has changed, a row with the time, the number of rows and the last, mean, minimum and maximum of each column is added to a report sheet. Run the command again to stop.
Project manager	Lists the projects of a folder and its subfolders, with their sheets, graphs, notes and dates.
Project searcher	Finds the projects whose sheet names, column names, graph names, titles, axis labels or notes contain a text.

Graph tools (Graph → Apps)

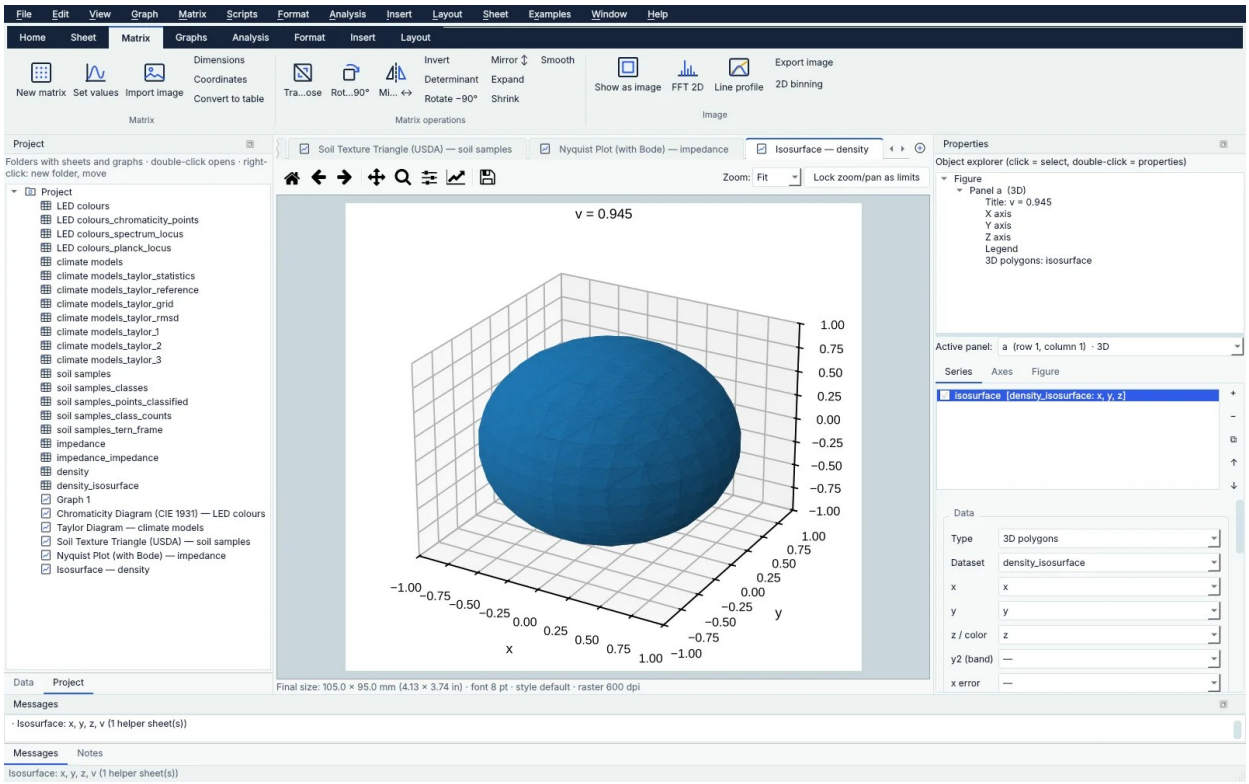
App	What it does
Graph maker	Click the columns in the order you want and choose any graph type, basic or Origin.
Batch plotting	Repeats the active graph for other sheets with the same column names, one new graph per sheet.
Common bar width	Gives every bar series of the graph, or of the active panel, the same width.
Layer stack manager	Reorders, hides and shows the panels of the graph. Insets stay with their parents.
Color editor	Makes your own colour maps from a list of colours (smooth or in flat bands). They are kept between sessions and appear, with their reversed "_r" forms, in every colour-map list.
LaTeX equation	Typesets LaTeX math, such as `\frac{a}{b}` or `\int_0^\infty e^{-x^2}dx`, on the panel.
Toolbar maker	Builds your own toolbar from any commands. It is kept between sessions.



Taylor diagram



USDA soil texture triangle


File Edit View Graph Matrix Scripts Format Analysis Insert Layout Sheet Examples Window Help
Home Sheet Matrix Graphs Analysis Format Insert Layout
New matrix
Set values
Import image
Dimensions
Coordinates
Convert to table
Matrix
Tra...ose
Rot...90°
Mi... ↔
Invert
Determinant
Rotate −90°
Mirror ↕
Expand
Shrink
Smooth
Matrix operations
Show as image
FFT 2D
Line profile
Export image
2D binning
Image
Project
Folders with sheets and graphs · double-click opens · right-click: new folder, move
Project
LED colours
LED colours_chromaticity_points
LED colours_spectrum_locus
LED colours_planck_locus
climate models
climate models_taylor_statistics
climate models_taylor_reference
climate models_taylor_grid
climate models_taylor_rmsd
climate models_taylor_1
climate models_taylor_2
climate models_taylor_3
soil samples
soil samples_classes
soil samples_points_classified
soil samples_class_counts
soil samples_tern_frame
impedance
impedance_impedance
density
density_isosurface
Graph 1
Chromaticity Diagram (CIE 1931) — LED colours
Taylor Diagram — climate models
Soil Texture Triangle (USDA) — soil samples
Nyquist Plot (with Bode) — impedance
Isosurface — density
Data
Project
Soil Texture Triangle (USDA) — soil samples
Nyquist Plot (with Bode) — impedance
Isosurface — density
Zoom: Fit
Lock zoom/pan as limits
v = 0.945
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
x
y
Final size: 105.0 × 95.0 mm (4.13 × 3.74 in) · font 8 pt · style default · raster 600 dpi
Properties
Object explorer (click = select, double-click = properties)
Figure
Panel a (3D)
Title: v = 0.945
X axis
Y axis
Z axis
Legend
3D polygons: isosurface
Active panel: a (row 1, column 1) · 3D
Series
Axes
Figure
isosurface [density_isosurface: x, y, z]
Data
Type
3D polygons
Dataset
density_isosurface
x
y
z / color
z
y2 (band)
x error
Messages
· Isosurface: x, y, z, v (1 helper sheet(s))
Messages
Notes
Isosurface: x, y, z, v (1 helper sheet(s))

Isosurface

Matrices and surfaces

A **matrix** is a grid of values z(x, y) with coordinates for its rows and columns — a map, an image, a 2D spectrum or a potential-energy surface. Matrices live in the project next to the sheets and have their own **Matrix** tab and menu.

Creating a matrix

- **Matrix → New matrix**: dimensions and the first and last x and y.
- **Matrix → Set values**: a formula of `i`, `j`, `x`, `y`, the cell value and other matrices, e.g. `exp(-(x^2 + y^2))` or `M1 - M2`.
- **Matrix → Import image**: an image becomes a matrix of intensities.
- **Sheet → Convert to matrix**: x, y, z columns on a regular grid become a matrix; **Matrix → XYZ** goes back.
- **Matrix → 2D binning**: counts two columns into a 2D histogram matrix.

Operations

Dimensions and coordinates, transpose, invert, determinant, rotate ±90°, mirror horizontally and vertically, expand (bilinear) and shrink (block mean), smooth, 2D FFT, a **line profile** along a line you choose, **Show as image** with a palette (grey, rainbow, custom) and **Export image**. Outputs are recalculated when the matrix changes. **Convert to table** writes the matrix as a table, as XYZ or as YXZ.

Scattered points → surface

Sheet → Surface from XYZ turns scattered x, y, z measurements into a surface, either by interpolation onto a grid (linear, cubic, nearest, thin-plate spline) or by fitting z = f(x, y) (plane, quadratic, 2D Gaussian or your own formula, with standard errors and R^2). It draws a contour, a map or a 3D surface with the measured points on top.

Graphs of matrices

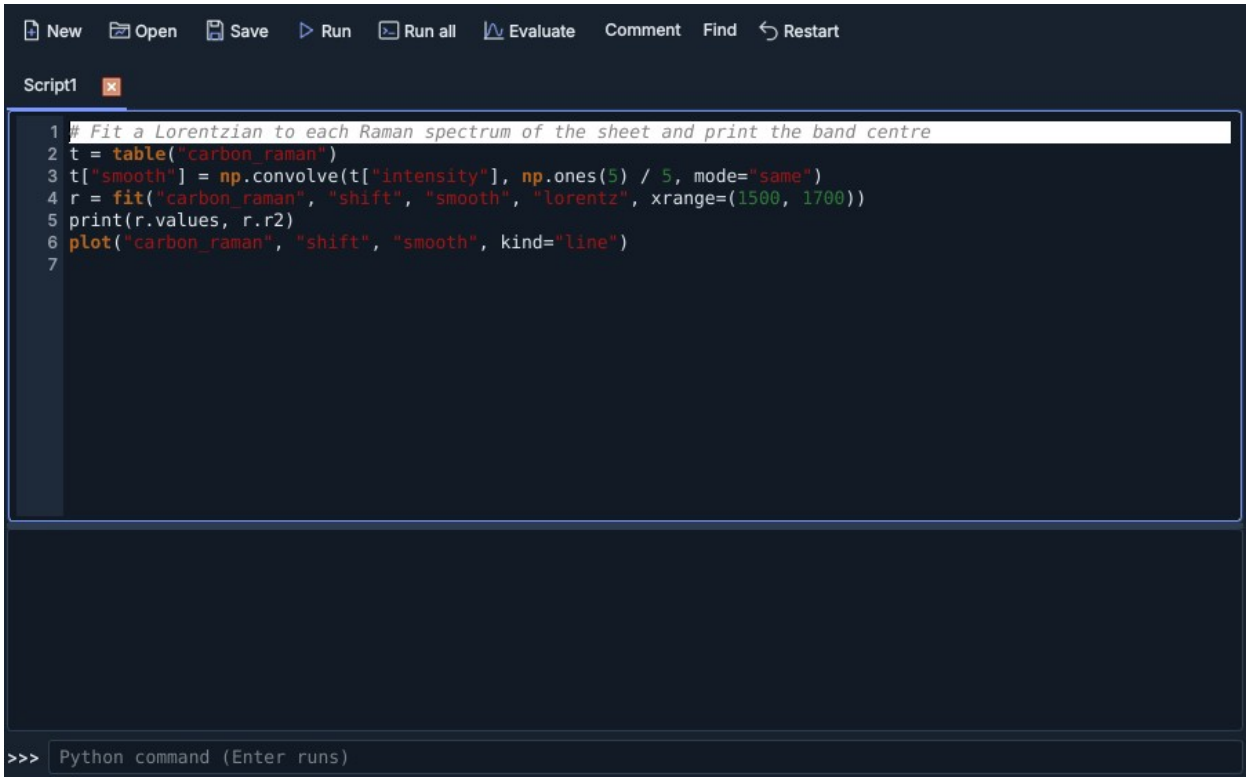
Contour (lines or filled), heat map (also with the values in the cells), 3D surface, mesh and the other 3D styles, contours projected on the floor and walls of a 3D box, and polar contours. SciDAVis and QtiPlot matrices open as matrices and are written back as matrices.

Scripts

Scripts → Script window (Ctrl+Shift+P) opens a window with several scripts in tabs, line numbers, syntax highlighting and a console. Scripts are Python and are saved in the project.

Key	Action
Ctrl+J	Run the current line or the selection
Ctrl+Shift+J	Run the whole script
Ctrl+Enter	Evaluate and write the value below the line (`#> value`)

The console below runs one command at a time and keeps the history. **Restart** clears the interpreter.



The script window

The API

The namespace already holds NumPy (`np`), pandas (`pd`), `math` and these functions:

Call	What it does
`table("name")`	A sheet; `t["col"]` reads a column as a NumPy array, `t["col"] = values` writes one (the table grows if needed).
`new_table("name", x=…, y=…)`	Creates a sheet from columns (or from a DataFrame).
`matrix("name")`, `new_matrix("name", z=…, x=…, y=…)`	Read or create a matrix; `m.values("sin(x)*cos(y)")` sets values by formula.
`fit(data, x, y, model, …)`	Least-squares fit (`model` is a built-in name such as `"gauss"` or a formula such as `"a*exp(-x/t) + c"`); accepts `p0`, `sigma`, `fixed`, `xrange`, `bounds`, `method`, `tol`, `maxfev`, `scale_errors`, `n_curve`, `curve_x`. Returns `values`, `errors`, `r2`, `report()`…
`plot(data, x, y, kind="line", new=False)`	Adds a curve to the active graph (or a new one).
`export("figure.pdf", dpi=300)`	Exports the active graph.
`note("text")`	Appends to the project notes.
`recalculate()`	Redoes the analyses that depend on changed data.
`tables()`, `matrices()`, `datasets()`	Names in the project.

```
# Fit every spectrum of a project and collect the peak centers in a table
centers, errors = [], []
names = [n for n in tables() if n.startswith("spectrum_")]
for name in names:
    r = fit(name, "x", "y", "gauss")
    k = list(r.names).index("xc")
    centers.append(r.values[k])
    errors.append(r.errors[k])
new_table("centers", sample=names, xc=centers, xc_err=errors)
plot("centers", "sample", "xc", kind="scatter", new=True)
```

Without the window

The same API works from the command line, for batch jobs and clusters:

```
scivelyx script analysis.py                                  # data created by the script itself
scivelyx script analysis.py --project project.scivelyx.json     # on an existing project
scivelyx script analysis.py --project p.scivelyx.json --save p2.scivelyx.json
```

Scripts run with your own permissions, like any Python program, so run only scripts you trust.

Coming from Origin, SciDAVis and QtiPlot

SciVelyx Graphs opens and saves the projects of the programs you already use, so years of finished work come along. Every import and export produces a **report** with three lists: what came across exactly (✓), what came across approximately (≈, with the reason) and what did not come across (□, with the reason). Nothing disappears without a warning.

Program	Open in SciVelyx Graphs	Save from SciVelyx Graphs
SciDAVis (`.sciprj`, `.sciprj.gz`, every version)	directly	`.sciprj` in the SciDAVis 2.9 format
QtiPlot (`.qti`, `.qti.gz`, 0.7 to 0.9.8)	directly	`.qti` in the 0.9.8 format
Origin (`.opj`, `.opju`)	through Origin itself (below)	a Python script that rebuilds the project inside Origin and saves `.opju`

QtiPlot 0.9.8 projects, which SciDAVis does not open, reach SciDAVis through SciVelyx Graphs: open the `.qti` and export it as `.sciprj`.

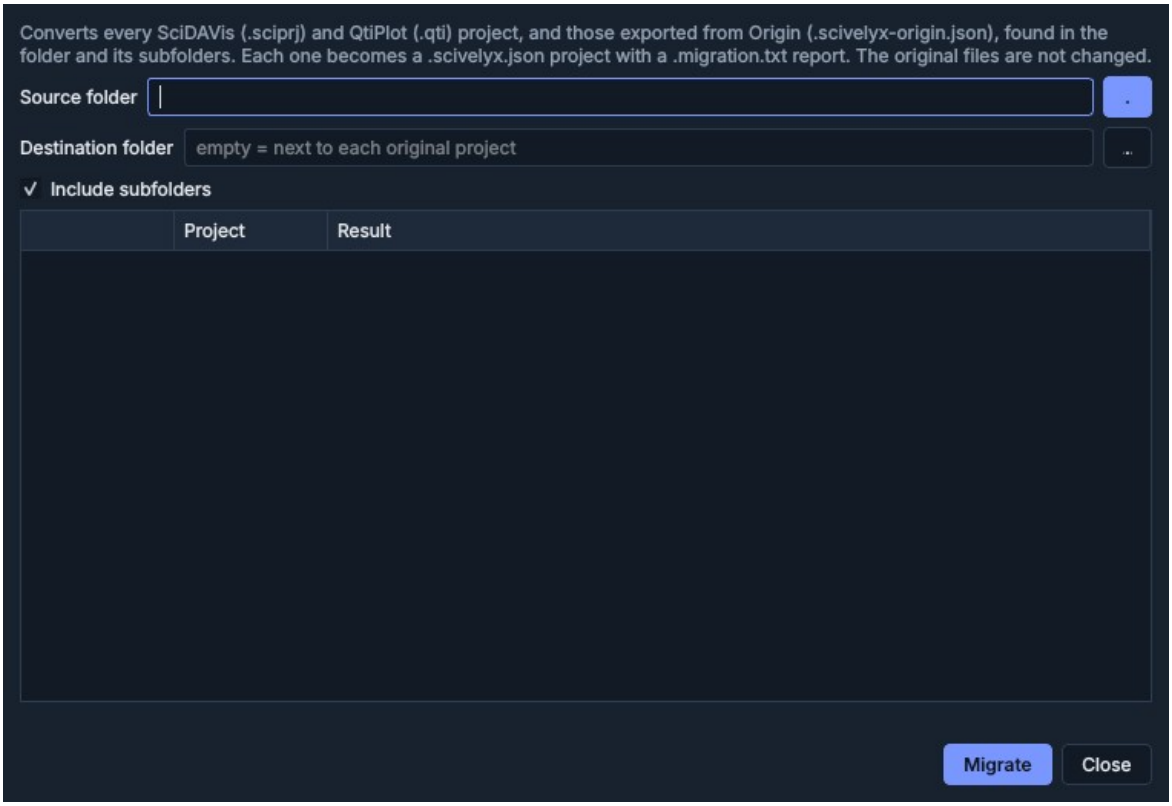


A migration report

One project

File → Open project (Ctrl+Shift+O) and choose the `.sciprj`, `.qti` or `.scivelyx-origin.json`. The original is never overwritten: the project opens untitled and is saved as `.scivelyx.json`.

A whole folder



Migrating a whole folder of projects

File → Migration → Migrate a folder of projects... converts every project in a folder and its subfolders into `.scivelyx.json`, each with its `.migration.txt` report. From the command line:

```
scivelyx migrate old_projects/ -o converted/
```

Origin projects

File → Migration → Script to export from Origin... (or `scivelyx origin-script`) saves a small Python script. Open the project in Origin 2021 or later and run the script in Origin's Python window (Connectivity → Python). It only reads the project and writes a `.scivelyx-origin.json` next to it; open that file in SciVelyx Graphs. The other way round, **File → Migration → Export project to...** with the `.py` type writes a script that rebuilds sheets, graphs and notes in Origin.

What is converted

- **Sheets:** values, text and date columns, designations, long names and units, comments, masks and formulas. A formula stays live only if it reproduces the saved values; otherwise the values stay and the formula goes to the column comment, and the report says why.
- **Matrices:** they arrive as matrices with their coordinates and go back as matrices.

- **2D graphs:** lines, symbols, bars, areas, stems, steps, histograms, boxes, pies, vectors, colour and contour maps, function and fit curves (with their parameters), error bars, right and top axes, log scales, limits, grids, titles with sub- and superscripts, fonts, legends, texts, arrows and several layers.
- **3D graphs:** surfaces of matrices, of functions and of x, y, z columns.
- **Notes, results log and folders.**

Known limitations

- Graph types that SciDAVis and QtiPlot do not have (polar, ternary, Smith, violin...) are exported to them as an image of the graph, with the PNG next to the project.
- Nested folders become one level in the project explorer.
- The Origin scripts use the official `originpro` API of Origin 2021 or later; reading `.opj` files without Origin is not offered yet.
- `.scitrace-origin.json` and `.sciplot-origin.json` files, written by the exporters of the program's earlier names, still open.

Export and publication

Size first

A figure is defined in millimetres and points, so what you see is what the journal prints. **Format → Journal** sets the width for common journals and presentations:

Preset	Width
Nature — 1 / 2 columns	89 mm / 183 mm
APS (PRB, PRL) — 1 / 2 columns	3 3/8 in / 7 in
ACS — 1 / 2 columns	3.25 in / 7 in
Elsevier — 1 / 1.5 / 2 columns	90 / 140 / 190 mm
RSC — 1 / 2 columns	83 mm / 171 mm
Presentation 16:9	254 mm

The figure dialog (double-click outside the panels) sets the exact size, the DPI, the base font, the math font (Computer Modern as in LaTeX, STIX…), Matplotlib style sheets and the colour cycle.

Formats

File → Export (Ctrl+E):

Format	Use
PDF, SVG, EPS	Vector files at the exact size, with TrueType fonts embedded and the text editable.
PNG, TIFF (LZW), JPEG	Images at the resolution you choose.
Copy figure (Ctrl+Shift+C)	Pastes into documents and presentations.
Export Python script…	A script that redraws the figure identically, pixel for pixel, from the project data.

File → Print..., **Print preview** and **Print all graphs...** send graphs to a printer or a PDF.

Several graphs on one page

Layout → Layout page (Ctrl+Shift+L) composes graphs from different tabs and texts on a page in millimetres: move and resize them, arrange them in a grid, align them, add panel letters, and export vector PDF, PNG or TIFF. The pages are saved in the project.

The project file

File → Save writes one `.scivelyx.json` with the graphs, the notes, the scripts, the layout pages, the column formulas and designations and the data — embedded, or by reference with a SHA-256 checksum. Projects saved by earlier versions always open in later ones, including the `.scitrace.json` and `.splot.json` files of the program's earlier names.

Command line

Every package contains `scivelyx`, the command-line side of SciVelyx Graphs: on Windows next to `SciVelyx-Graphs.exe` (or on the PATH, if you chose it when installing), on macOS in `SciVelyx-Graphs.app/Contents/MacOS/`, on Linux next to `SciVelyx-Graphs` in the folder or as `SciVelyx-Graphs.AppImage scivelyx …`.

Command	What it does
`scivelyx gui [project]`	Opens the window (with a project or data file).
`scivelyx render project.scivelyx.json -o figure.pdf [--dpi 300]`	Draws a project to PDF, SVG, EPS, PNG or TIFF.
`scivelyx quick data.csv -x T -y rho -o figure.pdf`	Plots columns of a data file (`--kind`, `--preset`, `--xlabel`, `--ylabel`, `-z` for maps).
`scivelyx github <url> [-r]`	Lists the tabular files of a GitHub repository or folder (`-r`: subfolders too).
`scivelyx migrate paths… -o folder`	Converts SciDAVis and QtiPlot projects (and Origin exports) to SciVelyx, with reports.
`scivelyx export-project project.scivelyx.json -o out.sciprj`	Writes a project for SciDAVis (`.sciprj`), QtiPlot (`.qti`) or Origin (`.py`).
`scivelyx origin-script -o export_from_origin.py`	Saves the script that exports `.opj`/`.opju` from inside Origin.
`scivelyx script file.py [--project p.scivelyx.json] [--save out.scivelyx.json]`	Runs a Python script with the scripting API.
`scivelyx selftest [--gui]`	Runs the self-test and writes the report and `selftest.zip`.
`scivelyx --version`	Prints the version.

`scivelyx <command> --help` lists every option of a command. In installations made from the source with `install.sh`, the old `scitrace` and `scitrace-gui` commands still work and start the same program.

Examples

```
# Redraw every figure of a thesis after changing the theme in the projects
for p in figures/*.scivelyx.json; do scivelyx render "$p" -o "pdf/$(basename "$p" .scivelyx.json).p

# Convert a whole archive of SciDAVis and QtiPlot projects
scivelyx migrate ~/old_projects -o ~/scivelyx_projects
```

FAQ and troubleshooting

General

Is SciVelyx Graphs a fork of SciDAVis or QtiPlot? No. It is written from scratch on PySide6, Matplotlib, NumPy, pandas and SciPy, and reads and writes their file formats. No code from QtiPlot, SciDAVis or LabPlot is used.

Does it need an internet connection? No. The network is used only when you ask for it, for example to read data from GitHub.

Which language is the interface in? English, the program's only language. This manual is in English and Portuguese.

I used SciPlot or SciTrace before. What changes? Only the name: the program was called SciPlot up to 0.14 and SciTrace from 0.15 to 1.0.0rc5. Old projects (`.splot.json`, `.scitrace.json`), themes, the fit-function library and the preferences are read and carried over, and in installations made with `install.sh` the old `scitrace` commands still start the program.

Does it cover everything SciDAVis and QtiPlot do? Yes. The menus of both and, since version 0.12, the options inside their fit, axis, curve and 3D dialogs. Version 0.17 added the last 21 items, such as 3D pies, logarithmic 3D axes, pictures as symbols, a log scale after an axis break, the Plot wizard, Append project and custom script actions. See Plot wizard, windows and presentations.

Installation

Windows says "Windows protected your PC". The test builds are not code-signed yet: click **More info → Run anyway**.

macOS says the app cannot be opened. Control-click the app in Applications → **Open** → **Open**; on macOS 15, System Settings → Privacy & Security → **Open Anyway**.

The AppImage does not start. Install `libfuse2`, or use the `.tar.gz` package.

On Linux (Wayland) or WSL, a double click on the title bar did not maximize the window. Fixed in 1.0.0rc6: the program now uses Qt's "adwaita" title bar, where a double click maximizes and restores. To use another one, set the environment variable `QT_WAYLAND_DECORATION`.

Working with the program

A formula column does not update. If you typed a value into it, it became an ordinary data column; set the formula again with **Sheet → Set values**.

The fit result does not follow my data. Check that **Analysis → Automatic recalculation** is on, or press **Recalculate now** (Ctrl+F5).

Some text shows a red warning in the text dialog. The math notation between `$…$` is invalid (for example an unclosed brace); the dialog keeps the figure unchanged until it is fixed.

Bold does not affect `$…$` math. Use the **B** button of the text dialog, which writes `\mathbf{}`.

My numbers use a decimal comma. To read such files, use **File → Import wizard** and choose *Decimal: comma* (*1,5*) (Worked examples, section 5). In formulas the decimal separator is always the point.

Reporting a problem

Open an issue in the downloads repository with the version (Help → About), the system, what you did and what happened, and — if you can share it — a small project or data file that shows it. Attach the `selftest.zip` of **Help → Run the self-test...** too.